

STRATEGIC CHOICE-MAKING
IN COMPLEX SPACES OF INNOVATION

BY

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ACKNOWLEDGMENT

No dissertation is done in isolation. It is shaped in early mornings of quiet determination, in charged conversations, and in the hands of those who lift us when the weight feels unbearable. It is never a solitary endeavor; it is shared by those who stand beside you, offering guidance, encouragement, and the quiet assurance that you are never truly alone. This is where these acknowledgments begin.

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AUTHORSHIP STATEMENT

I, XXXX XXXXXX, attest that the work in this thesis is substantially my own.

In accordance with the disciplinary norm of authorship in Design (see IIT Faculty Handbook, Appendix S), the following collaborations occurred in the thesis.

My adviser, Prof. Carlos Teixeira, contributed to the design of all experiments and assisted and guided in the interpretation of data, as is the norm for a Design supervisor.

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ABSTRACT

Institutions today stand at a critical juncture, facing *macro forces* such as digitization, shifting political climates, sustainability imperatives, and equity demands—all of which unveil deep imbalances in how resources *flow*. Against this backdrop, existing design practices cannot address the scale and scope of today’s complex challenges. This dissertation reframes design as a practice of *strategic choice-making*, emphasizing the capacity of redirecting *resource flows* to drive value and growth through large-scale transformations. It centers on the phenomenon of *infrastructure-platform mashups*—as a complex space of innovation—where platforms increasingly exhibit infrastructural characteristics and infrastructures adopt platform logics, revealing intersections where localized interventions reverberate system-wide. Through a multi-exploratory approach, the research demonstrates how carefully orchestrated design choices, implemented as strategic features, can mediate flows of resources while bridging institutional intent with broader societal goals. The main finding is a *Model for Strategic Choice-Making*, advancing design from a *form-function-context* paradigm to a *form-function-flow* perspective. By treating data as a designable medium, acknowledging both the disruptive and generative potentials of macro forces, recognizing how resources flow across overlapping institutional contexts, and elevating features to drivers of systemic impact, the model provides a scaffold for aligning local design choices with broader agendas in complex spaces of innovation. This dissertation reframes design as a capability for large-scale transformation, bridging disciplinary boundaries and responding to the urgent complexities of our interconnected world.

CHAPTER 1

NAVIGATING COMPLEXITIES THROUGH DESIGN

1.1 The Complex Challenges of the 21st Century

The 21st century finds many of our long-standing economic, social, and governance structures under increasing strain, as digitization, sustainability imperatives, and shifting political climates exert unprecedented pressures (UNDESA, 2021; WEF, 2022). Originally shaped by industrial-era logic, these systems are now challenged by macro forces such as equity concerns and de-globalization, collectively redefining how resources are generated, stored, and allocated (AlphaBeta, 2017; Sachs, 2015; Stiglitz, 2019). At the same time, public health achievements—driven in part by heavy investments in vaccines—have increased life expectancy worldwide, leading to rapidly aging populations and exposing infrastructures designed for 20th-century needs (UNDESA, 2023; WHO, 2024). Middle-class households, once considered economically secure, face shrinking job prospects and stagnant wages in many regions (OECD, 2019), highlighting systemic imbalances in how capital and opportunities flow. Meanwhile, advancements in renewable energy technologies, such as wind and solar, are driving cleaner and more cost-effective alternatives to traditional fossil fuels. However, the transition to sustainable energy systems is hindered by persistent challenges, including price volatility in fossil fuel markets, infrastructure limitations for integrating variable renewables, and significant disparities in access to financing and deployment of clean energy infrastructure, particularly in developing regions (AlphaBeta, 2017; IEA, 2021; IPCC, 2022). These intersecting pressures reveal that traditional, fragmentary and

isolated approaches to policymaking and resource allocation often fall short of meeting today's multifaceted demands.

Such complexities raise a vital question: *How might we establish more resilient, equitable, and sustainable ways to orchestrate resources on a large scale?* In response, this dissertation posits that *design—reframed as a strategic choice-making practice—can realign these resource flows* to help organizations and societies navigate their growing interdependencies. Rather than dwelling on crisis narratives, this view recognizes that demographic shifts, advancements in public health, and fluctuating energy markets can serve as catalysts for meaningful reform. By acknowledging both the disruptive and generative potentials of macro forces, we open the door to innovative solutions that span social, environmental, and economic spheres. This redefinition of design as choice-making moves us beyond reactive, short-term fixes toward a holistic approach that redistributes resources more wisely, fosters inclusive growth, and sets the stage for long-term sustainable transformation. The sections that follow will illustrate why this paradigm shift is imperative, with infrastructure-platform mashups serving as pivotal arenas in which design's strategic capacities can address the evolving challenges of the 21st century.

1.2 Transformative Forces and the Need for a Paradigm Shift

The systemic challenges outlined in the previous section do not arise in isolation; rather, they reflect a confluence of *transformative forces* that both overlap and at times come into tension with one another. In addition to—*digitization, sustainability, financialization, equity, and de-globalization*,—these forces encompass rapid

technological advancements (e.g., automation, AI), intensifying environmental disruptions (climate change, energy transitions, biodiversity loss), shifting demographic patterns (population growth, aging societies, large-scale migration), and diverging political and economic pressures (rising nationalism versus globalism, wealth inequality, trade imbalances). Although equity remains a crucial pillar for inclusive progress, recent political and economic shifts in certain regions have turned it into more of a point of friction rather than a universally championed force, revealing a contested territory of deeper tensions around how resources and opportunities are allocated. For instance, technological progress can both streamline resource management and exacerbate social inequalities, while global environmental objectives may clash with nationalist or protectionist policies. Together, they are profoundly reshaping how resources are generated, stored, and moved (Raworth, 2017; Sachs, 2015; Thackara, 2005), highlighting the need for strategic know-how capable of addressing both the opportunities and conflicts they create.

Digitization, for instance, accelerates connectivity and enables data-driven innovation, fostering unprecedented opportunities (Kenney & Zysman, 2016; UNDESA, 2021; Verganti et al., 2020; WEF, 2022). At the same time, it simultaneously intensifies surveillance mechanisms and entrenches market concentration, amplifying power imbalances and raising critical concerns about privacy, autonomy, and social equity (Zuboff, 2020). Meanwhile, sustainability urge us to align innovation with ecological constraints and the moral obligations to future generations (IPCC, 2022; Sachs, 2015). Financialization extends beyond traditional banking to permeate nearly all facets of global commerce, sometimes prioritizing returns over social well-being and attaching

monetary values to objects (Epstein, 2006; Ioannou & Wójcik, 2019; Stern & Siegelbaum, 2019; Tooze, 2018). Equity concerns highlight persistent disparities in access to healthcare, education, and economic mobility (Bonn et al., 2019; OECD, 2019). Finally, de-globalization—manifest in shifting trade relationships and nationalist policies—reveals both risks and opportunities for more localized, community-driven solutions (Rodrik, 2019; Sevinç et al., 2023). Collectively, these macro forces—whose visibility has been intensified in recent months in global politics, economic shifts, environmental crises, and social well-being—function as governing factors that demand more than incremental fixes. They point to a need for systemic, holistic approaches capable of addressing the interconnected nature of 21st-century problems.

In this emerging reality, design must evolve beyond aesthetics or user-centric practice into a practice of strategic choice-making situated at the intersection of these macro forces. Rather than treating technology, sustainability, finance, equity, and global-local dynamics as separate concerns, design can reframe them as interlocking forces that shape how resources flow. For example, rethinking energy infrastructures to accommodate cleaner sources not only reduces carbon footprints but also redefines financial models and equity considerations around access (IEA, 2021). By adopting a paradigm that views design as a mediator of resource flows, we can reconfigure products, services, platforms, and infrastructures to serve broader societal goals—be they environmental protection, social equity, or economic resilience. The next sections will dive deeper into how infrastructure-platform mashups create fertile ground for such holistic innovation, ultimately showcasing design's strategic capacity to guide large-scale transformation.

1.3 The Blurring Lines Between Platforms and Infrastructures

Modern life increasingly revolves around digital platforms that offer services once handled by traditional infrastructures (Plantin et al., 2018). As platform-based offerings scale, they begin to acquire the characteristics of infrastructure, providing essential “utilities” in communication, logistics, data management, and commerce. Conversely, existing infrastructures (e.g., transportation, energy grids) are being retrofitted with platform logic—often privatized and driven by market efficiencies (Graham & Marvin, 2002). This reciprocal transformation blurs the line between what was historically public or regulated and what is now privately managed, reshaping accountability and control over critical services. Design, conceived here as a strategic choice-making practice, plays an increasingly central role in negotiating these blurred boundaries. Through design decisions—about interfaces, data governance, or system architectures—platforms and infrastructures become mediated spaces where resource flows meet societal demands. As we will see, the concept of infrastructure-platform mashups marks a key site for innovation, demanding more integrative approaches that account for scale, complexity, and social impact.

According to Plantin et al. (2018), the contrast can be summarized this way: Infrastructures are heterogeneous, gateway-linked systems governed by negotiated standards; they are usually state-regulated or run as public monopolies, funded through taxes, subscriptions, or lifeline tariffs, and designed for long-term reliability and universal access. Platforms, by contrast, rest on a programmable core with modular complements; they impose their own APIs and standards, operate in competitive private markets,

monetize through subscriptions, pay-per-use, device sales, or advertising, and update frequently to stay ahead of rivals. Where infrastructures aim at public value and allow users to “opt out” only at significant social cost (e.g., going off-grid), platforms pursue private profit and invite users to “opt in,” later locking them into proprietary ecosystems. This divergence in architecture, governance, funding, temporality, and user agency underscores why the hybrid infrastructure–platform mashup has become a contested design space: each mashup inherits—and must reconcile—these competing logics.

Although (Plantin et al., 2018) does not explicitly account for decision-making structures and the role of design, these aspects are partially captured in referencing ‘Agency of Users,’ ‘Market Structure,’ and ‘Funding Structure.’ Each property influences who decides on the system’s direction, how design affordances are shaped, and ultimately which users or stakeholders stand to benefit the most. Table 1.1 presents *Decision-Making & Design* (see below).

Table 1.1. Decision-making & Design Properties in Infrastructures-Platforms

	Infrastructure	Platform
Decision-Making & Design	Often decided by administrative or regulatory bodies in the public interest. Design focuses on reliability and universal access.	Largely determined by private entities, emphasizing user experience, profit, and developer ecosystems. Design aims to balance monetization with user adoption.

1.4 The Evolution of Platforms and Infrastructures

Historically, infrastructures were considered public utilities managed at national or municipal levels, built to ensure universal access to water, electricity, transportation,

and communication (Chachra, 2023; Graham & Marvin, 2002). Yet, from the 1970s onward, a wave of privatization and deregulation began to transform how these utilities are governed. Market-driven strategies intended to boost efficiency often resulted in asymmetrical distribution of resources—favoring wealthy communities over low-income ones, thereby paving the way for platforms to substitute traditional infrastructures. This asymmetry distribution of resources manifested, for instance, in certain areas labeled as “food deserts,” few or nonexistent major grocery stores in lower-income areas paved the way for private platforms to step in to provide on-demand grocery delivery, another example, in the finance industry (Bataineh & Teixeira, 2024g), Yape arose to serve underbanked communities in Peru—where banks and public programs hadn’t bridged the gap—thus illustrating how asymmetrical resource allocation can spur private, platform-based solutions—substituting or supplementing public offerings that failed to scale equitably. Simultaneously, digital platforms emerged, extending the idea of an “operating system” from software to entire ecosystems of offerings (S. Chan & Klareld, 2022; Plantin et al., 2018). Today, platforms do more than merely connect buyers and sellers; they establish new forms of socio-economic infrastructure by managing data, shaping user experiences, and imposing their own rules of engagement (Srnicke, 2017). This evolution underscores the growing complexity in how critical services are created and maintained.

Historically, large-scale infrastructures—such as roads, water systems, or public transit—have always involved engineering, landscape design, and urban planning, shaping how communities accessed essential services. Today, infrastructure-platform mashups extend that legacy by integrating user-facing interfaces, data-driven systems,

and governance models. As a result, design choices—from interface design to system governance—now directly influence how resources flow across communities, markets, and nations—affirming an expanded role for design that treats platforms and infrastructures as deeply interconnected. (Plantin et al., 2018) further illuminate this transformation by highlighting the "*platformization*" of infrastructures and the "*infrastructuralization*" of platforms often stems from broader shifts in political economy commonly associated with deregulation and privatization. This reciprocal evolution not only reflects the integration of digital technologies into our societal fabric but also underscores a significant political and economic recalibration. The distinction between platforms and infrastructures, as encapsulated in Table 1.1, marks a critical juncture in understanding the intersection of the technological, political, and economic landscape.

1.5 The Societal and Business Impact of Platforms

Platforms bring powerful consequences for both society and business. On the one hand, Platforms enable unparalleled scalability, facilitate real-time data exchange, and leverage network effects to foster collaborative ecosystems that drive innovation, transforming industries and creating new value-creation models (Kenney & Zysman, 2016). On the other hand, they raise concerns over market concentration, undermine data privacy, and highlight an urgent need for regulatory oversight, given their far-reaching influence on consumer behavior and economic systems (Zuboff, 2020). For businesses, platform-based models can enhance efficiency and reduce barriers to entry, yet they also foster intense competition and raise questions about fair labor practices—particularly as gig-economy platforms redefine employment relationships. For society, the promise of

frictionless interaction and personalized services coexists with issues of algorithmic bias, monopolistic power, and the erosion of traditional local economies (Srnicek, 2017).

In this context, design emerges as a critical lever for social and economic alignment. By shaping platform governance, enhancing transparency, and refining user experience, design practitioners can steer platforms toward more equitable resource distribution, responsible data practices, and sustainable innovation. Yet, design's influence is not inherently benevolent; it can also be exploited to serve narrow, owner-centric interests in ways contrary to the public interests. This dual potential highlights the need to critically examine who sets the agendas, whose interests are prioritized, and how success is defined. The criteria for success are largely determined by the institutional context (Heskett, 2001)—*the frameworks, constraints, and enabling conditions, such as regulations, cultural expectations, and historical precedents, that shape the operational environment*. Similarly, Owen (2005) argues that institutional context both shapes and is reshaped by design-based decision-making. Together, these perspectives reveal that institutional intent—*the deliberate strategic orientation of institutions (governmental, non-governmental, or for-profit) to shape and evolve their structures, policies, and resource allocations in alignment with broader societal, economic, and technological goals*—and context are interdependent, creating a continuous cycle of transformation in which design plays a pivotal role in mediating and reshaping institutional evolution.

1.6 The Emergence and Impact of New Artifacts

Within infrastructure-platform mashups, new artifacts are emerging at the intersection of the algorithmic and the material, the user and the system, the essential and

the complementary (S. Chan & Klareld, 2022; Constantinides et al., 2018; Plantin et al., 2018) (see Fig. 1). From smart home appliances (e.g., voice-activated assistants) to ride-sharing platforms that integrate private and public transportation resources, these artifacts bridge physical infrastructures with digital intelligence. Their “embedded intelligence” allows for adaptive features and on-demand services once considered out of reach (Parker et al., 2016). Yet, these same capabilities raise critical questions around privacy, security, and the ownership of data flows (Zuboff, 2020). As design transitions to choice-making at scale, *artifacts* become mediators of complex resource flows—how energy is allocated, how information circulates, how services are delivered. In crafting these artifacts, designers and stakeholders can foreground transparency, inclusivity, and sustainability, setting new standards for how large-scale systems respond to community and individual needs. As a result, these artifacts represent a spectrum of functionalities and interactions that reflect the complexities of modern societal and technological needs

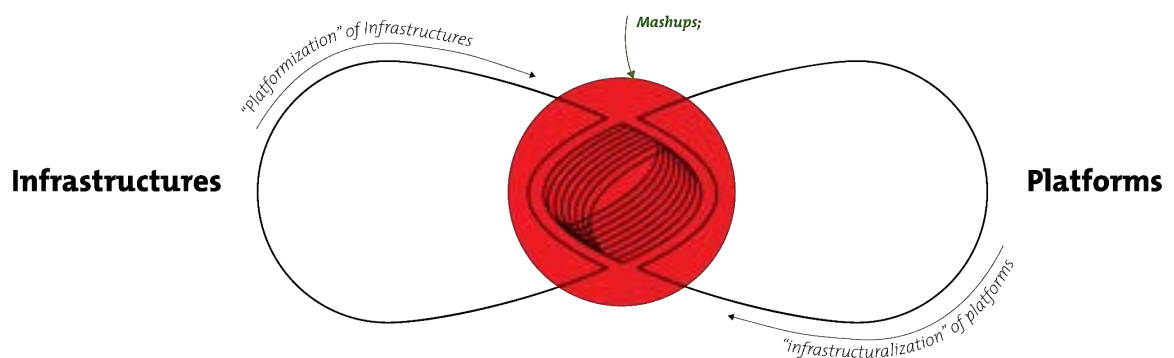


Fig 1.1. Infrastructures-platforms mashups

1.7 Examples and Challenges of the Emerging Artifacts

Real-world examples highlight both the promise and pitfalls of these emerging artifacts. Smart refrigerators, for instance, combine AI-driven ordering systems with traditional home appliances, blurring the boundary between essential household items and algorithmic commerce. Urban mobility solutions—like ride-sharing or bike-sharing platforms—integrate user data, GPS-enabled devices, and municipal infrastructure, offering tailored and efficient transportation that adapts to individual preferences and systemic requirements (Parker et al., 2016). However, this evolution also prompts critical concerns around *privacy* and data security, especially as personal data from home or mobility contexts becomes *commodified*. Additionally, questions of *equitable access* loom large, as affluent regions typically gain first access to these innovations. Policies and design interventions alike must address these disparities, ensuring that digital infrastructure can serve broader societal well-being. Thus, design approaches that balance innovation with ethical considerations and resource flow realignment become vital for navigating this complex terrain.

1.8 The Mashups in a De-Globalized World

In an era of *de-globalization*, marked by shifts in trade relationships and an emphasis on local economic resilience (Rodrik, 2019; Sevinç et al., 2023; Tricks, 2024), *infrastructure-platform mashups* take on additional significance. Platforms that once had global ambitions now adapt to serve regional or community-based models, potentially revitalizing local supply chains and cultural contexts. This localization can promote *customized* solutions that respond more accurately to specific environmental or societal

needs—an approach that may reduce dependency on global markets. Yet, the risk of fragmentation remains: regions or communities with fewer resources may be excluded from the benefits of platform-enabled offerings. By approaching design as strategic choice-making, policymakers, businesses, and civil society can shape infrastructures that foster inclusive growth, bridging digital divides and strengthening local ecosystems (Bonn et al., 2019). In doing so, resources—be they financial, human, or technological—are reallocated in ways that reflect local values, mitigate external vulnerabilities, and encourage collaborative innovation.

1.9 Beyond Financialization, Prioritizing Sustainable and Equitable Outcomes

The era of *financialization* has often prioritized short-term returns over long-term well-being, monetizing aspects of life once considered outside market logic, attaching monetary values to objects (Epstein, 2006; Ioannou & Wójcik, 2019; Tooze, 2018). While financial mechanisms can efficiently mobilize *capital*, they may also perpetuate *unsustainable practices* and *social inequalities* if not balanced with broader societal objectives (Meharry, 2022; Nogueira, 2019). For instance, when essential services like healthcare or education become fully *commoditized*, vulnerable populations risk being priced out (OECD, 2019). Design can act as a counterbalance by embedding social, ecological, and equity considerations into resource-flow decisions. This approach resonates with emerging frameworks—such as *Capital Design*—that examine how reorienting capital allocation can generate more equitable, sustainable outcomes (Ostapchuk, 2024). By focusing on *strategic choice-making*, design practitioners and stakeholders alike can integrate financial viability with sustainability principles and social

justice, ultimately producing more holistic and ethically grounded models for innovation and growth.

1.10 Summary

In summary, the 21st century challenges us to reconsider how we orchestrate resources in an era defined by digitization, sustainability, financialization, equity demands, and de-globalization. The *infrastructure-platform mashups* emerging from these macro forces illustrate the complexity of this new landscape—and the urgency for a design paradigm that operates well beyond aesthetic or incremental solutions. Viewing design as strategic choice-making underscores the discipline’s capacity to align diverse stakeholder interests, redistribute resource flows, and facilitate large-scale transformations that are equitable, sustainable, and resilient.

The chapters ahead build on this premise, delving deeper into the know-how, the case for design, and theoretical frameworks that reveal design’s role in shaping the future of innovation in an interconnected world. Chapter Two demonstrates the knowledge gap in design, delving deeper into design precedents and theoretical framing, proposing a shift from a legacy paradigm of *form-function-context* models towards *form-function-flow* and establishing the foundation for SCMM’s flows-oriented logic. Chapter Three outlines the exploratory and empirically investigates the resource flow reorientation in mashups, while Chapter Four further situates the research explorations and descriptively discusses the data sets gathered from each research stream and the application of the methodology. Chapter Five focuses on discussing the findings and demonstrating the concept of flow as a discovery, grounding features, and the understanding of products and services and

discusses the original contribution of taking flow to a strategic level. Chapter Six Introduces and consolidates the Strategic Choice-Making Model as the main finding as a framework for a systems-centric practice. Chapter Seven explores the implications of the SCMM and further outlines where it matters, while Chapter Eight offers a conclusion that design, guided by SCMM, serves as a vital mediator reorienting resource flows long-term value, affirming the dissertation's central thesis and opening future pathways. By clarifying both the dissertation's structure and the boundaries of its focus, this summary highlights how each chapter contributes to redefining the ways in which we design and steward resources for equitable, sustainable futures.

CHAPTER 2

TOWARDS A STRATEGIC CHOICE-MAKING PRACTICE: ENABLING LARGE-SCALE TRANSFORMATIONS

This chapter explores how design can evolve into a strategic choice-making practice, particularly by focusing on *features* as choices that balance institutional purpose and macro forces to orchestrate transformation at scale. Drawing on interdisciplinary theories—such as *object-oriented ontology (OOO)*, *care-centered design*, and *hyperobjects*—and integrating the concept of *infrastructure-platform mashups*, the discussion highlights how *data* now lies as a designable medium, enabling intelligence-driven insights for strategic decisions. By foregrounding *purpose- and agenda-driven* approaches, design can more effectively address *resource flows* in complex systems across multiple levels and stakeholders, utilizing diverse forms of intelligence. This approach ultimately fosters sustainable, equitable, and value-rich outcomes while addressing the gaps in current design practices. Chapters 4 to 6, through the multiple research streams and the findings, demonstrate how reframing products and services into specific “features” can expand the choice set available to institutions, prompting them to incorporate broader considerations such as equity or sustainability. Grounded in these principles, this chapter establishes the conceptual foundation for how design can harness macro forces and institutional purpose, and develop the *know-how* to influence large-scale resource flows.

Traditionally, products and services are viewed as holistic solutions made up of multiple “features,” yet these features are rarely considered as discrete units in a systemic

context. In this study, however, we treat features as basic units of value within a larger system—an intentionally designed mechanism that performs specific tasks at the intersections of resource flows. By isolating these intersection points, we see that design at the feature level can regulate a system’s behavior far more effectively than focusing solely on entire products or services. This vantage arises from different interpretations of features: rather than minor add-ons or user-facing conveniences, they become strategic connectors through which institutional purpose and macro forces are enacted. If flows indeed run through features, then reorienting them can transform broader systems, because any shift in how a feature operates can alter the system’s resource flows—and ultimately, its long-term trajectory. Framing features as these pivotal intersections thus underscores their potential for steering large-scale change in complex environments.

2.1 Form, Function, and Context: A Legacy Paradigm

Design, as a practice, has historically sought coherence between form and function, viewing form as a solution to function and function as the defining requirement for form. In this traditional perspective, an object’s design emerges from an ensemble wherein form and function must fit together harmoniously. The desired property of this ensemble relates to some particular coherence between form and function that defines the specific ensemble (Alexander, 1964; Dilnot, 2011; Heskett, 2001; Kohler, 1938). The classic example is a chair—which can be a dining chair, wheelchair, office chair, or park bench—each variation reflects different demands on form and function. Likewise, a graphic sign is designed so its form (layout, typography) fits the functional demands of informing a driver in motion. Yet form and function constitute only the ensemble’s

internal coherence. Context is the third defining entity, as changes in production or usage can drastically alter how well form and function fit (Koffka, 1935; Ozenfant, 1952). A kettle remains coherent when it efficiently boils water within a particular context; a vehicle transforms from a generic car into a sports car or SUV when adapted to specific driving environments or cultural preferences. This coherence has depended highly on the context of use and production (Collins, 1998; Heskett, 2001; Holm & Larson, 1953; Potter, 2017; Teixeira, 2002). Context is any part of the world that imposes demands on the form-function ensemble (Alexander, 1964), making it an essential defining entity alongside form and function.

Although this *threefold interplay* of form, function, and context worked well for many challenges of the 20th century, it increasingly proves insufficient for tackling the *systemic complexities* of the 21st. As problems grow more intertwined—spanning economic, ecological, social, and technological domains—designers must move beyond static notions of context to handle dynamic resource flows and macro-level shifts

2.2 Beyond the Traditional Lens: Confronting Complex Problems

As problems become more *multifaceted* and *systemic*, a purely form-function-context lens can overlook broader systemic forces at play. While human-centered design, design thinking, and double diamond models introduced valuable ways to frame “wicked problems” (Buchanan, 1992, 1998; J. K. H. Chan, 2016; Churchman, 1967), they typically limit the problem space to usage and production contexts. Incorporating *pragmatism*—referring philosophically to knowledge that is shaped by real-world experience and iterative experimentation—into design for complex systems (Di Salvo,

2009) has further expanded how designers uncover and challenge underlying assumptions, facilitating a deeper engagement with complex societal challenges. Designers approaching complex systems pragmatically see each intervention as a hypothesis; emerging through reflective engagement, tested, and refined in practice, continuously reshaping both the design process and its underlying assumptions (Dalsgaard, 2014; Wakkary, 2021), resonating with reflections on actions leading to deeper insights and adaptation by Dewey (L. A. Hickman & Alexander, 1998). While this recognizes complexity, it often lacks a deeper engagement with interconnected contexts such as *–natural, social, financial, cultural, political, manufactured, human, and digital–* (Nogueira et al., 2019).

To address these *larger-scale* challenges, design must redefine its paradigms to recognize that *resource flows* reflect the power dynamics (Meharry, 2022) by multiple actors and to acknowledge how *resources flow* across multiple systems, levels, and stakeholders, often enabled by diverse forms of intelligence. A new narrative beyond *production and consumption* emerges here, infrastructure-platform mashups now create the conditions transcending the typical usage-production contexts. “Strategic Choice-Making” emerges as a paradigm not only focusing on coherence between form and function but also integrating *resource flows* and *systemic implications* of design choices. While some might argue that designers have always considered context; however, “context” in earlier models largely centers local conditions of usage or production contexts, whereas today’s conditions sprawl across interconnected contexts through multiple resources (Nogueira et al., 2019) across macro forces like sustainability, equity, or financialization– factors that require broader, more systemic perspectives.

2.3 Elevating Ensembles into Archetypes: From Contexts to Flows

Contemporary design must shift from merely adapting ensembles (form-function-context) to a perspective grounded in *the flows of resources* that pass through these ensembles. This reflects a recognition that, traditionally, design treated context as a static, human-centered backdrop for design, whereas flows represent continuously evolving processes and exchanges involving both human and non-human agents. As designers account for complex systemic challenges, they move beyond seeing an “ensemble” of form and function in isolation. Viewing the ensemble through a systems lens that foregrounds resource flows effectively transforms it into an “archetype”: a strategic configuration of *resources*—natural, social, financial, cultural, and more—combined with *forms* and *functions* in ways that address interconnected challenges.

This reframing emphasizes the *agency of flow*—the notion that resource flows themselves exert influence on how design solutions function and evolve. As Meharry (2022) observes, resource flows often reflect underlying power dynamics among multiple actors; in other words, flows carry a form of agency that actively shapes outcomes rather than serving as passive background conditions. Essentially, archetypes capture the potential of multiple choices operating together within a system. By reorienting resource flows (for example, shifting capital or labor allocations), designers can spark emergent value across entire ecosystems.

In practice, reframing an ensemble as an archetype relies on understanding how each feature of a design intersects with and redirects flows. Each feature can be seen as an intervention point that captures or modulates a particular flow within the system. For

instance, an electric vehicle (EV) charging station with bi-directional charging capability illustrates this principle: such a feature not only charges a car but also allows energy to flow back into the grid, actively redirecting an energy stream and thus reshaping the larger system dynamics, and as a result accelerating the effort towards energy transition. This shift in perspective does not abandon traditional design considerations like aesthetics or usability; rather, it expands them to account for how these flow-oriented configurations generate economic value and growth (Romer, 2021) within the institutional context (Heskett, 2001). This logic emphasizes *strategic intent* in shaping how resource flows are directed, harnessed, or reconfigured through design choices. Rethinking the ensemble as an archetype positions designers to orchestrate choices, often data-driven, that create value and growth within complex contexts. By emphasizing *flows* over *static usage or production contexts* as fixed attributes, designers become strategic enablers increasingly capable of mediating resource flows, thereby catalyzing sustainable and equitable transformations.

Yet such an expansion can introduce concerns about scope and feasibility. Nonetheless, *ignoring resource flows risks overlooking important externalities*, such as environmental impacts or social inequities, that arise in large-scale systems. In fact, a purely context-bound approach may blind design of the consequences of an intervention beyond its immediate use and production scenarios. Treating the ensemble as an *archetype* highlights the importance of multi-system interactions, especially as design confronts the realities of *infrastructure-platform mashups* that extend well beyond a single product or use case. This multi-system perspective highlights that what we consider “context” is no longer a stationary setting but is continually reshaped by

intersecting flows. Embracing flows as a fundamental lens enables designers to navigate and even steer these ongoing transformations, rather than simply responding to a static context.

2.4 Orchestrating Complexity: The Need for Strategic Choice-Making

With the potential for designers to evolve into *strategic orchestrators* of complex systems, the design practice must undergo a profound shift. Multi-system integrations on multiple levels—often involving conflicting stakeholders—are enabled by the technological advances embedded in everyday products and services. No longer purely physical or purely virtual, these *hybrid spaces* weave together technology, society, economics, and culture, forming a tapestry of intricate interdependencies that existing design models, once groundbreaking in emphasizing empathy, iterative prototyping, and user experience—struggle to navigate the complex, multi-dimensional resource flows that the strategic choice-making paradigm seeks to orchestrate. Highlighting the need for nuanced design models that address macro-level goals while also navigating blurred lines between infrastructures and platforms.

Amid this shift, transformative forces such as digitization, sustainability, equity, financialization, and de-globalization (Raworth, 2017; Sachs, 2015; Stiglitz, 2019; Thackara, 2005; Tricks, 2024) reshape the role of designers as change agents. These forces transcend trends, functioning as governing factors that set new ethical, economic, and ecological parameters. Addressing such intricate terrains of the mashups where traditional infrastructures—like transportation or energy—intertwine with digital

platforms, necessitates strategic choice-making—an approach capable of factoring in the multi-nodal realities of 21st-century design.

2.5 Integrating Interdisciplinary Insights: Data, Agency, and Distributed Intelligence

An interdisciplinary lens enriches design’s capacity to strategically shape resource flows in complex systems, positioning data as a designable medium. Concepts like *hyperobjects* (Morton, 2013) referring to entities such as climate change or capitalism that are massively distributed in time and space, rendering them nearly incomprehensible at human scales, *actor-networks* (Latour, 1992) posits that both human and non-human entities interact in networks that shape social realities, while *object-oriented ontology* (Lemke, 2021) asserts that all objects possess an existence independent of human perception. Together, those concepts challenge anthropocentric views by granting agency to *non-human entities*—ranging from algorithms to physical artifacts—that shape socio-technical-economic-ecological systems. While critics may dismiss these philosophical stances as abstract, focusing on *features* at intersections reveals their practicality in handling intangible resource flows and platform logic. When design integrates these theoretical insights into its practice, it creates conditions that optimize resource flows and system coordination, leading to enhanced resilience and equity.

For example, Morton’s hyperobjects are distributed phenomena across time and space, like climate change or capitalism, highlighting how large-scale issues surpass immediate human experience. Meanwhile, Harman’s OOO suggests that objects—be they products, services, or features—exist and interact in ways not always in relation to

human experiences. This also highlights how data becomes a strategic resource: its collection, interpretation, and deployment inform the formed choices.

Embracing these philosophical concepts allows design to recognize the *distributed nature of agency*—including leveraging data as a designable medium to influence coordinating, shaping, or rerouting resource flows. Data becomes a strategic asset and a designable medium, reinforcing the notion that *features* when deliberately structured, can orchestrate large-scale transformations. For instance, in the EV Charging Stations case study (Bataineh & Teixeira, 2024j), a feature of *real-time energy monitoring* in the charging system can serve as a cornerstone for the energy transition. When adopted by institutional fleets—such as school buses—this feature enables the optimization of charging schedules, improves vehicle availability during peak demands, adapts to price fluctuations and extreme weather conditions, and ultimately minimizes carbon footprints. Additional examples of feature-driven transformations are discussed in Chapter Five. This directly supports the dissertation’s central thesis: features serve as pivotal mediators that reorient resource flows and enable strategic transformations.

2.6 Understanding Flows in Infrastructure-Platform Mashups: Reframing the Design Lens

A deeper exploration of *infrastructure-platform mashups* illustrates how resource flows, often governed by *technological advancements*—such as artificial intelligence, automation, data integration, and sensor networks—transcend the boundaries of individual systems. These mashups (Constantinides et al., 2018; Plantin et al., 2018) combine traditional *infrastructures*—such as transportation or energy grids—with *digital*

platforms, forming new social, economic, and ecological dynamics. Examining *flows* rather than static artifacts reveals that design is not simply front-end work—referring to the shaping of concepts and user interfaces—(Cooper et al., 2014) but a *strategic practice* that orchestrates how streams of resources flow across boundaries. (L. Hickman, 2001) notes that some critics warn that analyzing flows adds complexity to the design, however, overlooking convergent streams underestimates the potential of features to direct or redirect them. By centering on features as the channels where capitals (Nogueira et al., 2019) converge, designers can harness or redistribute flows toward sustainable and equitable ends. This *flow-focused* approach underscores design’s role in enabling large-scale transformations—a foundation of strategic choice-making in multifaceted contexts.

2.7 Reorienting Boundaries Ontologically: Beyond Anthropocentrism

Object-oriented ontology (OOO) and care-centered design prompt a reorientation of design boundaries, emphasizing non-human agencies and broader ethical responsibility for ecological and societal implications (Haraway, 1991; Puig de la Bellacasa, 2017). Understanding *designed agents* as hyperobjects invites a deeper exploration into how *objects* interact and exist within networks, leading us to OOO as a foundational lens. OOO offers a fresh perspective on understanding the essence and interconnectedness of objects (products, services, and features), challenging conventional hierarchies between humans and non-humans. By prioritizing objects' reality beyond human perception, OOO encourages a non-hierarchical view of existence, where all objects, regardless of their relation to humans, possess inherent value and agency.

Harman's concept of OOO and Lemke's technologies—(1) technologies of production; (2) technologies of sign systems; (3) technologies of power; and (4) technologies of the self, inspired by Foucault's writings (Lemke, 2021)—provide frameworks to appreciate the complexities of human and non-human associations, moving beyond traditional anthropocentrism. This interdisciplinary approach enriches the design discourse by promoting a holistic understanding of object interactions, crucial for addressing the intertwined challenges of technology, society, and the environment.

Treating products, services, or features as having intrinsic value (Lemke, 2021), reframes them from static tools into actors within complex ecosystems (Gottlieb, 2022). Critics might argue that foregrounding care or ecological agendas could detract from functionality or market viability. However, by adopting an ontological approach, designers anticipate *systemic consequences* and shape features that responsibly manage resource flows (Escobar, 2018; Nogueira et al., 2019). This boundary reorientation strengthens design's strategic role: rather than confining itself to user-centric requirements alone, design accounts for the interplay between human and non-human entities, linking *macro forces* with *institutional purpose*. Aligning design outcomes with ecological and social considerations thus positions design as a *transformative practice*—capable of large-scale interventions guided by strategic choice-making.

2.8 Agenda-Driven Approaches: Leveraging Data and Purpose to Reshape Resource Flows

Confronting the 21st-century complexity demands of infrastructure-platform mashups calls for a practice of *agenda-driven, intelligence-based choice-making*, with

data and *purpose* anchoring the design process. Organizations increasingly assert that they are “intelligence-driven,” yet frequently lack the structured methodologies to translate data insights into strategic design decisions (Wamba et al., 2017). When data is wielded effectively, it spotlights user patterns, uncovers hidden correlations, and supports a more *equitable distribution* of resources.

By centering design around explicit agendas (e.g., sustainability, social justice, or economic resilience), *features* become critical junctions where data analytics inform strategic decisions. Far from overshadowing user-centric values, data amplifies them, ensuring that intelligence-based choices remain ethical, grounded, and equitable. Scholars such as (Whitney, 2015) envision an era of sustainable choice guided by platform logic, while (Heskett, 2001) highlights the designer’s role in shaping flexible systems based on complex technologies. (Teixeira, 2017) and (Nogueira, 2019) highlight the imperative of integrating artificial and multi-systems through digital features that address complex shared challenges and strategically realign agents and agencies. Yet, as (Martin, 2020) notes, many existing design practices lack the organizational frameworks to capitalize on intelligence-driven concepts fully. An agenda-driven approach ensures that intelligence-driven design is not only data-informed but also aligned with macro forces, strategically shaping features at the intersection of resource flows to reorient them to align institutional purpose with macro forces to drive large-scale transformation.

2.9 Design for Purpose: Bridging Activism and Strategic Implementation

Despite references to purpose-driven or mission-oriented design (Co, 2020; Gerlach, 2020; Lutz, 2022; Mazzucato, 2020), the field often underestimates the *strategic*

vantage needed to integrate human, non-human, and macro-level agendas. Activism, social design, and participatory methods excel at highlighting moral or cultural dimensions, yet they may lack the design capacity and know-how capabilities for enduring systemic change and effectively integrating social justice, environmental sustainability, and economic equality (Wendt, 2017)). Rethinking purpose as a *multi-agenda* framework—one that includes institutional purpose, macro forces, and the potential of features to realign resource flows—expands design’s capacity from *activism* toward *transformative strategy*.

By situating “purpose” at the intersection of diverse stakeholders, design transcends the incremental focus of user-centered or philanthropic approaches. Features become *intelligent agents* that align resource flows with broader values (social justice, environmental integrity, economic viability), an essential stance in the *infrastructure-platform mashups* shaping the 21st century. This dissertation —see Chapters 4, 5, and 6—suggests that only by developing such *purpose-driven, strategic* competencies can design orchestrate large-scale transformations and fulfill its evolving role in society.

2.10 Transforming Design for a Sustainable Future: Beyond Profit-Centric

Models

Current design practices, often driven by financialization and market competition (Chapman, 2021; Wizinsky, 2022), are ill-equipped to address the complex challenges of the 21st century. As (Martin, 2020) observed, this narrow focus on profit maximization neglects broader societal and environmental concerns. While past industrial shifts caused significant disruptions, the current transition to multi-intelligence-driven infrastructure-

platform mashups has a far wider reach, impacting diverse forms of capital and both human and non-human agents (Nogueira et al., 2019). By focusing on features as key intervention points within these mashups, designers can proactively guide resources towards sustainable outcomes, responding to macro forces like sustainability and technological advancements. This approach allows design to transcend profit-driven models and adopt holistic, systemic solutions, effectively addressing broader concerns without diluting its focus. This strategic shift, echoing (Teixeira, 2017) vision of design shaping alternative futures, is essential for navigating the complex interdependencies of our time.

2.11 Conclusion

Strategic choice-making is essential for design to address the 21st century's complexities. Features, where data, purpose, and macro forces converge, are the key to reorienting resource flows and driving systemic transformation. Integrating interdisciplinary perspectives like object-oriented ontology and care-centered design, along with agenda-driven approaches, empowers intelligence- choices. This shift enables proactive shaping of the future, moving beyond traditional design models to recognize the interplay of multiple intelligences, stakeholders, and systems on multiple levels that characterize the 21st century.

Advancing from a *form-function-context* focus to a *form-function-flow* focus enables a multi-agenda, multi-capital approach to equip design to catalyze innovations that align institutional goals, address macro forces, and promote sustainable futures.

Looking ahead, Chapter Three will detail the methodological approaches underpinning these insights, detailing the empirical approaches that later validate the strategic choice-making paradigm and illustrating how designers can leverage features to navigate and *influence resource flows* for large-scale transformations.

CHAPTER 3

RESEARCH METHODOLOGY

Design research that engages with intricate socio-technical-ecological (Nogueira, 2019) assemblies requires a methodology capable of addressing multiple layers of complexity. This dissertation focuses on infrastructure-platform mashups, where institutional purpose, macro forces, and systemic resource flows through features intersect to drive large-scale transformations. In such contexts, a rigorous approach is crucial for explaining not only *what* is studied but *how* and *why* it is studied in a certain way. Drawing insights from earlier chapters, this chapter clarifies the philosophical stance, research questions, methods, and frameworks that shape the examination of design as a strategic choice-making practice. The overarching goal is to present a coherent path that connects theoretical constructs—such as multi-intelligence, objects, and macro forces—with the iterative, practice-based dimension of design research. In doing so, this chapter serves as the critical methodological link between the theoretical reframing of design practice presented in Chapter 2 and the subsequent empirical explorations. It explains how foundational philosophical concepts and the theoretical reframing of design have been translated into a practice-based research design, which in turn produces empirical findings.

As this chapter bridges our theoretical reframing (see Chapter 2) and empirical findings (See chapters 4 & 5), this dissertation develops a new practice—strategic choice-making—grounded in Research through Design as the main research approach. Our original contribution is codified in a visual model (see Chapter 6) that captures how

features at the intersections of resource flows drive systemic transformation. This approach is operationalized through three research explorations (see Chapter 4): a set of Macro Assessment Case Studies (see Appendix A), a Semester-Long Workshop Series (see Appendix B), and Three Controlled Experiments with Senior Practitioners (see Appendix C). Together, these inquiries validate our research questions and demonstrate that integrating an iterative research approach reveals actionable insights for capturing the Strategic Choice-Making Model (see Chapter 6).

3.1 The Research Approach

In framing design as a strategic choice-making practice, this dissertation adopts a perspective aligned with Research through Design (RtD), an approach that treats the act of designing itself as a generative form of knowledge production (Zimmerman et al., 2007). It is a methodology that emphasizes iterative, practice-based inquiry to generate knowledge in ways that traditional methods cannot. RtD has emerged as a new tradition of research—rooted in design—bridging the gap between theory and practice by engaging designers directly with real-world challenges. As (Zimmerman et al., 2007) argue, RtD transforms the design process into a laboratory for continuous experimentation and reflection, while (P. J. Stappers et al., 2014) highlight its unique capacity to produce insights through creative action. This approach is particularly significant for addressing complex systems, as it enables the co-creation of knowledge that is both empirically grounded and deeply informed by theoretical reframing.

Whereas conventional research often looks backward to analyze existing phenomena, design practice looks forward, proposing future realities that do not yet exist

(P. J. Stappers et al., 2014). This tension between understanding “what is” and envisioning “what might be” is further complicated by the multi-faceted nature of platform-infrastructure mashups, which involve technical, economic, political, and societal factors. Consequently, the methodological framework must be robust enough to capture the interplay of these forces while remaining sufficiently agile to integrate emergent insights—particularly as new prototypes or experiments reveal unanticipated constraints or opportunities. By foregrounding the methodological rationale and specific tools used—such as Linkography for design-trace analysis (Goldschmidt, 2014; Hatcher et al., 2018)—this chapter clarifies the step-by-step processes undertaken to ensure that the knowledge generated is both empirically grounded and future-oriented.

In doing so, the research builds on a rich tradition of scholarship examining the interplay between design practice and research (Archer, 1995; Cross, 1982), extending it into the emergent territory of resource flows. Much research is trying to generate new insight from the past or present with the hope of putting the knowledge generated to use at a later stage, whereas design aims at constructing a possible future that may not exist yet (P. J. Stappers et al., 2014). Each methodological choice aims to illuminate how design can move beyond producing standalone products to shaping system-wide conditions.

This chapter, therefore, sets the stage for a multi-exploration inquiry that combines macro assessment case studies (see Appendix A), workshop series (see Appendix B), and controlled experiments (see Appendix C), each contributing a different lens to the central thesis. The result is a methodology that accounts for the complexities of embedding institutional intent through features for large-scale transformations while also providing a

clear roadmap for how these features can mediate resource flows at the intersection of systemic flows. This methodological framing directly builds on the contextual complexities introduced in Chapter 2 and paves the way for the empirical explorations detailed in Chapter 4.

3.2 Hypothesis and Research Question

At the heart of this dissertation lies the hypothesis that features can mediate between institutional purpose and macro forces, reorienting resource flows toward large-scale transformation. Such a hypothesis positions design as strategic choice-making, where tangible design decisions—at the feature level—have systemic implications. The rationale behind this claim stems from prior research—particularly from the doctoral program at the Institute of Design (Meharry, 2022; Nogueira, 2019)—suggesting that small but carefully orchestrated design outcomes can generate significant shifts in how resources flow in complex systems. By viewing these outcomes as deliberate “choices” rather than random outputs, the research proposes that design can reorient resource flows, potentially making socio-technical systems more equitable, sustainable, and resilient.

From this hypothesis emerges the central research question: *How can design offer strategic choice-making in infrastructure-platform mashups to drive value creation for large-scale transformations?*

Unlike conventional research approaches that often emphasize past or present, this design-centric research approach probes the dialectic between the “what is” and the “what could be” (P. J. Stappers et al., 2014). The question also underscores the integration of institutional intent with prevailing macro forces (e.g., financialization,

sustainability, digitization), it foregrounds the multi-level interactions that shape design outcomes. This vantage aligns with Research through Design (RtD), wherein the production of artifacts and prototypes is both a pathway for generating empirical knowledge and a vehicle for shaping future realities (Zimmerman et al., 2007).

Moreover, approaching design as strategic choice-making recognizes that such choices rarely occur in a vacuum. They are influenced by existing dynamics, governance models, and shifting regulatory frameworks. By capturing how macro forces act as both constraints and drivers of feature-level innovation, the research question opens a portal to understanding systemic changes and why certain interventions succeed or fail to gain traction (Hjorth & Bagheri, 2006). This perspective on nested scales of action—feature, institutions, macro force—provides an analytical lens through which the dissertation’s multi-exploratory methodology can be applied. Ultimately, the hypothesis and question together offer a conceptual springboard for the ensuing investigations, bridging theory and action to produce actionable knowledge regarding how and why design can steer ecosystems toward preferred states.

3.3 Research Design

This dissertation adopts a multi-exploratory research design that leverages both theoretical constructs and practice-based interventions. Rather than relying on a single method, the study combines macro assessment case studies, workshops, and controlled experiments to capture the explored interactions that shape feature-level interventions. This multi-pronged approach recognizes that infrastructure-platform mashups involve an evolving interplay of institutional purpose and macro forces. By examining how various

participants—from graduate-level students to senior practitioners—engage with design interventions, the research design yields insights into both the theoretical underpinnings and practical dynamics of strategic choice-making.

3.4 Multi-Exploration Structure

3.4.1 Macro Assessment Case Studies

To lay a foundation, the dissertation analyzes existing (products and services) that are viewed as either platforms or infrastructures—from controlled-environment agriculture to embedded finance—adopting a systems point of view to identify leverage points where features can shift resource flows. These case studies—detailed in Appendix A and structurally elaborated in Chapter 4—were structured to serve as the input for the strategic choice-making practice. This research exploration highlights current challenges, stakeholder relationships, and the macro forces—sustainability, financialization, digitization—driving or constraining design. As well as mapping the opportunity space. Those cases allow tracking of how systemic factors shape or limit feature-level innovation and clarify where design can intervene and why some leverage points prove more impactful than others.

3.4.2 Workshops Session Series

Building on the macro assessment cases, the semester-long workshop series—detailed in Appendix B and structurally elaborated in Chapter 4—serve as “living laboratories” where participants iterate prototypes to embed institutional purposes into tangible feature designs. Designed in the spirit of Research through Design (RtD), these

workshops meld theoretical models with hands-on experimentation, allowing ideas to be tested, refined, and sometimes discarded based on emergent insights. The goal of the exploration was to help better construct the know-how of design knowledge for the strategic choice-making practice.

3.4.3 Controlled Experiments with Senior Practitioners

The controlled experiments—detailed in Appendix C and structurally elaborated in Chapter 4—delve deeper into how professional design practitioners conduct their practice through a specific institutional context and enter the mashups space, confirming whether strategic choice-making indeed drives resource-flow realignment for large-scale transformation. Understand how institutional context shapes the choice and the decisions being made.

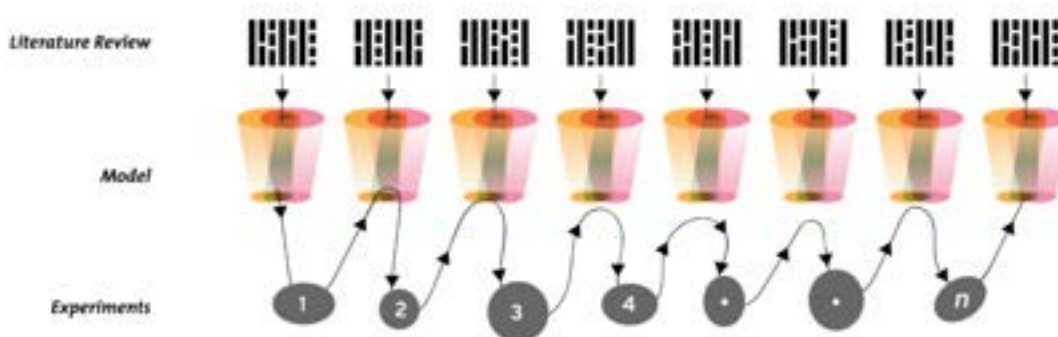


Fig 3.1 The Iterative Research Approach

The iterative cycle adopted through the RtD approach reflects the nested nature of the dissertation's investigation: macro-level analyses highlight systemic openings and constraints, workshops delve deeper into feature-based experimentation, and practitioner

experiments investigate how institutional context shapes choices and decisions related to the infrastructure-platforms mashups. By continually cross-referencing data points and insights from each exploration, the research design maintains both breadth and depth, ensuring that its conclusions address the complexities of platforms-infrastructure mashups while still advancing the theoretical discourse on design's capacity to reorient resource flows for large-scale transformation. Figure. 3.1 illustrates how the research will be guided through a process of exploration—at a meta-level— leading the reasoning through a sequence of individual grounded explorations.

3.5 Methodology

This dissertation's multi-exploration research design centers on Research through Design (RtD) as a guiding framework for uncovering how features can strategically reorient resource flows within infrastructure-platform mashups (P. J. Stappers et al., 2014; Pieter Jan Stappers, 2013; Pieter Jan Stappers & Giaccardi, 2017; Zimmerman et al., 2007). Rather than following a linear process, RtD capitalizes on iterative prototyping—from conceptual sketches to full-scale artifacts—to probe technical, societal, and policy constraints (Gaver, 2012). Each design intervention acts as a powerful investigative mechanism, uncovering system-level challenges, addressing user needs, and identifying transformative opportunities for innovation (Cross, 1982; Dorst, 2011). Alongside RtD, the study employs Linkography (Goldschmidt, 2014; Hatcher et al., 2018) to map how ideas evolve or converge, documenting emergent insights or constraints. Throughout the early workshop series, Linkography served as a key tool for tracing the development of ideas, design actions, and decisions. By analyzing the

connections between design steps, the research illuminated the process of embedding strategic intent into features and the pathways through which innovative archetypes could emerge. Linkography was adopted to develop a sensibility of features connecting when choices reoccur or get extrapolated through a specific time frame, to offer recognition of features in the design process and less about the evidence (see Fig 3.2 as it illustrate design moves over a ten minutes time frame). This clarifies the impact of individual design decisions while also informing broader discussions on how features mediate institutional goals and macro forces.

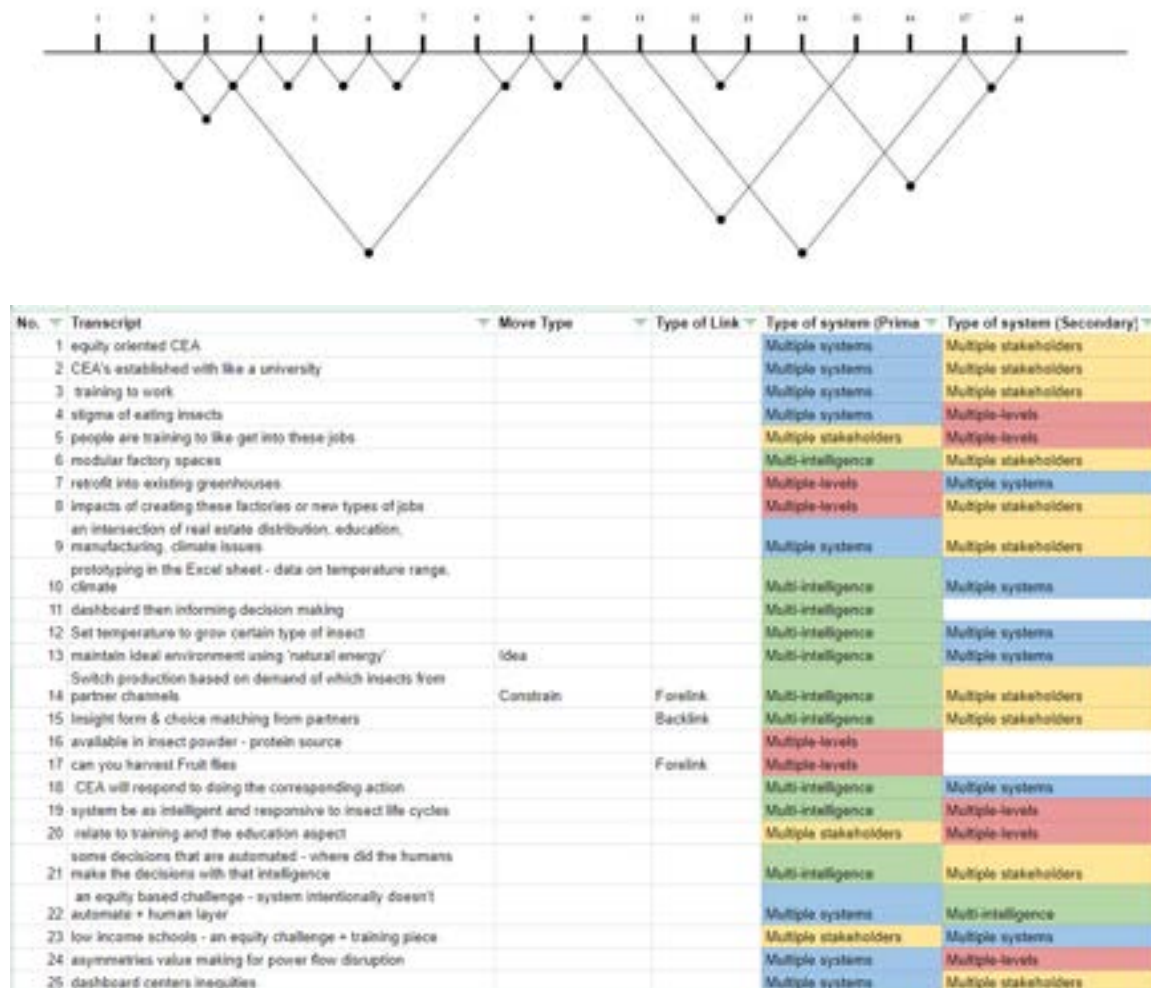


Fig 3.2 Ten minutes Linkography diagram and Moves List.

3.6 Conclusion

The multifaceted approach outlined in this chapter underscores the dissertation's commitment to Research through Design (RtD) as both a methodological justification and a practical toolkit for exploring how features can strategically reorient resource flows in infrastructure-platform mashups. By intertwining iterative prototyping, systematic reflections, and purpose-driven interventions, the methodology positions design as a generative force that not only reveals hidden complexities but also shapes future

possibilities (P. J. Stappers et al., 2014; Pieter Jan Stappers, 2013; Pieter Jan Stappers & Giaccardi, 2017; Zimmerman et al., 2007). Methods such as Linkography (Goldschmidt, 2014; Hatcher et al., 2018) add transparency to this exploration, capturing the creative trajectory of design ideas and helping researchers trace how strategic intent materializes in the form of tangible features.

The macro assessment case studies, workshop sessions, and controlled experiments collectively provide a triangulated view of how institutional purpose and macro forces converge at the feature level as it will be discussed in the following chapters. Through this multi-exploratory strategy, the dissertation captures insights into both the theoretical constructs and the practical realities of implementing these constructs in ever-evolving socio-technical environments. The following chapter will delve into the application of the methodology and the framework outlined in this chapter, illustrating how the design know-how knowledge was structured to further yield the data sets to support the hypothesis and respond to the research question.

CHAPTER 4

EXPLORING CHOICE-MAKING FOR TRANSFORMATIONS

This chapter builds on the methodological foundations laid out in Chapter 3, guiding the research on *strategic choice-making* in the context of *infrastructure-platform mashups*. It reframes the research problem around the nexus of intelligence features, multi-system flows, and purpose-driven innovation. It presents a set of macro assessment case studies, a semester-long workshop series, and controlled experiments conducted to articulate the rationale behind focusing on features at intersections of flows to drive large-scale transformations. The following sections detail how these research explorations unfold to examine how features—situated at the intersections of resource flows—are envisioned and shaped to validate the strategic choice-making model, emphasizing the tools, frameworks, and data-collection approaches adopted. In doing so, this chapter also shows how the research findings were codified into a new practice, including a visualization that captures the core elements of this design approach.

4.1 Framing the Exploration for Transformation

This dissertation introduces *strategic choice-making* as a design practice, specifically in the context of *infrastructure-platform mashups*, where opportunities for large-scale transformation are immense and complex. In a world where choices and decisions are increasingly shaped by macro forces—such as sustainability, equity, and digitization—institutions face the challenge of aligning their purposes to drive meaningful outcomes. The fundamental proposition of this dissertation is that design,

conceived as a strategic choice-making practice, can reorient resource flows and unlock value at critical intersections. Features, designed with purpose, serve as strategic assets: they bridge institutional intents with systemic macro forces to catalyze large-scale transformations.

A central challenge lies in navigating these intersections to create value, which raises the question: *How can design offer strategic choice-making in infrastructure-platform mashups to drive value creation for large-scale transformations?*

A hypothesis about the role design can play in expanding the institutional decision-making capabilities to address growth within and beyond the institutional context was formed, and this hypothesis was tested through targeted research activities, forming a hypothetical model to guide broader understanding, application, and further research. These activities include:

1. *A set of Macro Assessment Case Studies*
2. *A Semester-Long Workshop Series* (Spring 2023, 2024)
3. *Three Controlled Experiments* with Senior Practitioners

Together, these inquiries sought to explore how *features* at intersections of flows are envisioned and shaped to validate the strategic choice-making model. While the methodological approach—particularly the use of Research through Design (RtD)—was laid out in Chapter 3, this chapter focuses on the application of that approach.

4.2 A Set of Macro Assessment Case Studies:

To establish a strong foundational understanding, five distinct cases were examined: controlled environment agriculture (Bataineh & Teixeira, 2024h), EV

charging stations (Bataineh & Teixeira, 2024j), embedded finance (Bataineh & Teixeira, 2024i), 911 emergency response system (Bataineh & Teixeira, 2024g), and in-flight meal services (Bataineh & Teixeira, 2024k). These cases share a common theme—the pursuit of value creation, integrating sustainable and equitable solutions into complex systems. Each case represents a segment of a broader ecosystem undergoing large-scale transformation, where traditional practices are being reimagined in response to the urgent need for sustainability, efficiency, and equity. From optimizing food production within controlled environments to rethinking EV charging stations as enablers of the energy transition and redesigning in-flight meal services to improve waste management while enhancing the user experience, these cases demonstrate the potential for innovation at the intersection of multi-systems across multiple levels, involving diverse stakeholders through multiple intelligence. Examined at the feature level, these cases were chosen to provide inputs for the Strategic Choice-Making Model (SCMM), offering macro-level assessments of the current state of everyday infrastructure by deeply analyzing both the problem space and the opportunity space. This approach emphasizes the potential for systemic transformation while uncovering barriers, such as the overemphasis on financialization, which often restricts the capacity for strategic choice-making. See Figure 4.1 of the Macro Assessment Case Studies structure.

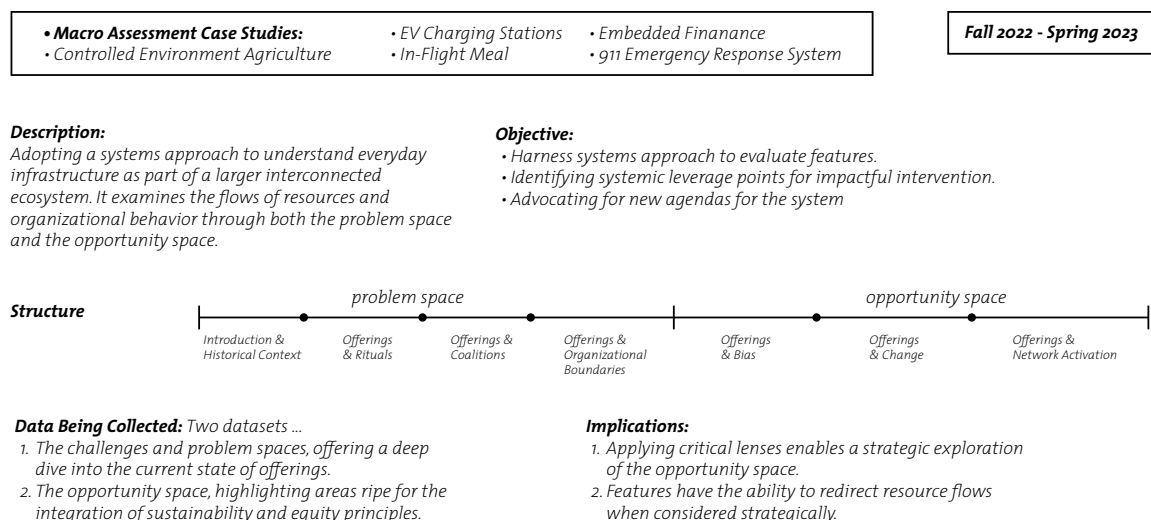


Fig 4.1. The Macro Assessment Case Studies Structure

Adopting a systems thinking approach allowed for a holistic examination of these infrastructures, situating them within larger interconnected ecosystems. This perspective was critical for identifying leverage points where design interventions could generate the most significant impact. Each case study was structured to explore key dimensions, including the historical evolution of the system, the rituals and routines embedded within it, the coalitions formed through the integration of services, and the shifting organizational boundaries necessitated by interdependence. Additionally, the analysis examined how offerings and their design inadvertently reinforce biases and inequities, highlighting the importance of integrating equity principles into systemic transformations. The research also reflected on the disruptive nature of introducing new infrastructures, considering how these changes impact existing systems, amplify biases, and promote shifts in societal norms. Lastly, the cases emphasized the importance of

network activation, where rethinking offerings becomes essential for addressing societal demands for equity, diversity, inclusion, and sustainability.

To support this analysis, multiple tools and frameworks were employed to understand the systemic dynamics of everyday infrastructures. The Action Situation framework (Bataineh & Teixeira, 2024a; Ehn, October 1-4 2008; Nogueira, 2019; Ostrom, 2011) was used to map activities influenced by resource flows and the underlying infrastructures enabling these flows. Causal Loops & System Dynamics Maps (Bataineh & Teixeira, 2024d; Hjorth & Bagheri, 2006) visually illustrated the relationships between system components and their interactions, helping to identify leverage points. The Anatomy of Infrastructure framework (Bataineh & Teixeira, 2024b; Nogueira et al., 2019) supported an understanding of the interconnectivity between agents and their agency within a system, enabling insights into how multi-level integration and feature interactions drive systemic impacts. The Archetype (Bataineh & Teixeira, 2024c) of Infrastructure framework provided insights into the core logic of seamless combinations of products, services, methods, communication channels, and data ecologies that contribute to common purposes. Tools such as Intersectional Stories (Bataineh & Teixeira, 2024f; Neimand et al., September 29 2021) helped communicate systemic issues like racism and classism by analyzing user experiences through their lenses. Finally, the Equity Framework—Dark Patterns & Hot spots—(Bataineh & Teixeira, 2024e; Meharry, 2022) was employed to identify asymmetries of power within the systems, exposing dark patterns and hotspots that amplify inequities and influence power dynamics.

The overarching objective of these case studies was to harness systems thinking to critically evaluate and reimagine everyday infrastructures. This multi-faceted approach focused on identifying systemic leverage points for impactful interventions and fostering sustainable and equitable systems. By offering macro-level assessments of the current state of everyday infrastructure, the research highlights both the problem and opportunity spaces of these systems. This approach challenges the limitations imposed by financialization as the dominant lens for decision-making, advocating instead for an opportunity space that weaves equity and sustainability into the process of navigating choices for value creation. Ultimately, these case studies establish new agendas for systemic transformation and provide the foundational context for subsequent stages of the research. Serving as the gateway of the strategic choice-making model.

4.3 A Semester-Long Workshop Series: Translating Macro Findings into Design Experiments

Building on the systems-level insights gathered through the *macro assessment case studies*, two *semester-long workshops* (see Fig 4.2 and Appendix B) served as a “space of experimentations” to translate the macro-level findings into simulating *real-world design experiments*. Masters-level design students and professional collaborators worked iteratively to craft feature concepts that respond to the challenges and opportunities identified in the macro assessments, the project briefs, and the design prompts shared with them.

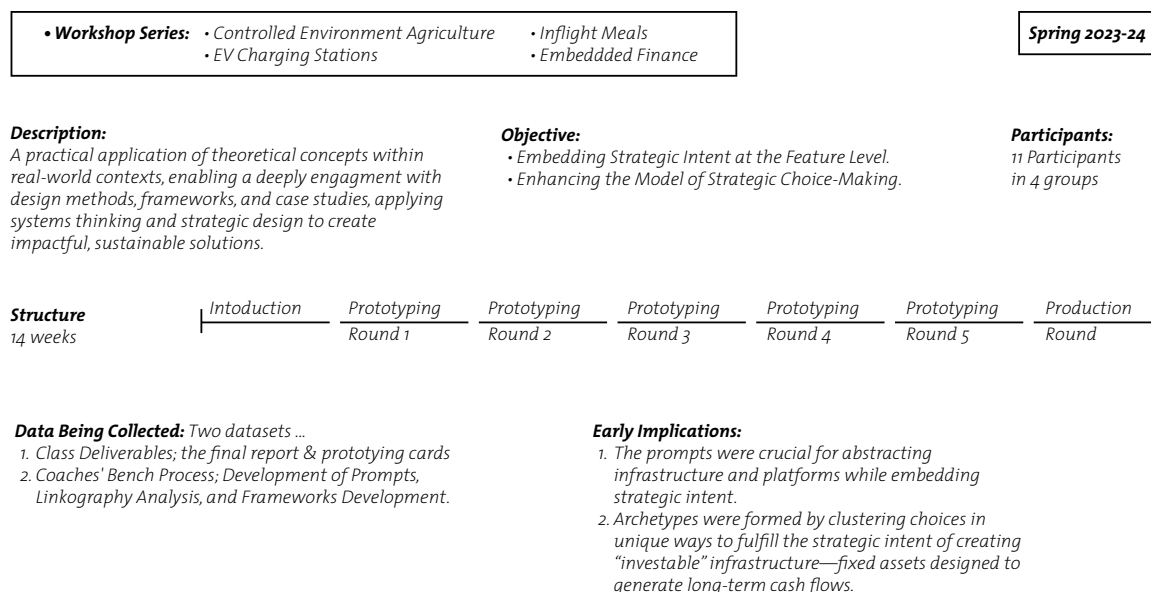


Fig 4.2. The Workshop Series Structure

The workshops, conducted in Spring 2023 and Spring 2024, aimed to embed institutional intent—which refers to the deliberate strategic orientation of institutions toward shaping, sustaining, and evolving their structures, policies, and resource allocations in alignment with broader societal, economic, and technological goals—into everyday infrastructures, from EV Charging Stations to In-Flight Meals Services. Each round of iteration emphasized iterative prototyping and critical evaluation of the task at hand—each round would prompt participants to evaluate a different institutional intent through a different macro force— Participants began by reviewing relevant macro-assessment findings to anchor their designs in existing systemic tensions and responding to the task at hand. Design actions were positioned as key inquiry points to generating and articulating knowledge guided by RtD principles (Gaver, 2012; P. J. Stappers et al., 2014; Pieter Jan Stappers, 2013; Pieter Jan Stappers & Giaccardi, 2017; Zimmerman et al., 2007); prototypes were conceived, tested, and refined, focusing on *features* at the

intersections of *flows* as the *unit of analysis*. Through multiple rounds of iterative prototyping and envisioning future archetypes (a set of features combined with intent), the workshops fostered a comprehensive reflection on design actions and their outcomes, explicitly considering the institutional intent of value creation within the infrastructure-platform mashup space. By linking design explorations with institutional objectives—where the identification of problems and opportunities directly reflects strategic goals—this new approach established the foundational practices of strategic choice-making. It is worth noting that both workshops were in collaboration with institutional partners from investment, engineering, business, and academic backgrounds and offered guidance and feedback to align prototypes and simulate real-world constraints. See Figure 4.2 of the workshop structure.

In Spring 2023, the workshop focused on two case studies: *EV Charging Stations* and *Controlled Environment Agriculture (CEA)*, both framed as everyday infrastructures within broader ecosystems. Participants were tasked with designing future archetypes of these infrastructures, to transform them into sustainable, equitable, and, most importantly, investable infrastructures. The process incorporated systems thinking, design frameworks, and multiple iterative rounds of prototyping, which encouraged participants to critically evaluate how strategic intent could be embedded into the design of features, products, and services. The research team was recording findings and enhancing the strategic choice-making model further along the process.

Building on this foundation, the Spring 2024 workshop continued under the same methodological structure, focusing on two new projects: Embedded Finance and In-Flight Meal Services. These projects were similarly framed as everyday infrastructures, and

participants were again asked to develop future archetypes at the feature level, applying the conceptual model of Strategic Choice-Making (SCMM). The iterative process began with structured design briefs and prompts, guiding participants to envision archetypes that are equitable, sustainable, intelligent, and investable. After each prototyping phase, participants clustered the features into one or more archetypes based on strategic intent, exploring how design can reorient resource flows to meet societal and environmental goals. This practice-driven approach provided an opportunity to validate and refine the SCMM, identifying how choice-making and decision-making are shaped within and around organizational contexts.

Throughout these workshops, Linkography (Goldschmidt, 2014; Hatcher et al., 2018) served as a key analytical tool for tracing how design choices evolved, as mentioned in Chapter 3. By mapping design moves, the research team could identify when strategic intent was embedded effectively into feature designs and when constraints—from policy to institutions—necessitated pivots. Frameworks and tools similar to those used in the macro assessment case studies were used generatively to further guide participants to weigh systemic tensions of how *flows* are reoriented. The iterative process revealed how *features*, strategically envisioned and prototyped, could orchestrate resource flows that meet *broader societal goals*.

Ultimately, these workshops clarified how to embed strategic intent into infrastructural designs, resulting in prototypes that were innovative, sustainable, and aligned with macro forces. By focusing on features at intersections of multiple systems, participants demonstrated how strategic choice-making navigates complexity to deliver value in the infrastructure-platform mashup space. Such insights enriched the Strategic

Choice-Making Model (SCMM), offering tangible examples of how design fosters large-scale transformations.

4.4 Three Controlled Experiments with Senior Practitioners

To stress-test some critical moments in SCMM, three controlled experiments (see Appendix C) were conducted with senior practitioners—*design professionals with leadership experience*—. While the semester-long workshops highlighted exploratory prototyping, these experiments were structured in a more “lab-like” format to learn how experts navigate macro forces through their institutional context under structured prompts and scenarios through the features and the archetypes given to them.

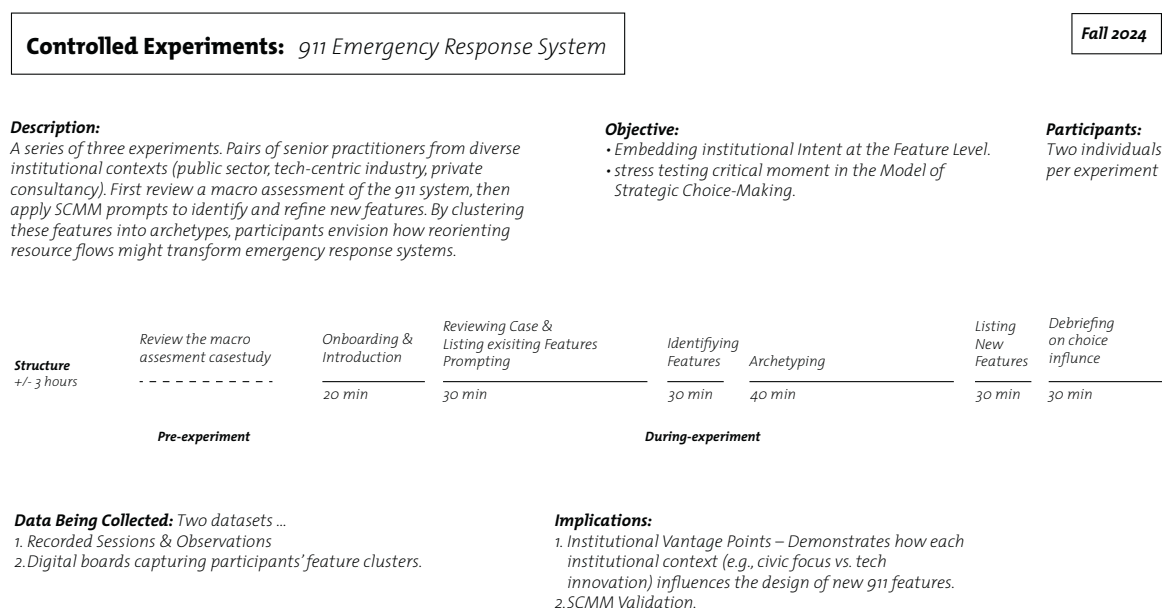


Fig 4.3. The Controlled Experiments Structure

Building on the central thesis that design, conceived as a strategic choice-making practice, can reorient resource flows and thus foster large-scale transformations, these

experiments specifically focus on the process by which senior practitioners embed institutional purpose into feature-level decisions. They were designed to validate the distinction between choice-making and decision-making practices. In our setup, the participants were first provided with a wide range of feature-level choices—generated through a phase we refer to as choice-making. They then applied their institutional context to select among these options, effectively embedding institutional intent into their final decisions and simulated real-world professional decision-making. This process was not focused on demonstrating the choice-making capability as much as confirming that subsequent, context-driven decision-making is essential for translating those choices into actionable outcomes. They also test whether the Strategic Choice-Making Model (SCMM) provides a coherent framework for guiding strategic conversations about macro-level challenges. Unlike the master-level students in previous workshops, participants here bring decision-making capabilities and broader institutional authority, allowing for a realistic simulation of organizational constraints and interplay.

The experiments were asked to reimagine the 911 emergency response system as an everyday infrastructure going through transformation, a domain characterized by high stakes, multiple stakeholders, and pressing macro forces. Before each experiment, participants reviewed a “macro assessment” case study of the 911 system, acquainting themselves with the problem space (e.g., resource inefficiencies, equity gaps, communication silos) and the opportunity space (e.g., advanced data analytics, improved dispatch, new stakeholder partnerships). This aligns with the SCMM’s emphasis on mapping both problems and opportunities at a systemic level and situates participants to

consider how institutional viewpoints might shape the design of intelligence features within a critical, time-sensitive context.

To capture a broad range of institutional perspectives, each experiment was conducted with a different group representing a different institutional context:

1. An in-house *tech-centric* consulting team, where participants tend to prioritize data-driven solutions and algorithmic scalability;
2. A traditional *innovation-focused* consultancy, whose practitioners emphasize design methodologies, user experience, and creative ideation;
3. A *civic-centric* innovation team, oriented toward policy imperatives, public-sector governance, and community-focused impact.

The research team initiated the experiment by introducing the SCMM, explaining its core idea of expanding or reorienting resource flows through feature-level interventions. Participants then situated their view of 911 emergency services within their institution's specific context—in relation to the platforms-infrastructure mashups. Following this, they were provided a guiding brief highlighting a hypothetical (yet realistic) challenge in 911 emergency response—often combining issues of operational strain, equity deficits, or cross-agency data sharing. Alongside the brief, participants received a set of ten prompts, each proposing distinct orientations of resource flows and strategic intents, such as emphasizing equity-driven resource deployment or leveraging platform-based analytics.

Participants identified archetypes and features from a set of ten different archetypes—like Wearable Emergency Alert Technology, Community-Driven Training Programs, or Multilingual Support Systems—each archetype contained 10 different

features combined to make up the archetype. Each group selected the archetypes/features that resonated with their institutional vantage point. They then clustered these features into new or reconfigured archetypes, effectively simulating the SCMM's movement from broad problem recognition to purposeful feature design. By mapping the resulting archetypes onto infrastructure-platform mashups, they visualized how the process has helped them rethink their positioning and expand the type of choices within their institutional context.

The final step of each experiment included a reflection and debrief: participants discussed how the SCMM's structure influenced their decision-making, if it imposed discipline problem-solving, and to what extent it helped them “imprint” institutional purpose onto features. While choice-making expands possibilities, decision-making validates and structures these choices within institutional contexts (Heskett, 2001; Owen, 2005). These sessions describe how senior practitioners—unlike graduate-level design students—might grapple with the macro forces and institutional purpose through reorienting resource flows.

4.5 Research Framework

The three streams of inquiry outlined in this chapter converge to construct a multifaceted view of strategic choice-making in infrastructure-platform mashups. Each stream addresses a distinct purpose, yet all coalesce around the question: *How can design offer strategic choice-making in infrastructure-platform mashups to drive value creation for large-scale transformations?* They build a contextual understanding of how features at the intersection of multiple flows can be deliberately shaped to embed strategic intent.

Ultimately, this research framework ensures that “choice-making” is both a descriptive phenomenon (documenting how features are shaped) and a prescriptive practice (guiding what features might be). It also adds design capabilities to institutional decision-making processes.

By weaving these streams together—*namely, the macro assessment case studies, the semester-long workshop series, and the three controlled experiments with senior practitioners*—address a gap in the design knowledge —*the need for navigating macro forces and institutional purpose to systematically drive infrastructure-platform mashups toward large-scale value creation and transformations through featuring*—the research transcends disciplinary silos and theoretical speculation, moving toward empirical validation and systemic considerations at the feature level. Through this convergence, each approach yields complementary insights:

- The macro assessment case studies produce contextual awareness of multi-layered complexities and resource flows;
- The workshops capture how the design practice prototype, cluster, and embed purpose into features through iterative inquiry of choice-making in institutional decision-making processes;
- The controlled experiments with senior practitioners stress-test the SCMM in a controlled setup of specific institutional context and decision-making authority.

The next chapter will present and analyze the data sets generated from these three streams, focusing on critical key findings that anchor the Strategic Choice-Making Model:

1. Conceiving features at the intersections of macro forces, systemic resource flows, and institutional purpose drive strategic value creation.
2. Choice-making practice expands the possibilities of value creation within the infrastructure–platform mashups.
3. Institutional context shapes choices and decisions.

CHAPTER 5

HOW FLOWS-ORIENTED CHOICES ARE SHAPED

This chapter builds on the empirical findings gathered through the macro assessment case studies, semester-long workshop series, and controlled experiments to illustrate the broader implications of a flow-oriented design approach. Earlier chapters established a conceptual and methodological foundation, highlighting the relevance of macro forces, purpose-driven value creation, and institutional intent as key catalysts for reshaping resource flows. The following pages synthesize these insights and address three interrelated key findings: *Conceiving features at the intersections of macro forces, systemic resource flows, and institutional purpose drive strategic value creation; Choice-making practice expands the possibilities of value creation within the infrastructure–platform mashups; Institutional context shapes choices and decisions.*

The discussion synthesizes the datasets drawn from the multiple explorations throughout the research, generalizing the findings beyond a single institutional setting, and situating them in a dialogue with the central thesis of this dissertation. By examining the interplay of macro-level forces and institutional purpose, the findings affirm that the design of everyday features goes well beyond mere artifact improvement. Instead, deliberate framing of the design problem, with structured system-centric frameworks and strategic intent, collectively enables a choice-making practice to reconfigure flows for sustainable, equitable, and future-oriented outcomes.

5.1 Framing the Exploration for Transformation

This section addresses how features might be conceived at the intersections of macro forces, systemic resource flows, and institutional purpose—a recurring theme throughout the data gathered across all research explorations of this dissertation. As discussed in earlier chapters (particularly Chapter 2), current design practices often prove insufficient for orchestrating large-scale transformations. Across the various research streams (see Chapter 4), it was evident that unless micro-level considerations of features, products, and services are deliberately framed within a broader multi-stakeholder, multi-level, and multi-intelligence context, design frequently falls short of catalyzing meaningful large-scale impact. By contrast, the data showed that an expanded systems-thinking lens reveals deeper leverage points for transformation. Rather than treating features as discrete endpoints, the data suggest adopting a broader perspective that regards design elements as nodes within vast and intricate networks of policies, stakeholders, and resource pathways.

A recurring observation, highlighted especially in the macro assessment case studies, is that *macro forces*—including sustainability, digitization, equity, financialization, and deglobalization—subtly but powerfully shape the constraints and opportunities surrounding any design intervention. A relevant observation of this dynamic emerged in the Controlled Environment Agriculture (CEA) project. Prior to the semester-long workshop, participants received the CEA macro assessment case detailing both the *problem space*—ranging from climate vulnerabilities to the restrictive financing models often tied to industrial-scale agriculture—and the *opportunity space*, such as the potential for high-yield vertical farms and local community partnerships. While teams

initially gravitated toward modest features, such as specialized lighting controls or nutrient-monitoring dashboards, the integration of systems-thinking frameworks revealed how macro forces intersect with local agricultural practices, enabling CEA to address more than just operational efficiencies. The workshop’s archetypes, for instance, showed that reimagining data collection, waste recycling, and distribution channels can reorient power dynamics between farmers, investors, and governing bodies. One archetype (see Figure 5.1) even bundled sensor-driven yield optimization with a cross-subsidization plan to route a portion of produce revenue toward equity-focused community-supported agriculture programs.

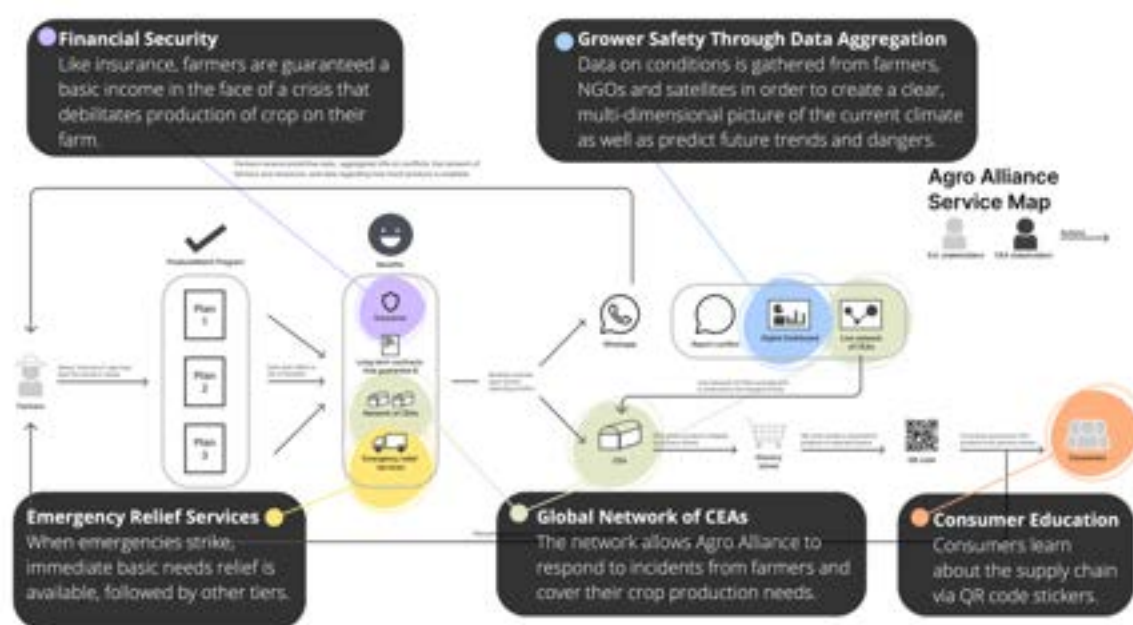


Fig 5.1. Agro Alliance Archetype

This reframing aligned broader institutional agendas—namely, a desire for both robust sustainability and financial viability—with concrete, feature-level innovations. Rather than focusing only on immediate efficiency optimization, these concepts tackled deeper questions of resource distribution and community empowerment: local

governments could leverage job creation potential, investors gained a new asset class in sustainable food systems, and growers benefited from aggregated data and shared logistics. As a result, features once limited to surface-level prototypes (e.g., improved humidity controls) evolved into strategic nodes for redistributing financial resources and achieving equity objectives. The CEA example thus illustrates how adopting a systemic and flow-focused approach can expand the role of design from a tool for incremental improvements to a catalyst for large-scale transformation across agricultural, governmental, and investment domains.

A central insight from this work is that applying systems-centric design frameworks and guides—such as Action Situations (Bataineh & Teixeira, 2024a), Causal Loops & System Dynamics (Bataineh & Teixeira, 2024d), Hot Spots & Dark Patterns (Bataineh & Teixeira, 2024e), Intersectional Narratives (Bataineh & Teixeira, 2024f), Archetypes (Bataineh & Teixeira, 2024c), or the Anatomy of Infrastructures (Bataineh & Teixeira, 2024b)—helps define the interplay of resource flows and the technological advancements embedded in everyday infrastructures. Through these lenses, a feature like an in-flight meal selection system is no longer just a convenience; it represents an intersection where financial investment, environmental considerations, and cultural values intertwine, generating value-driven choices. Workshop participants who employed these frameworks began to see design choices as opportunities to redirect resource flows—financial, human capital, or data—toward outcomes aligned with institutional objectives like equity or sustainability. In this way, the features themselves become gateways for orchestrating wider social, economic, and technological impacts.

Such findings underscore the importance of *declaring the strategic intent* and *acknowledging institutional purpose* early in the design process. While many user-centered methods emphasize near-term usability or aesthetics, the in-flight meal macro assessment and the workshop report illustrate how embedding institutional agendas like—public health standards to net-zero commitments—can significantly expand design horizons. In the macro assessment, researchers documented not only the logistical intricacies of meal preparation and distribution but also the broader systemic forces at play, including airline branding strategies, evolving passenger expectations, and external sustainability pressures. Initially, participants encountered tension between fulfilling immediate user needs (e.g., taste preferences, packaging convenience) and organizational goals like cutting carbon emissions or minimizing food waste. Yet, the workshop sessions revealed that aligning feature ideation with these higher-level agendas—such as an airline’s net-zero pledge—ultimately prompted new alliances with suppliers, waste-management firms, and even adjacent travel services.

Similarly, the inflight meal case study highlights that *explicitly* aligning innovation with organizational imperatives—such as net-zero pledges or public health standards—can enhance both user-centric outcomes and broader systemic gains. Merely focusing on immediate passenger needs (e.g., taste, convenience) can overlook the deeper resource flows and strategic agendas that define airline operations. In the workshop, *declaring strategic intent* early on helped teams reimagine onboard catering, linking food-waste reduction efforts to corporate ESG metrics—leveraging the embedded intelligence and the data ecologies—, and exploring localized partnerships that benefit not only farmers and airline suppliers but could also support sustainable urban

ecosystems (See Figure 5.2) highlighting the leverage points identified. While skepticism might contend that tying design choices to large-scale agendas risks diluting practical features, but the workshop data demonstrate the opposite: agenda-driven choices encouraged novel coalitions—like waste-management collaborations or carbon-offset meal upgrades—that furthered business objectives while satisfying passenger expectations. Ultimately, viewing in-flight meals as nodes within a resource-flow ecosystem enabled design teams to transcend incremental fixes and propose features that meaningfully advance airlines’ operational efficiency, reputational gains, and environmental commitments.

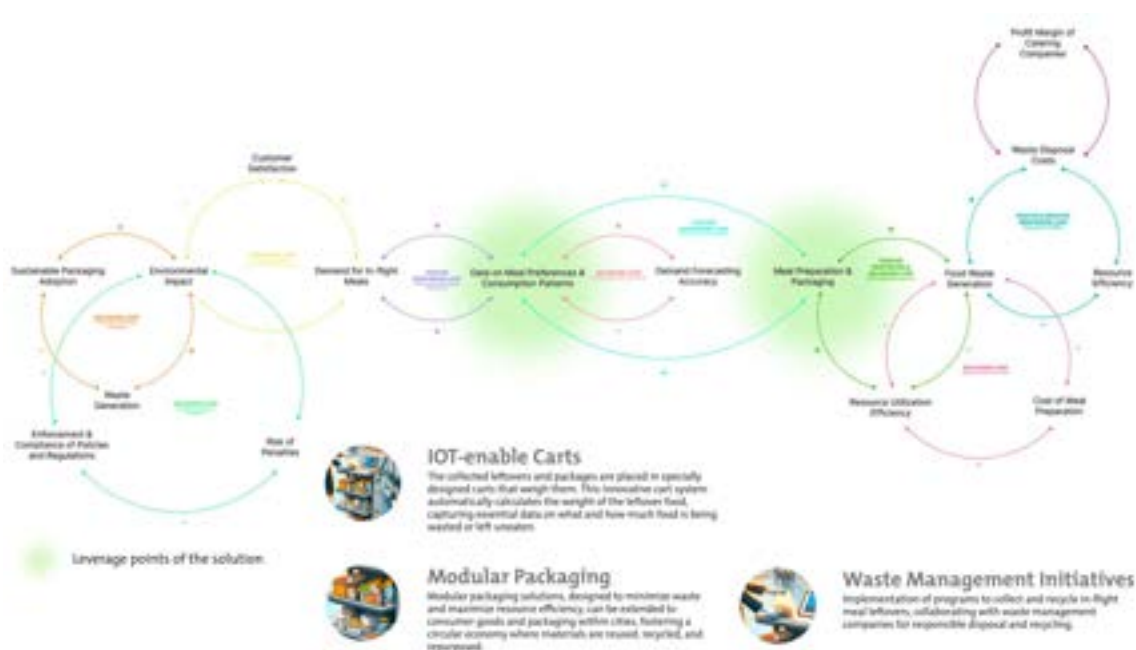


Fig 5.2. Features and Causal Loops of the Waste Management Archetype

Taken together, these insights affirm that *features can be conceived as strategic levers* where macro forces, systemic resource flows, and institutional purpose intersect. By intentionally mapping such flows and acknowledging how multiple stakeholders

operate at various levels, designers are empowered to generate choices that extend well beyond an app screen or product interface. Far from being a theoretical construct, this approach surfaced consistently in the data—design practices that robustly integrated systems-centric frameworks were more likely to yield features with institutional traction, enable new coalitions, and advance the organization’s long-term goals. Consequently, *macro-awareness* and *institutional alignment* become fundamental, rather than supplementary, to design practice. They constitute the scaffolding through which features evolve into pivotal instruments in orchestrating large-scale, beneficial change.

5.2 Generating Choices in the Infrastructure-Platform Mashups

This section examines how design teams *generate* and *orchestrate* strategic choices within the complex space of innovation defined in Chapter One as platform-infrastructure mashups. Whereas the previous section addressed how features are conceived at the intersection of macro forces, systemic resource flows, and institutional purpose, can catalyze transformation, the focus here shifts to operationalizing the process of generating such choices in practice. The question becomes how to integrate macro-level imperatives and institutional agendas into micro-level features—detailing choice-making in practice to enable large-scale transformation. Building on recurring insights across the research streams, one central finding is that *declaring strategic intent* and *acknowledging institutional purpose* early in the design process sets a clearer agenda for choice-making. This recognition surfaced the need for a structured mechanism—Prompts (see Figure 5.3)—that narrows context, constraints, and relevant macro forces while abstracting the function of underlying infrastructures and platforms. Rather than

constraining creativity, these prompts were structured to focus design efforts on explicit strategic intents, thereby initiating the solution space with an agenda.

Spring 2023 | Sustainable Solutions Workshop

Controlled Environment Agriculture

Prompt; Prototyping Round 3/4

Prompt 3.3

Infrastructure is always changing, yet two things remain true: humans have always needed to **generate, move and store** things; and infrastructure emerges from the interaction of human and non-human agents.

With this in mind, develop a concept for **investable infrastructure** in Controlled Environment Agriculture **for land-based protein**.

Consider focusing on one, or a combination of, the three core human needs and agents that necessitate infrastructure. Explain and describe which aspect you incorporated, and why. You must define and explain where to play and how to win. Your investable infrastructure must be aligned with the 3D forces shaping infrastructures (**digitize, decentralize, and decarbonize**) as well as the 3 fundamental functions of infrastructure (**generate, move, and store**).

Fig 5.3. Prompt CEA 2023

Through the data gathered, the prompts played a crucial role in triggering the process of designing choices within platform-infrastructure mashups. While participants in the research initially struggled to integrate multiple transformative forces—such as digitization, sustainability, and financialization—and gravitated towards generating choices that are centered on convenience and efficiency, for example, the teams participating would start to generate choices related to user interface refinements or payment conveniences in the case of embedded finance, but once considering the multiple prompts shared with them (See Figure 5.4), they began exploring features related to “community supporting profiles or alternative ID systems” to enable financial

agency, and “Bank on Wheels” to democratizing access to finance. The prompts started to play a crucial role in embedding strategic intention into the features being designed, even at an archetypal level. Each new feature concept was evaluated not only for functionality but also for alignment with higher-level policy, social, or economic goals. Thus, agenda-driven prompts become a linchpin for generating choices, ensuring that even seemingly narrow interventions can leverage multi-capital resources and resonate with institutional strategies—a theme that resonates with the discussion on choice-making and decision-making in Chapter 6.

- Sustainable Finance for Underserved Communities:**
Prompt: *How can Credicorp leverage embedded finance to design sustainable financial solutions that not only cater to underserved communities but also align with environmental goals, creating a more inclusive and ecologically responsible banking experience?*
Considerations: Consider green financing options and eco-friendly investment portfolios. Data centers where data is stored contribute significantly to sustainability metrics as well. Offer financial services that support sustainable practices in underserved regions, promoting both environmental and financial inclusion.
- Decentralizing Financial Services for Broader Access:**
Prompt: *What role can decentralized, blockchain-based financial services play in Credicorp's mission to broaden financial access, especially in remote or underbanked areas, and how can these technologies be used to enhance transactional efficiency and trust?*
Considerations: Investigate blockchain for transparent and secure transactions. Explore mobile-based decentralized banking solutions to reach remote customers; i.e. payment through text, QR codes ... etc. Focus on building trust and understanding of decentralized services.

Fig 5.4. Prompt Embedded Finance 2024

Equally crucial in shaping choices is the archotyping process, which clusters and synthesizes individual features into coherent strategic bundles. This important step echoes the notion that when *features, products, and services are conceived at the intersections of macro forces, systemic resource flows, and institutional purpose*, there

needs to be a mechanism in place to cluster them at an archetypal level because of the fragmentation risk if addressed in isolation. Data from the sustainable solutions workshop and the controlled experiments consistently show that after teams generated a breadth of ideas prompted by macro-level agendas, they then grouped these concepts by overarching strategic intent, forming archetypes. These archetypes illuminated patterns tying user demands, institutional mandates, and resource flows together, enabling design teams to envision how discrete feature ideas might collectively transform entire ecosystems. The workshop data highlights that an archetype (See Figure 5.5) contextualizes each feature within a broader design vision, scaffolding it from the macro to the micro—for instance, the Democratizing Access archetype fits within the broader ecosystem envisioned of Equitable Financial Future and has four different features such as Digital Twins or Adaptive Defi—allowing teams to assess which bundled solutions can simultaneously address local constraints and achieve large-scale institutional objectives; a multi-level approach to the solution space. In this way, archotyping ensures that micro-level prototypes are not just scattered innovations but cohesive, strategy-aligned interventions that can reorient resource flows for systemic impact.

Both prompting and archotyping were findings that have major implications for the strategic choice-making hypothesis. Those findings were gathered through the “coaches bench process,” where ongoing analysis, observations, and further development of the tools and the frameworks were conducted with the instructor and the researchers. To further inform archotyping, the research team developed two frameworks to determine the function of features in the mashup space and how these features can be strategically positioned within these spaces. Figure 5.6 leveraged strategic thinking as a framework to

strategically position features from a design point of view at the intersection of systems.

In contrast, the “Features Clustering” framework in Figure 5.7 of platforms-infrastructure functions was developed to help cluster features based on their function. It facilitates better alignment of features functioning in platform-infrastructure mashups, reflecting the complexities of modern societal and technological needs.

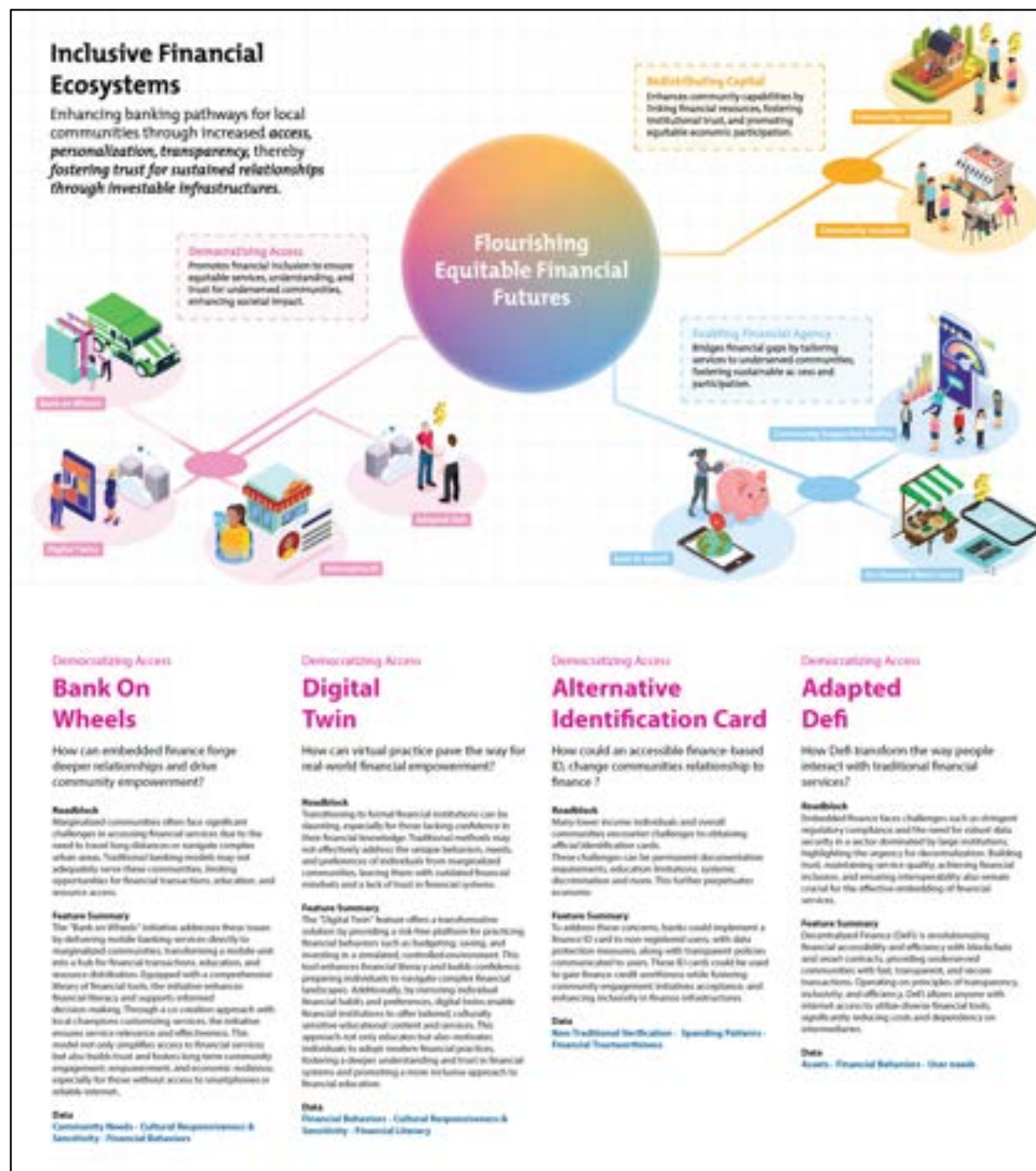


Fig 5.5. Embedded Finance Archetype 2024

Creating a dashboard to monitor insect protein powder generation.		
	Where to Play	How to Win (Value Proposition)
Multiple Systems	Extend to offering on-the-job training for incarcerated people in prison, guaranteeing positions and competitive salaries once they are released	Building a highly smart (automated control and operations management of CEA) dashboard centers inequities
Multiple Levels		
Multi-intelligence	Focus on Insect-based protein powder	Establishing intelligent systems that - 1. Using data on optimal insect growth to influence the CEA climate 2. Using sensors to detect current weather info (temperature, light levels, humidity?), comparing requirements for insect production within the CEA, 3. Automating most sustainable practice for the CEA to conserve energy
Multiple Stakeholders	Expand to emerging markets - 1. Distributing the protein powder to pet food companies, grocery stores, local restaurants, museums, institutions, schools, enterprises(airlines) for revenue generation 2. Distributing the protein powder to prison and food insecure communities	Putting into place 1. Insight form & choice matching from partners 2. Automated decisions - consciously thinking of where humans can make the decisions with that intelligence (an equity based lens)

Fig 5.6. Strategic Thinking in the Mashups Framework

		Functions of Platforms (Features)		
		Monitor (to display)	Control (to add variables)	Manage (to act)
Infrastructure functions	Store			
	Generate	Creating a dashboard to monitor insect protein powder generation.	Special constraints 1. Temperature control to grow a certain type of insect 2. retrofit into existing greenhouses by adding modular factory spaces 3. A dashboard to be intelligent and responsive to insect life cycles & control the generation	
	Move		4. Automated responses to enable insect protein distribution.	

Fig 5.7. Features Clustering Framework

Multiple data points (see Figure 5.8)—from the workshop materials— together were layered and cross-tabulated to trace how features are developed through the strategic choice-making model. This layering of findings aids in further refining the model, providing a structured approach to feature development. A detailed description of the model can be found in Chapter Six.

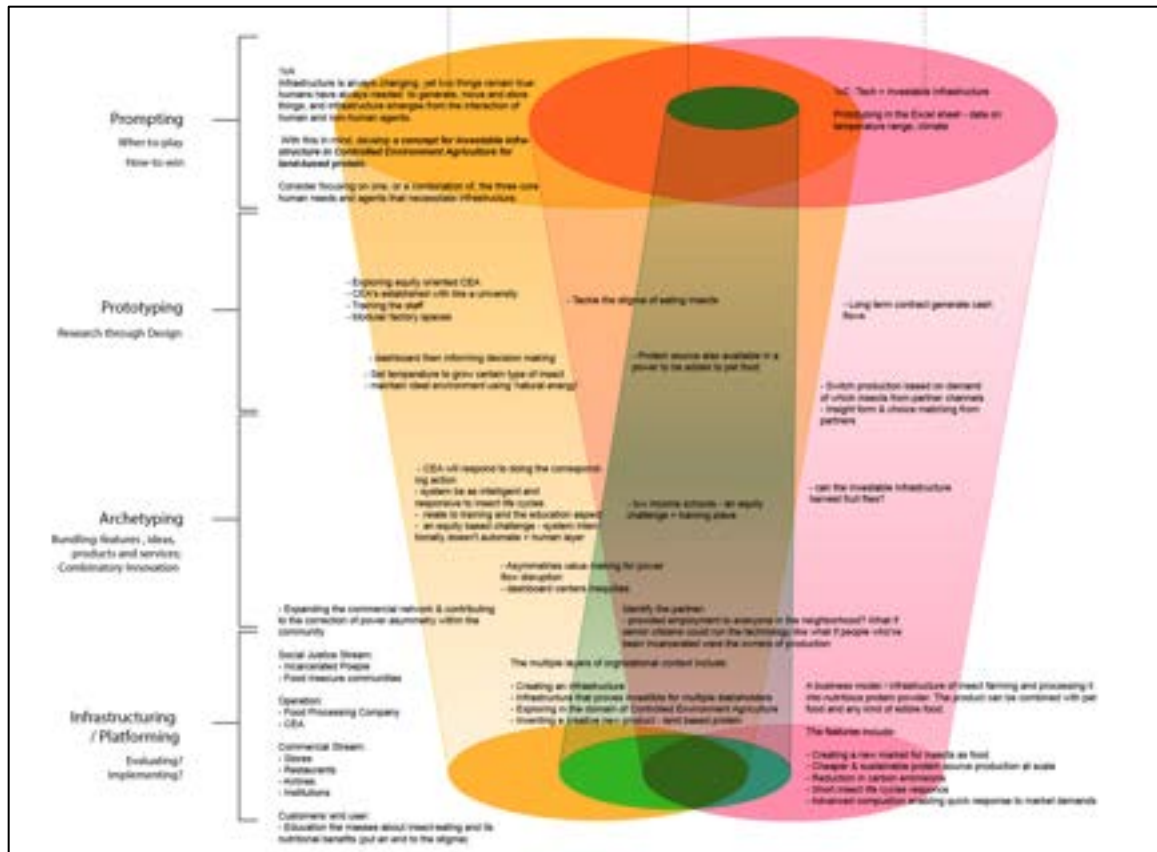


Fig 5.8. Controlled Environment Agriculture Case Layered

Taken together, *agenda-driven prompts* and *archotyping* emerged as interlocking mechanisms for generating and orchestrating choices in platform-infrastructure mashups that truly integrate macro imperatives with micro-level features. Far from being abstract exercises, *prompts* guide design teams to explore solutions whose scope naturally extends beyond immediate user experience to address institutional and societal agendas while archotyping transforms this abundance of feature concepts into unified strategic bundles. The data suggested that an iterative prototyping cycle confirmed that incorporating macro-level constraints and institutional goals early on broadened, rather than narrowed, the solution space. As a result, the evolving prototypes not only satisfied local user

demands but also contributed to more ambitious transformations, aligning with the Strategic Choice-Making Model (SCMM) that will be further detailed in Chapter 6.

5.3 Institutional Context and the Shaping of Choices

Institutional context shapes how choices and decisions are made, particularly in the context of infrastructure-platform mashups. As outlined in Chapter 1 and further expanded in Chapter 2, the blurring characteristics of these mashups demand a reframing to navigate the institutional tensions emerging. The findings from the controlled 911 experiments in this section illustrate how institutional context can act as both enabler and constraint, establishing the boundaries within which design choices either thrive or stall. For instance, the interplay between public and private logic creates a dynamic context where traditional infrastructure-based services converge with platform-driven innovations, opening new avenues for innovation while introducing tensions that require strategic negotiation.

One key implication of these experiments is that institutions inherently influence how choices are prioritized and framed. As established by the macro assessment cases (Section 5.1) and the archetype/prompt approaches (Section 5.2), macro-level forces permeate each institution's DNA, steering teams toward particular resource flows and agendas. The data showed, for example, that the in-house tech-centric consulting team gravitated toward data-intensive or efficiency-focused features (e.g., location-based dispatch), while a civic-centric innovation team emphasized public-value archetypes (e.g., cross-agency data integration or privacy safeguards). These divergent directions highlight that choice-making is governed by the institutional context each team

represents. The institutional context serves as a mechanism for teams to refine broad, sometimes abstract, choices into grounded themes aligned with their institutional mandates.

Traditional emergency services like 911 have historically operated in a public framework that emphasizes universal access, systems-centric, and for the public interest. However, the growing integration of technologies developed mainly by the private sector illustrates further the blurring boundaries of institutional contexts as highlighted in both Chapters One and Two, thus reshaping the meaning and purpose of these services. As shown in Figure 5.9, participants in the controlled experiments initially positioned 911 on the “infrastructure” side of the mashups spectrum. Yet after reviewing the archetypes and the features—Such as real-time location tracking, multilingual triage, or predictive analytics,—911 evolved into a platform-like offering in the mashups, necessitating private-sector participation and opening the door to new forms of monetization or data sharing. This shift sparked debate among participants regarding equity, data ownership, and regulatory oversight. In effect, the “blended ownership” of 911 highlights how public and private logic interweave in the mashup space, redefining what an “essential service” might look like and for whom it is designed.

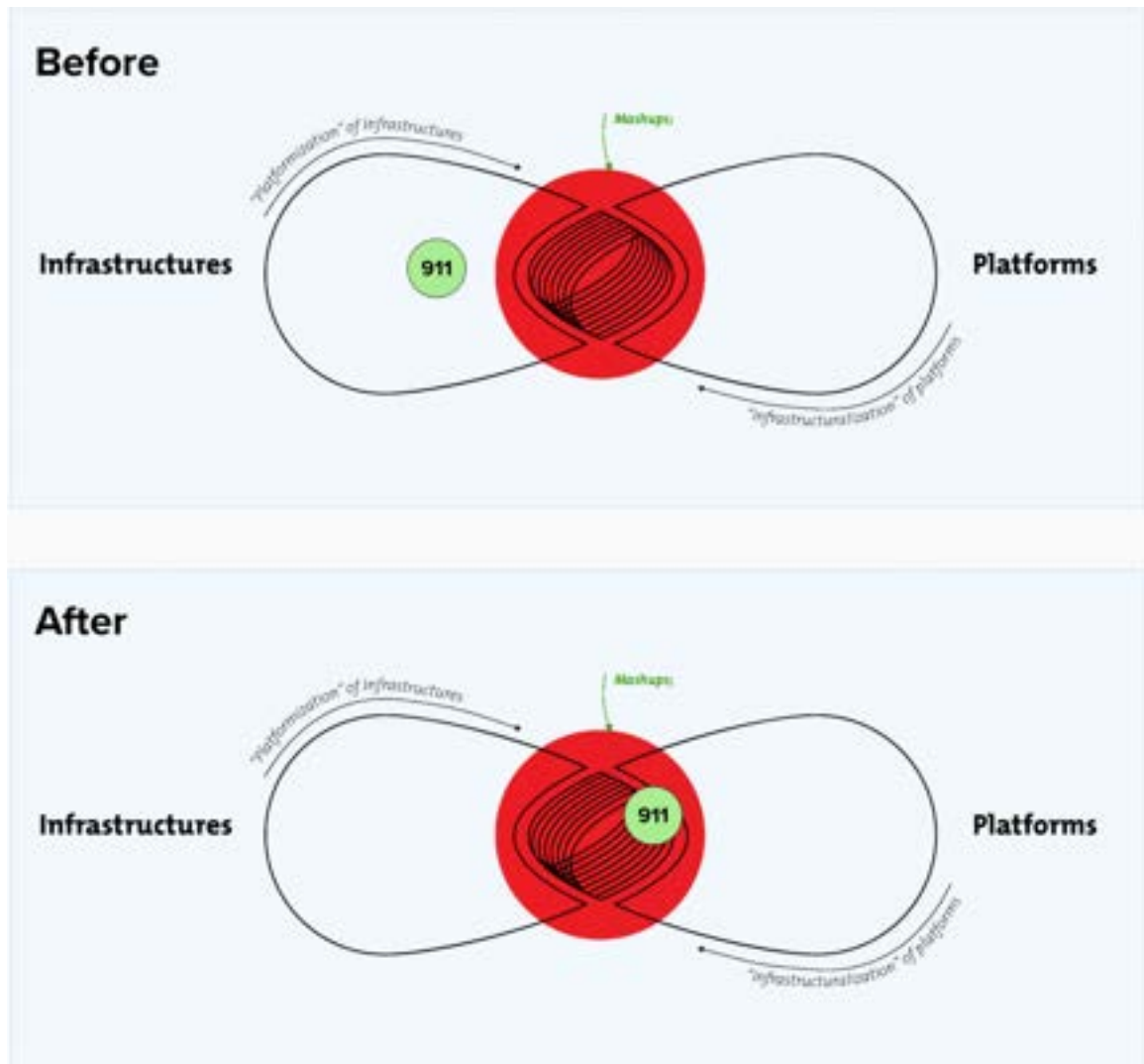


Fig 5.9. 911 Positioning in Infrastructure-Platforms Mashups

The findings from the controlled experiments further highlight the importance of institutional context in governing the feasibility of design choices. Regulatory frameworks and policy mandates serve as critical gatekeepers, ensuring that innovations align with ethical standards, data privacy requirements, and public accountability. On the one hand, these constraints may limit the speed and scope of private-sector innovation. On the other hand, they also safeguard public welfare by embedding normative

principles—e.g., equity and universal access—into the decision-making process, further influencing the convergence of choices and decisions. For instance, the location-based dispatch prioritization feature championed by the tech-centric team raised significant concerns from the civic-focused team, who questioned sensitive data use and potential biases in responding. Meanwhile, the cross-agency data integration feature—viewed favorably by the civic team to enhance accountability—faced disinterest from the tech team, which saw regulatory hurdles as exceeding its institutional context. The fact that both features are interdependent, yet the convergent choices through both institutional contexts revealed a tension of an institutional “missing link” further suggests the need for a proactive multi-stakeholder collaboration at the feature level to respond to the complex socio-technical challenges in the platform-infrastructure mashups.

Furthermore, these findings suggest that a “hybrid institutional context” driven by proactive multi-stakeholder collaboration is increasingly necessary to support new, platform-like features for emergency services. Although innovations like real-time location tracking, predictive analytics, or multilingual support can improve response quality, they also risk fragmenting access or adding hidden costs if driven by purely private logic. The institutional context thus shapes not only what is possible but also who benefits, compelling design teams to integrate equity considerations (e.g., open access, cross-subsidies) into any new offerings. By doing so, they mitigate the risk of excluding underserved populations—a core public concern for an emergency service with a universal mandate.

the controlled experiments show that institutional context mediates the divergent and convergent phases of choice-making, as proposed in Sections 5.1 and 5.2. The

controlled experiments through the 911 topic reveal how prompts, archetypes, and macro forces collide with different institutional imperatives, yielding varied outcomes. By spotlighting critical moments when public and private perspectives conflict or converge, we see how institutional context becomes the arena in which design's broader potential is either realized or constrained. This understanding sets the foundation for Chapter 6's outlining the strategic choice-making model, elucidating precisely how institutions, along with macro forces and design frameworks, co-create and mediate the divergent–convergent interplay of choice-making, shaping the transformative offerings in the infrastructure-platform mashups.

5.4 Critical Insights for Flows-Oriented Choices in the platform-infrastructures mashups

To summarize, in this chapter, we explored how design can strategically reorient resource flows by aligning features, macro forces, and institutional purpose—thereby addressing the knowledge gap identified in Chapter 2 and reaffirming the dissertation's central thesis and hypothesis. Three important insights stand out:

1. *Reframing Macro Forces*: Rather than treating macro forces as external constraints, they can ignite feature-level innovations capable of large-scale transformations if reframed as strategic levers for choice-making.
2. *Agenda-Driven Prompts and Archotyping*: Structured agenda-driven prompts, synthesized into coherent intentional archetypes, enable the generation of choices that embed strategic intent that is equitable, sustainable, intelligent, and investable.

3. *Institutional Context is Reciprocal*: Institutional intent not only shapes features but is itself *reshaped* by novel archetypes, highlighting the strategic choice-making impact in addressing growth well beyond the immediate institutional context. Point 3 in section 6.1 of chapter Six, elaborates more on how the Strategic Choice-Making Model addresses growth opportunities in the Infrastructure-Platform mashup space.

Building on these findings, the next chapter details how the Strategic Choice-Making Model proposed provides a structured approach for a practice that navigates macro forces and institutional purpose to drive large-scale transformation.

CHAPTER 6

THE STRATEGIC CHOICE-MAKING MODEL

This research identified a model for designing features at the intersections of complex systems, culminating from the theoretical groundwork, methodological insights, and empirical findings presented in previous Chapters 1–5. By taking into account the central thesis—*that features in complex spaces of innovation can mediate between institutional purpose and macro forces to enable large-scale transformations*—and addressing the core research question—*How can design offer strategic choice-making in infrastructure-platform mashups to drive value creation for large-scale transformations?*—this chapter defines the *Strategic Choice-Making Model (SCMM)*. It situates the model in the broader context of complex spaces of innovation, namely *the infrastructure-platform mashups*, illustrating how strategic choice-making illuminates the systemic implications of features, facilitates design at these complex intersections, and opens up growth opportunities both within and beyond institutional contexts.

6.1 Addressing Complexity through Strategic Choice-Making

As previous chapters, specifically Chapter Two, pointed out, design’s traditional focus on *form-function-context* is often limited in capturing the breadth of interconnected contexts and systems resources that drive today’s complexities (Alexander, 1964; Koffka, 1935; Kohler, 1938; Nogueira et al., 2019). In response, the *Strategic Choice-Making Model (SCMM)* addresses the limitations of linear design models by incorporating systemic resource flows and purposed-driven agendas, building on the central thesis call

for design as the mediator for large-scale transformations. Enabling designers and institutions to navigate and influence complex systems more effectively. This model serves multiple purposes:

1. *The systemic implications of features:*

- *Enabling the Activation of an Agenda (Strategic Intention):* While existing design practices and models (e.g., human-centered design, design thinking, the double diamond, and the innovation funnel) Typically spur the understanding of the problem space, often limited to the context of production and use, and the goal is often capital-centric, allowing the generation of cash flows and promising efficiencies and convenience. As Charles Owen (Owen, 2005) argues, current decision-making models often focus on optimizing known conditions rather than exploring future potentialities and navigating the unknowns. As underscored in Chapters One and Two, these approaches can overlook broader sustainability and equity concerns. Centering resource flows at the feature level, the SCMM integrates strategic intent (Poggenpohl, 2017) to ensure deeper engagement complex nature of challenges in the infrastructures-platforms mashups.
- *A Multi-Dimensional Perspective:* By centering on multiple flows of resources—financial, manufactured, human, social, etc.—aligning choice-making with institutional context, the SCMM acknowledges non-human agency (Lemke, 2021; Morton, 2013)

and advances a systems lens that integrates long-term impacts. This systems approach, as highlighted by Heskett, aligns decision-making with broader societal and institutional goals, ensuring a deeper engagement with systemic and long-term considerations while maintaining economic and strategic relevance (Dilnot, 2011; Poggenpohl, 2017). Chapters Four and Five showed how embedding strategic intent at the feature level shapes systems that address more than immediate capital gains, ensuring deeper engagement with sustainability and equity.

2. *Designing features at intersections:*

- *Navigates Choice-Making and Decision-Making through the Institutional Context:* Existing design models often put an emphasis on the practice of choice-making and overlook the decision-making process and the organizational context in which those choices will be implemented. According to Heskett, institutional structures heavily influence design choices by shaping economic and strategic parameters (Poggenpohl, 2017). The SCMM ensures alignment between choice-making (exploring a breadth of possibilities) and the convergence of choices within and beyond the institutional context.
- *Holistic View of Multi-system Integration:* The SCMM facilitates a holistic approach that considers multiple systems, stakeholders, and embedded intelligence, ensuring a holistic view and

understanding of its implications, addressing the gap indicated in Chapter Two as a result. Owen's (Owen, 2005) perspective on balancing exploration and refinement further reinforces the need for an integrated approach that accounts for institutional constraints and opportunities. The fluency of design in choice-making and the capabilities of confronting the unknowns urges a systemic understanding in navigating the complexities of the institutional context (Dilnot, 2011), ensuring strategic coherence in decision-making to prototype large-scale transformation.

3. *Growth Opportunities Through Features:*

- *From Merely Focusing on Products and Services to Shaping Archetypes:* Existing design models envision products and services in silos; however, confronting the challenges presented through the mashups necessitates a broader contextual understanding, further echoing Whitney's notes, on the transition from an economy of scale to an economy of choice (Whitney, 2015). While examining products and services in silos serves as an exercise for improving the quality, performance, and experiences, they often fall short of envisioning the broader context of systems in which these products and services operate.
- *Leveraging Multi-System Integration:* Highlighting features as strategic assets for innovation that operate at the nexus of multi-system integration, accommodating diverse and conflicting

stakeholder needs on multiple levels and the multiple intelligences embedded, leverage its ability to drive growth opportunities.

Viewing features, products, and services at an archetypal level through institutional contexts provides opportunities for growth by shaping economic and cultural value creation (Dilnot, 2011).

The evolving landscape of design practice, influenced by the transformative forces shaping complex infrastructures—platform mashups, calls for a new, comprehensive model—one that not only guides designers through the intricacies of the 21st century but also empowers them as effective change agents. By building on and expanding the contributions of (Owen, 2005), who has explored how design can structure choices for decision-making, and (Heskett, 2001), whose work underscores the value of design in institutional contexts, this model introduces new dimensions that integrate resource flows with strategic value creation. In doing so, it offers a framework that bridges established theories with emerging practices, ensuring that design interventions are both visionary and grounded in real-world constraints.

6.2 The Space of Choices

The SCMM (See Figure 6.1) signifies a departure from linear, compartmentalized thinking to a more dynamic, systems-oriented approach, an essential evolution highlighted in Chapters 1 and 5, demonstrating the arguments about reframing design from an object-centric (S. Chan & Klareld, 2022; Di Salvo, 2009; Lemke, 2021; Morton, 2013) to flows- and systems-centric practice. Satisfying the knowledge gap discussed in Chapter Two. The model integrates choice-making and decision-making within the

institutional context, making it an essential tool for strategic decision-making. Echoing the notion of balancing exploration and refinement, ensuring that strategic decisions are both adaptive and aligned with the institutional context (Dilnot, 2011; Owen, 2005; Poggenpohl, 2017; Whitney, 2015).

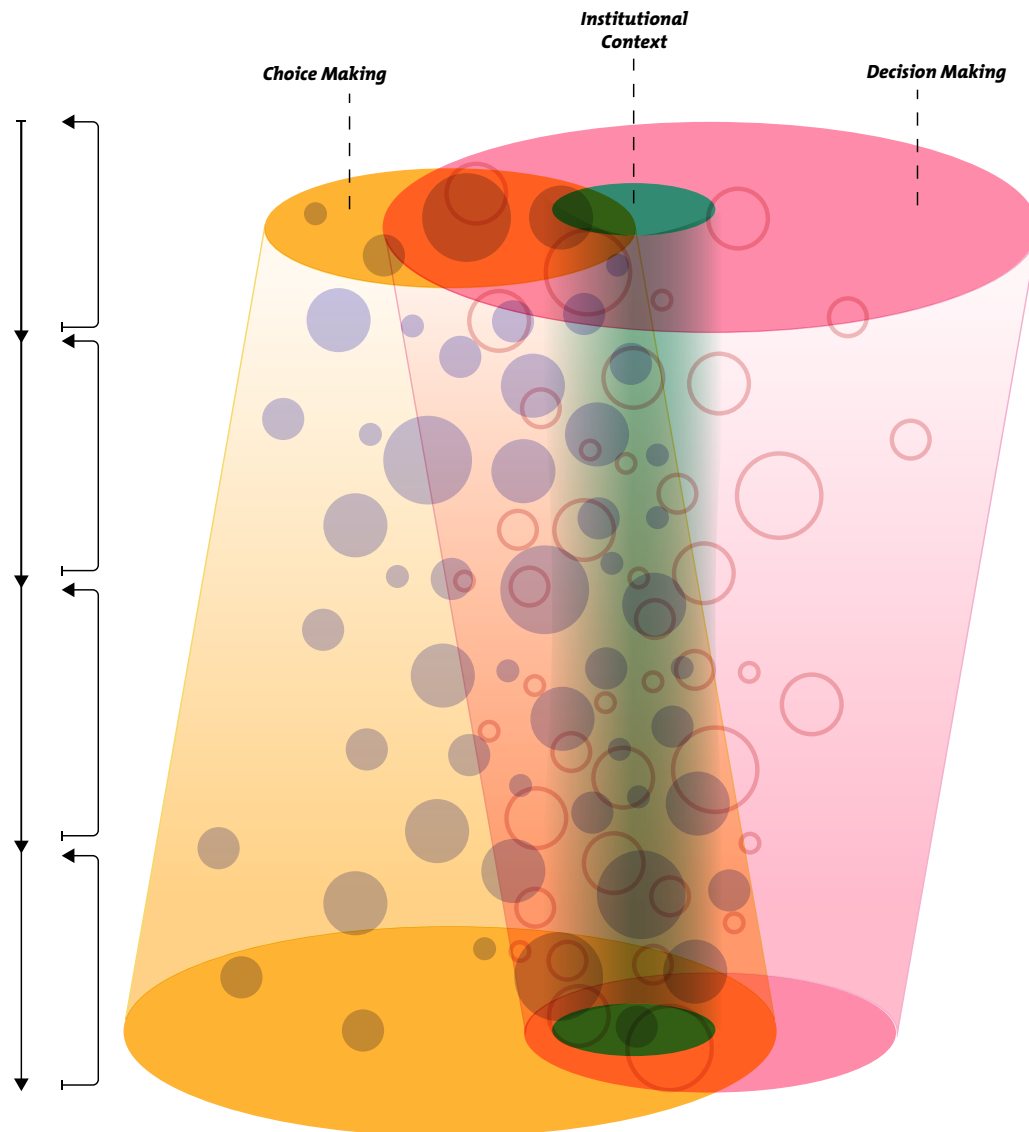


Fig 6.1. The Strategic Choice Making Model Visualized

Visually, the SCMM is represented by two funnels:

- *Choice-Making Funnel*: Expansive, welcoming a breadth of choices, fostering divergent thinking that encourages the exploration of a wide array of possibilities.
- *Decision-Making Funnel*: Narrows more pointedly, signifying convergent thinking where choices are critically evaluated and refined.

These funnels, while intersected, also intersect within the *institutional context*, where institutional intent filters and nurtures emerging choices, which in turn influence institutional strategic direction.

As Owen (Owen, 2005) highlights, the integration of divergent and convergent thinking processes is critical in managing complex choices effectively, ensuring iterative exploration and adaptive decision-making. This model ensures that strategic decisions are not only aligned with the institution's intent but are also adaptive to the dynamic interplay of choices, thereby enhancing overall efficiency and effectiveness.

6.3 The Choices Within the Space

The top-down gradient descent of the SCMM's funnels illustrates how choices evolve from abstract, broad, low-fidelity beginnings to concrete, high-fidelity choices. This iterative refinement process mirrors Whitney's (Whitney, 2015) discussion of choice-making capabilities, such as abstraction, reframing, visualization, and probabilistic thinking, which allow for informed yet flexible strategic directions. The dynamic interplay between design intent and institutional context ensures that emerging choices resonate with systemic goals (Dilnot, 2011; Poggenpohl, 2017).

The varied circles within the funnels, representing individual choices such as features, products, and/or services encapsulated into archetypes, depict strategic choice-making's possibility-rich nature. This visual metaphor conveys the process of developing and refining choices from their inception to their final implementation through the institutional context. Aligning it with Heskett's emphasis on the designer's need to engage with institutional stakeholders to align design choices with organizational priorities (Dilnot, 2011; Poggenpohl, 2017). In line with Chapter 2's emphasis on a broader context, these circles capture the rich tapestry of influences—policy, economic shifts, environmental constraints, and cultural factors—that shape the evolution of ideas (Escobar, 2018; Heskett, 2001; Lemke, 2021; Nogueira et al., 2019).

6.4 Divergent and Convergent Processes

The SCMM's structure emphasizes divergent and convergent thinking in a simultaneous yet iterative manner—echoing insights from the workshops and controlled experiments (Chapters Four and Five)—fostering innovation and strategic alignment within institutions. Unlike traditional models, where these phases occur sequentially (Buchanan, 1992; Churchman, 1967; Dorst, 2011), the SCMM integrates them simultaneously, ensuring a continuous interplay between exploration and decision-making. Further echoing Owen's notes on balancing both the exploitation (divergent) and refinement (convergent), ensuring a wide range of possibilities before committing to a focused decision (Owen, 2005).

- *Divergent Thinking (Choice-Making Funnel)*: Encourages wide-ranging possibilities and potential reconfigurations of resource flows. Chapters 2

and 5 demonstrate how early, diverse explorations can reveal overlooked opportunities.

- *Convergent Thinking (Decision-Making Funnel)*: Critically refining, narrowing, evaluating, and synthesizing these possibilities to align with institutional imperatives and feasibility constraints. Reinforcing Chapter Two's discussion on the exploration of possibilities for strategic alignment and coherence.

This continuous interplay between expansion and focus transforms broad-spectrum, low-fidelity concepts into concrete, high-fidelity solutions that respond both to immediate needs and macro-level institutional agendas.

6.5 Structure and Key Properties of the Model

The Strategic Choice-making Model (SCMM) is structured around four iterative stages—Prompting, Prototyping, Archetyping, and Regenerative Ecosystem (Infrastructuring/Platforming)—each reflecting an iterative, layered approach uncovered in Chapters 3, 4, and 5. Encapsulating the generation, evaluation, and implementation of choices (See Fig 6.2.).

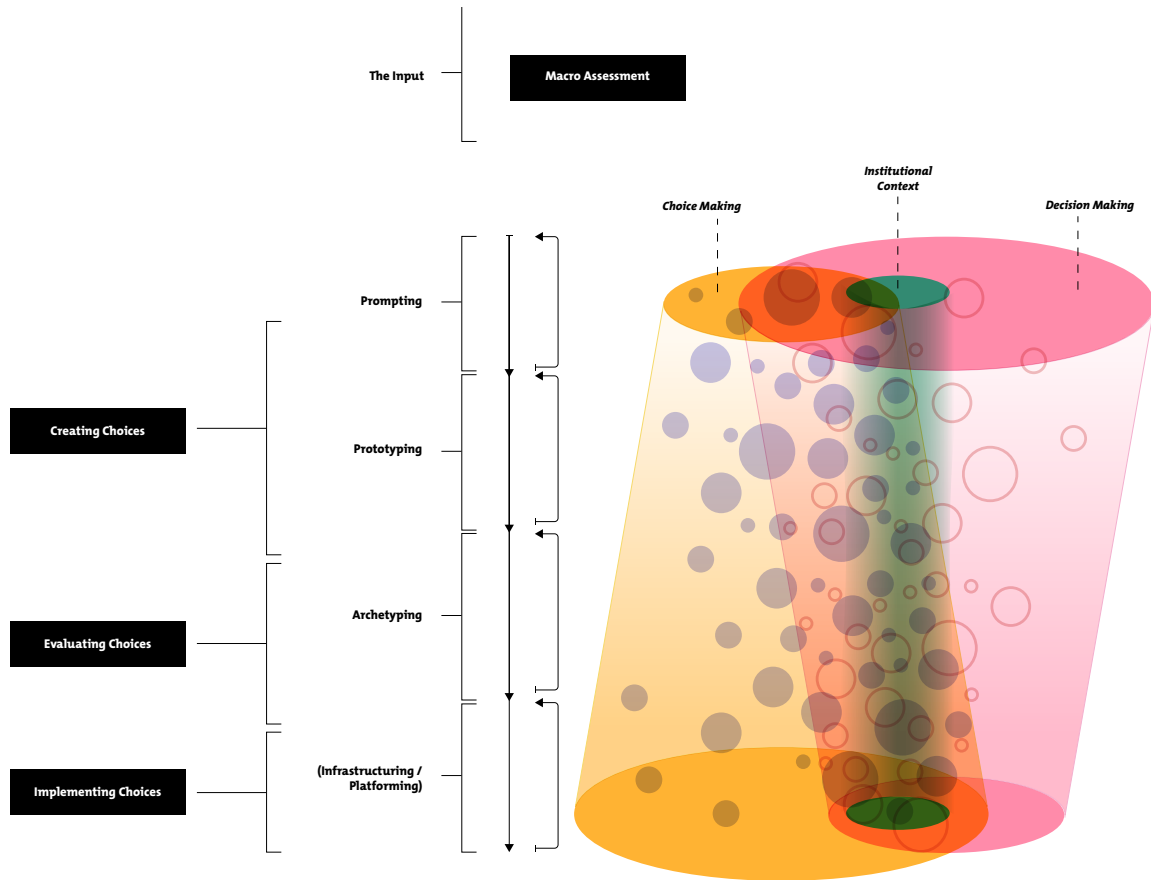


Fig 6.2. Strategic Choice-Making Stages

The input for the SCMM is a macro assessment of the current situation of the archetype, deeply analyzing both the problem space and the opportunity space. Leveraging a systems approach to understand and incorporate the multi-system flows and bridging choices with the macro forces and the institutional purpose, setting the stage for the new agendas at hand as well as the context for subsequent stages of the model.

1. *The Prompting Stage*: This stage initiates the design process by setting the strategic intent and aligning with broader institutional and societal goals. It involves defining foundational questions and understanding the broader

context in which the design activity will take place. Key properties of this stage include:

- *Agenda Setting*: Establishing the strategic intent and goals for the design process (Owen, 2005), linking to macro forces highlighted in Chapter One and the knowledge gap in Chapter Two.
- *Strategic Alignment*: Ensuring alignment with broader institutional and societal objectives (Dilnot, 2011; Lafley & Martin, 2013; Poggenpohl, 2017).
- *Contextual Understanding*: Integrating a deeper understanding of the broader context, including—societal, technological, economic, etc.—factors (Dilnot, 2011; Nogueira et al., 2019; Poggenpohl, 2017).
mirroring Chapter Two's emphasis on multi-resource flows.

2. *The Prototyping Stage*: In this stage, strategic intentions are translated into tangible prototypes. This iterative process encourages exploration and refinement, engaging multiple stakeholders to ensure alignment with the strategic goals. Key properties include:

- *Iterative Development*: Encouraging iterative cycles of prototyping and refinement (Galey & Ruecker, 2010; Owen, 2005; P. J. Stappers et al., 2014; Pieter Jan Stappers, 2013).
- *Stakeholder Engagement*: Incorporating various stakeholders' perspectives and insights into feature prototypes (Dilnot, 2011; Heskett, 2001; Owen, 2005; Poggenpohl, 2017; P. J. Stappers et al., 2014).

- *Practical Application:* Translating strategic ideas into tangible prototypes that can be tested and evaluated (Turner, 2014).
3. *The Archetyping Stage:* This stage involves clustering and refining prototypes into strategic archetypes. These archetypes represent patterns and themes that align with the strategic intent and emphasize new combinations that can drive economic value and growth. Key properties include:
- *Pattern Recognition:* Identifying recurring patterns and themes or systemic opportunities (Owen, 2005; Whitney, 2015), addressing the gap Chapter Two noted in focusing primarily on local form-function coherence.
 - *Strategic Synthesis:* Bundling various features into cohesive, strategic archetypes that would potentially drive large-scale transformation (Dilnot, 2011; Poggenpohl, 2017; Romer, 2021).
 - *Formation of New Combinations:* Creating innovative combinations of—features, products, or services—enables novel approaches to growth with intentionality (Owen, 2005; Romer, 2021), echoing Chapter Two’s insight that design must confront multi-level resource flows (Nogueira et al., 2019).
4. *The Regenerative Ecosystem Stage:* The final stage consolidates and integrates the developed archetypes into the institutional contexts, constructing supportive systems and platforms to ensure their robustness and sustainability. Key properties include:

- *System Integration*: Embeds archetypes into the institutional context, linking infrastructures and platforms and ensuring their coherence with existing systems (Dilnot, 2011; Owen, 2005; Poggenpohl, 2017). This is consistent with Chapter Two’s argument for bridging micro- and macro-level design.
- *Institutional Alignment*: Ensuring these systems fit institutional strategy and governance structure (Dilnot, 2011; Owen, 2005; Poggenpohl, 2017).
- *Creation of Regenerative Ecosystems*: Establishing sustainable, adaptive ecosystems capable of evolving amid changing macro forces (Dilnot, 2011; Owen, 2005; Poggenpohl, 2017)—aligning with Chapter Two’s call for large-scale, purpose-driven design transformations.

6.6 How the SCMM Addresses Growth through Featuring

The following sub-questions—responses to the knowledge gap introduced in Chapter Two highlighting design’s untapped potential for large-scale systemic innovation, and further examined in Chapters 4 and 5—clarify how strategic choice-making at the feature level can generate growth opportunities within infrastructure-platform mashups.

6.6.1 How can strategic choice-making facilitate the design of features at complex intersections?

- *Prompting*: By setting a clear strategic intent and aligning it with transformative forces—such as sustainability, equity, financialization,

deglobalization, and intelligence—the Prompting stage ensures that design activities are grounded in broader institutional and societal goals. This early agenda-setting helps designers understand the complex intersections and prepare for informed decision-making.

- *Prototyping*: Through iterative development, features become tangible objects of inquiry. This iterative process validates feasibility and stakeholder alignment, refining how features function at the nexus of multiple systems.
- *Archotyping*: Through clustering and refining of prototypes into strategic archetypes to handle diverse systems demands without losing coherence. By merging multiple features into coherent archetypes, designers can expand or redirect resource flows for systemic impact.
- *Regenerative Ecosystem*: Integrating newly formed archetypes into an institutional context cements their viability at scale. This ensures the designed features persist and adapt, meeting both micro- and macro-level demands.

6.6.2 How does the SCMM enhance understanding of the systemic implications of features?

- *Prompting*: Emphasizes contextual understanding of macro forces and resource flows, considering implications of features across systems. This echoes Chapter Two emphasis on seeing artifacts as nodes in vast networks (Latour, 1992; Lemke, 2021; Morton, 2013)
- *Prototyping*: Provides iterative feedback loops and refinement, bringing concerns of agency, institutional context, and ecosystem concerns into

clearer focus. This reveals real-world constraints and synergies (bringing various stakeholders' perspectives), deepening awareness of how features shape and is shaped by broader systems—an essential approach for addressing hyperobjects or large-scale phenomena (Morton, 2013).

- *Archotyping*: Reveals how framing new combinations helps interlocking individual features to produce holistic transformations. This archetype-level view illuminates where resource flows might be reoriented for intentional outcomes, aligning with Chapter Two perspective on forging systemic shifts through design-level interventions.
- *Regenerative Ecosystem*: Ensures that formed archetypes are not only technically integrated but aligned with institutional context and systemic integration, forming robust structures that endure and evolve alongside shifting macro forces—responding to Chapter Two critique that design often lacks a path to sustainable, large-scale impact.

6.6.3 How does the SCMM create growth opportunities within and beyond institutional contexts?

- *Prompting*: By setting strategic intent that aligns with transformative forces, institutions can identify new arenas of potential growth.
- *Prototyping*: Encourages exploration and refinements, discovering new features-product-service mixes, or cross-sector collaborations that can unlock novel revenue streams and value creations—echoing Chapter One on the intersection of macro forces.
- *Archotyping*: Groups potential choices into coherent visions that serve multiple stakeholders, generating added value at scale. These archetypes

can define new business models, cross-sector partnerships, or community-based ecosystems.

- *Regenerative Ecosystem*: Once institutionalized, these archetypes sustain long-term growth by evolving with external changes—macroeconomic, regulatory, or technological—solidifying their viability and extending their impact beyond the immediate institutional context.

6.7 The Strategic Choice-Making Contribution to the Field of Design

By integrating choice-making (design’s generative exploration) with decision-making (institutionally driven convergence), the SCMM offers a multi-layered model that closes a key knowledge gap (Chapter Two) and extends design’s influence across complex systems’ resource flows. Empirical work in Chapters Four and Five validated the model’s potential to align micro-level feature generation with macro-level strategic imperatives across institutional contexts, directly addressing the dissertation’s central thesis.

- *Shifting Toward Systems Orientation*: This model shifts the design focus from the legacy paradigm of form-function-context to a systems-oriented one of form-function-flow, enabling designers to consider multi-system dynamics to enable large-scale transformations.
- *Collaboration and Stakeholder Integration*: Foregrounding the mediation capability of design (Bertola & Teixeira, 2003), facilitating large

transformations among multi-stakeholders—public institutions, private enterprises, or civic groups—who jointly shape large-scale systems.

- *Value Creation and Resilience:* By embedding institutional alignment early and focusing on intentional and value-driven outcomes, the SCMM underscores the design’s potential to catalyze sustainable growth and equitable transformations.

Thus, the *Strategic Choice-Making Model* stands as a significant advancement in design practice, addressing the intersection of design, strategy, and systems in complex spaces of innovation, namely the infrastructure-platform mashups, and reinforcing design’s critical role in shaping resilient, agenda-oriented outcomes.

CHAPTER 7

TOWARD SYSTEMIC IMPLICATIONS: DESIGN, INSTITUTIONS, AND BEYOND

This chapter shifts focus from describing what the Strategic Choice-Making Model (SCMM) *is* to examining *why it matters*. We have argued that legacy design practice—centered on form, function, and context—often fails to grapple with the large-scale complexities of the 21st century. In response, a system-oriented practice comprising interdependent *form–function–flow* elements was proposed, highlighting the importance of multi-resource flows in design. Contemporary macro forces are blurring traditional boundaries: phenomena of infrastructuralization and platformization (as introduced in Chapter One) are accelerated by technological advancement embedded in everyday offerings, creating hybrid spaces that blur public vs. private services, user-centric vs. system-centric mindsets, what is essential vs. what is complementary, and the algorithmic vs. the material. These new *infrastructure–platform mashups* can yield both intended and unintended consequences in how resources are produced, distributed, and exploited. At the same time, they present immense opportunities for design to embed principles of sustainability and equity at a systemic level rather than as afterthoughts.

The SCMM views design as a *choice-making practice*—a shift needed to better position design capabilities in contexts of resource allocation, institutional purpose, and strategic decision-making. It was developed in answer to a key institutional challenge: organizations aspire to respond to complex systemic issues through their offerings but often lack a mechanism to make purposeful choices in complex spaces of innovation, leading to a paradox of decision and choices. SCMM fills this gap by providing a

structured approach for generating intentional offerings amid uncertainty. The model is broadly applicable across sectors: it can guide governments, NGOs, international development agencies, and private firms alike in making their voices heard amid transformative macro forces. In the following sections, we consider the systemic implications of SCMM—how this *flows-oriented* design logic can transform the evolution of design practice, design education, and institutions, and how design itself emerges as a mediator of multi-stakeholder coalitions.

7.1 The Evolution of The Design Practice

One immediate implication of the SCMM is the expanded role of designers as *systemic change agents*. The model stresses that designers are not simply tasked with crafting visually appealing or functionally efficient artifacts; rather, they become stewards of complex resource flows that span social, economic, technical, and ecological domains. The research in Chapters 4 and 5 showcased how, in multi-stakeholder workshop settings, designers armed with strategic choice-making prompts could navigate broader institutional and societal parameters than conventional design allows. By integrating equitable principles, environmental stewardship, and community considerations into micro-, feature-level choices, design shifts from a reactive problem-solving discipline to a proactive practice of *orchestrating large-scale resource allocations*. This expanded mandate means each design intervention is conceived not just in terms of user experience or form, but also in terms of how it redirects flows of resources within a system. Designers, in effect, serve as *system orchestrators*, mediating between institutional intentions and macro-level forces through the features they shape,

elevating designers from a *reactive discipline* to a *proactive mediator* of large-scale resource allocations.

To clarify the shift, we can compare key aspects of legacy and emerging design practice:

- *Focus Shift – Static Context to Dynamic Flows:* Earlier practice framed context as a set of given conditions and constraints of usage and production. Now, emphasis is on *flows*—continuous exchanges of resources—acknowledging that contexts are ever-changing rather than fixed points.
- *Role Shift – Reactive Problem-Solving to Proactive Mediation:* Traditionally, designers addressed problems within narrow scopes (often after issues arose). Under a flows-oriented approach, designers act as mediators of multi-capital flows, deliberately orchestrating how different forms of capital (financial, social, environmental, etc.) flow and interact.
- *Scope Shift – Isolated Artifacts to Interconnected Systems:* Legacy design often optimized a single product or interface for user satisfaction. The evolving practice broadens the scope to consider impacts across infrastructure–platform systems, meaning each feature or artifact is designed with its system-wide positionality in mind.

These shifts demand a reconceptualization of the designer’s role. Chapters 4 and 5 illustrated this expanded role: multi-stakeholder workshops, guided by strategic choice-making prompts, empowered designers to navigate broader institutional, ecological, and social parameters. Decisions made from a design perspective are now recognized as decisions about distributing resources and opportunities at scale. A seemingly small

design feature might determine how energy is allocated in a neighborhood, how users share data on a platform, or how benefits and burdens are divided among communities. Embracing this mediator role enables designers to intentionally steer outcomes toward sustainability and equity, rather than treating such outcomes as externalities beyond the purview of design.

Equally important is the methodical process that SCMM embeds in the design practice, from the earliest *Prompting* stage to the *Platforming/Infrastructuring* stage, SCMM-based design processes collect data, engage various stakeholders, and conduct repeated evaluations, ensuring each design decision resonates with strategic intentions. *This systematic method fosters a broader mindset, making every design step an opportunity to shape or redirect resource flows.*

This evolution highlights today's infrastructure–platform mashups as contested territories where changes can propagate rapidly and unpredictably. Moving from a legacy setting, relying on the context of production and usage solely to an active web of platforms and infrastructures – a transit app tied to a city's transit infrastructure, or an e-commerce platform entwined with global supply chains. In such scenarios, a design intervention is not a lone insertion into a static backdrop; it is a disturbance or enhancement to a living system of complex flows. SCMM's flows-oriented logic explicitly recognizes that strategic design choices both shape and are shaped by these complex mashups. Each design choice must account for how an underlying infrastructure might constrain what is possible (for instance, an electric grid's capacity will shape what an EV charging service can do), even as that choice aims to influence how the infrastructure is used (e.g. shifting when and how vehicles draw power from the grid).

Designers thus operate in a continuous dialogue with the systems around them: design decisions influence system behavior, and system dynamics inform design decisions. This two-way interplay is a hallmark of form–function–flow thinking, positioning design as an active agent within systems rather than an external solver of discrete problems.

7.2 The Evolution of Design Education

If design practice is to evolve in this systemic direction, design education must also adapt. Traditional design training has, as noted in Chapters 2 and 5, rarely prepared students to handle the political, economic, and infrastructural dimensions of contemporary design problems. Studio curricula have historically emphasized form-giving and user-centered innovation within relatively narrow contexts, leaving a gap in the ability of new designers to contend with multi-layered systems or to anticipate the downstream impacts of their work. ***Conventional programs rarely equip students to grapple with policy constraints, multi-stakeholder environments, or the ethics of large-scale interventions.*** This gap means that as design’s role broadens, many practitioners remain ill-equipped to exercise the kind of strategic choice-making that SCMM envisions.

To bridge this gap in design education, a broader suite of competencies that equip future designers to better initiate and operationalize a *flow-oriented practice* of strategic choice-making to become *systemic thinkers and strategic facilitators* is needed. Key skill areas include:

- *System’s Centric Futuring:* The ability to anticipate long-term systemic changes and unintended consequences. Designers should be comfortable using strategic

futuring techniques and embedding them into their prototyping process, equipping future practitioners to embed future-oriented thinking into prototypes and assess how today's choices shape tomorrow's systems.

- *Multi-Capital Fluency*: Proficiency in recognizing and working with multiple forms of capital (e.g., financial, social, environmental, human capital). A flows-oriented curriculum prepares designers to navigate and influence diverse forms of capital and to design interventions for greater sustainability and equity. This means understanding how a single design feature can impact not just one metric of success, but an array of outcomes – from carbon footprints and community well-being to profit margins – and learning to balance these in the design process.
- *Coalition Building*: The capacity to communicate, work, and collaborate interdisciplinarily. SCMM projects demand that students practice negotiation, co-creation, and leadership. This entails the role of design as a knowledge broker to align different expertise around a shared vision.
- *Prototyping with Flows in Mind*: Beyond products and interfaces, designers must prototype systems' interactions and resource flows. The SCMM calls to treat each prototype as a node in the wider system, revealing how changing one feature might redirect financial, material, or information flows.

Integrating these competencies into design education can *accelerate design's shift from object-centric problem-solving and human-centered solutions to systemic stewardship, evolving the design curricula.*

Studio practice—the cornerstone of design education—also needs to be reimaged to reflect the open-ended, multi-scalar nature of contemporary challenges. Rather than

working on isolated, user-centered briefs, students should engage with complex spaces of innovation and prototype at the intersection of overlapping flows—financial, natural, political, or social. Cross-department collaborations allow students to co-create with peers from other fields while integrating SCMM’s frameworks for systemic resource flow and strategic intent. Instructors can deepen these experiences by embedding live stakeholder input, applying system-centric frameworks and tools, and facilitating mentorship from multidisciplinary faculty teams. Together, these shifts make the studio a testing ground for the flows-based mindset, equipping future practitioners not only to design artifacts but to navigate and reshape complex systems through strategic, collaborative intervention. Ultimately, evolving the studio practice in this way helps students learn by doing in complex environments, making the academic studio a safe proving ground for the flows-based design mindset before they enter professional practice.

A parallel shift in how student work is assessed—moving beyond polished artifacts and user feedback to emphasize systems thinking, long-term foresight, and collaborative process. Evaluations should account for how well students map and address resource flows, articulate multi-capital impacts, and align their design work with institutional and societal goals. Assessment becomes more formative and reflective, with checkpoints featuring the application of a system-centered framework, and in-progress prototypes that reveal strategic development over time. Incorporating peer and stakeholder input allows educators to gauge coalition-building and communication effectiveness. This reframes design excellence to include systemic stewardship and strategic intent as central criteria alongside creativity and usability.

A flows-based design education inherently breaks out of the silos that have traditionally separated designers from other experts in fields like engineering, public policy, and data science. Through joint courses, mixed-program studios, and exposure to diverse perspectives, future practitioners learn to translate knowledge across domains and lead multi-stakeholder teams—mirroring SCMM’s coalition-building ethos. This commitment to cross-disciplinary learning must also extend beyond academia into professional settings, where executive programs, workshops, and online modules help mid-career practitioners adopt flow-based thinking. These platforms not only upskill designers with tools like SCMM but also create vital feedback loops between practice and pedagogy. Ultimately, evolving design education is about building a talent pipeline of system integrators and strategic choice-makers who bring systemic foresight, multi-capital fluency, and coalition skills into institutions, shaping a new culture of design that favors long-term, holistic transformation over reactive, siloed solutions.

7.3 The Evolution of Institutions

Strategic Choice-Making Model (SCMM) enables institutional transformation that aligns an organization’s purpose with larger macro forces beyond short-term financial goals and operational goals. In essence, the model provides a strategic framework for institutions to continually realign *what they do* (their offerings and internal processes) with *why they do it* (their fundamental purpose in a changing context). In other words, ensuring an alignment between the institutional context and what is possible with the institutional intent and the institutional priorities. This elevates institutional strategy

from reacting to day-to-day pressures toward proactively shaping long-term, systemic outcomes.

One of the fundamental shifts SCMM encourages is moving from a reactive linear mentality to a proactive, systemic “flows” strategy. Legacy institutions often operate in a linear fashion, tackling challenges in isolated, sequential steps and focusing on single metrics (often financial). In contrast, SCMM embeds flows-thinking – an approach that views value creation as a continuous flow across interdependent resources. This means decisions are made with awareness of how resources flow through and beyond the immediate institutional context. Such thinking is inherently systemic and integrative. It echoes shifts in modern strategic value creation towards reconfiguring roles and relationships among an assemblage of actors and agents to co-create value in new combinations. In other words, strategy expands from optimizing internal units to orchestrating a broader ecosystem of agents. As complexity grows, a continuous realignment is required that SCMM operationalizes through its iterative phases. By embedding this agile, ecosystem-oriented approach, institutions can anticipate and shape changes rather than merely react, aligning their evolution with emergent trends (e.g., digital transformation, climate resilience, inclusive growth) in a deliberate way.

The macro assessment case studies (Chapters 4 and 5) demonstrated how organizations applying SCMM uncover opportunities beyond immediate capital gains—such as forging community relationships or adopting more sustainable supply chains. By recognizing social, cultural, and ecological capital alongside financial capital, businesses reduce external risks and nurture ecosystems around their offerings. *This multi-capital*

perspective goes beyond standard corporate social responsibility, anchoring strategic design decisions in a broader sense of value creation.

Institutional Context Is Reciprocal and Fuels Growth. A key insight here is that institutional intent does not merely shape features; it is, in turn, reshaped by the novel archetypes that arise during the SCMM's choice-making process. An example comes from the macro assessment case study on embedded finance: initially, an institution behind a digital wallet feature was driven by user adoption metrics, yet when SCMM prompted them to integrate local credit unions and address data privacy concerns, they revisited their broader strategic role. Over time, this spurred more inclusive lending practices and a deeper commitment to community ecosystems. The archetype development effectively crystallized a new strategic intent ("banking the unbanked" in this case), which then informed subsequent prototyping and infrastructuring efforts. In summary, the embedded finance example demonstrates how SCMM enables institutions to pivot their strategy: a once purely operational project becomes a catalyst for larger institutional mission change. By anchoring design features in macro-level purpose, SCMM helped the institutions shift from incremental service improvement to a transformative business model – one that is purpose-driven and system-aware.

In this sense, once an institution invests in a new archetype, that archetype redefines how the organization interprets its mission, allocates resources, and measures success. This reciprocal loop triggers growth opportunities that extend far beyond near-term profit or operational efficiency—often revealing unexpected collaborations, alternative revenue models, business models, or new stakeholder alignments. A question for future inquiry follows naturally: *How might institutions leverage newly emergent*

archetypes to systematically reshape their strategic goals and governance models, creating multi-capital benefits for both the organization and its broader ecosystem? By exploring this line of research, scholars and practitioners can deepen our understanding of how reciprocal interactions between institutional intent and the SCMM unlock novel pathways for sustainable, large-scale transformation.

Adoption trajectories, however, diverge sharply between legacy and greenfield contexts. Mature agencies in the United States exemplify the legacy case: entrenched silos, organizational inertia, and ageing technological infrastructures inhibit systemic experimentation. Here, SCMM functions as an incremental antidote. Early-stage cross-silo prototypes expose shared value, archetypes legitimize new narratives that cut across departments, and platform thinking builds the business case for modernizing infrastructure. In contrast, newly developed bodies, such as smart-city authorities in GCC states or emergent Latin-American conglomerates, are unburdened by history, long policymaking cycles, and bureaucratic burdens. They can embed flow-thinking from inception, structuring themselves as platform orchestrators with offerings in the mashup space and adopting multi-capital metrics without a costly unlearning phase. The contrast is telling: where legacy institutions must unlearn and retrofit to evolve, newly developed institutions can instantiate SCMM-aligned structures outright. Nonetheless, both stand to gain from SCMM's strategic blueprint. In legacy cases, SCMM provides a gradual road map to transform linear thinking into platform thinking, helping break down silos and justify modernization by highlighting long-term value. In newly developed cases, SCMM offers a ready-made playbook to institutionalize innovation and adaptability as core tenets, ensuring the institutions remain flexible as they grow.

Flow-oriented, purpose-driven institutions also become stronger partners: the same capabilities that align internal strategy with macro forces prepare them to join cross-sector coalitions, where multi-capital value is co-created and amplified. In sum, SCMM cultivates organizations that are internally coherent and externally synchronized, moving beyond legacy constraints to help tackle 21st-century challenges at system scale.

7.4 Design as a Mediator of Coalitions

Large-scale systemic transformations rarely hinge on solitary organizations; they emerge from coalitions of diverse actors. Crucially, these coalitions are not just ad-hoc collaborations but institutional contexts in their own right, each defined by a dominant structuring logic that shapes how change can unfold. This section reframes design's mediating role through the Strategic Choice-Making Model (SCMM) in terms of three distinct coalition-based institutional contexts: (1) alignment and vision, (2) trade-offs and negotiation, and (3) capital and investment. Each type provides a structural lens for how institutions coordinate action, negotiate compromise, or mobilize resources to drive large-scale transformations. In other words, each coalition form frames specific possibilities (and limits) for institutional transformation, and design interventions via SCMM engage with each context accordingly to steward change.

7.4.1 Coalitions for Alignment and Vision

Coalitions built on alignment and vision are driven by a shared purpose or set of values that unite otherwise disparate stakeholders. In this context, multiple institutions (public agencies, NGOs, private firms, community groups, etc.) rally around a common

vision—for example, a sustainability goal or a city-wide innovation agenda—creating a coordinated front. The dominant logic here is one of *shared values and coordinated action*: stakeholders should set aside minor differences to pursue an overarching goal that inspires collective commitment. This alignment-based institutional context enables strong coordination; it provides a clear direction and mutual trust that can accelerate decision-making and resource allocation. However, its effectiveness also hinges on sustained consensus – the coalition’s strength is maintained only so long as members remain bought into the unifying vision, meaning deep fractures can emerge if that consensus falters.

Within an *alignment-and-vision coalition*, design (*through SCMM*) acts as a catalyst for *articulating and reinforcing the shared agenda*. SCMM-driven design processes help stakeholders *envision and prototype the future they collectively seek*, making the abstract vision tangible. For instance, early design outputs serve as strategic “touchpoints” that allow diverse parties to align their resource contributions—financial, natural, technical, digital, or more— toward the agreed goals. By *giving form to the shared vision*, design provides a common reference point around which the coalition can coordinate. Moreover, the design’s role as a *knowledge broker* (Bertola & Teixeira, 2003) becomes evident in this context: by translating between different perspectives, it helps maintain coherence across the coalition. In essence, SCMM engages the alignment coalition by creating artifacts and frameworks that keep everyone focused on the same big-picture objectives, thereby mediating the coalition’s efforts and sustaining its unified momentum toward transformation.

7.4.2 Coalitions for Trade-offs and Negotiation

Not all coalitions start with full alignment—many form out of necessity among actors with divergent interests, requiring negotiation and compromise. Trade-offs and negotiation coalitions are institutional contexts defined by pluralistic stakeholders and the need to balance competing priorities. Here, the dominant structuring logic is one of negotiated compromise: progress is achieved by bargaining, deliberation, and incremental consensus-building among parties (for example, multiple municipalities and corporations agreeing on regional climate action plans, each conceding something). This coalition context frames transformation as a joint problem-solving exercise where no single actor's vision prevails outright; instead, institutions coordinate through formal agreements, policy compacts, or mediated discussions that surface and address conflicts. The possibilities in this context lie in its inclusivity and resilience – solutions tend to have broad buy-in because they have been co-authored through compromise. On the other hand, the need to accommodate different interests can limit the scope of change to what is mutually acceptable, potentially yielding more incremental or “lowest common denominator” outcomes than a singular bold vision might have produced.

Design, via SCMM, engages these negotiation-centric coalitions by providing impartial processes and tools to navigate complex trade-offs. In practice, SCMM convenes relevant stakeholders in a productive negotiation setting around the choices and decisions to be made. Through facilitated sessions, design helps surface each party's values and constraints and makes the necessary trade-offs. System-centric frameworks become mediums for discussion – stakeholders can literally see the implications of different choices. By iteratively prototyping potential solutions, the design process allows

the coalition to converge on options that everyone can support. The outcomes of these negotiations are often functional prototypes or pilot projects that embody mutual commitments reached by the group. In this way, design mediates among differing institutional logics, ensuring that stakeholders' voices are incorporated while guiding the group toward a coherent path forward. The SCMM's mediating role is to keep the dialogue constructive and goal-oriented, so that even as compromises are made, the resulting designs still push the system forward. Ultimately, in a trade-offs coalition, design's intervention helps transform adversarial bargaining into creative co-creation, yielding solutions that, while moderated by compromise, are collectively owned and thus more likely to be implemented and sustained by all parties.

7.4.3 Coalitions for Capital and Investment

A third form of coalition is one structured around the orchestration of finance and resources: the capital and investment coalition. In this institutional context, stakeholders (e.g. government agencies, investors, banks, development funds, philanthropies, large corporations) come together primarily to mobilize and direct financial capital in service of a transformative initiative. The dominant logic of this coalition type is capital orchestration – aligning financial flows, investments, and economic incentives across institutional boundaries to enable large-scale change. Such coalitions often materialize in scenarios like public–private partnerships for infrastructure, multi-agency investment compacts for innovation, or global funding alliances for sustainable development. The coalition's structure is essentially a financial architecture: agreements on co-investment, risk-sharing, and returns define the collaboration. This context frames transformation in

terms of what gets financed. The possibilities here are significant – with substantial capital commitment, projects that seemed unattainable can be realized, and systemic change can be scaled rapidly. At the same time, the limits of this context stem from financial discipline and interests: initiatives must appeal to funders’ criteria (return on investment, risk mitigation, strategic alignment) to receive backing, which can narrow the focus to solutions that meet those economic or political thresholds.

SCMM extends design’s mediating role into the realm of strategy and finance in capital-driven coalitions. Designers engage with investors and funders by translating visionary concepts into credible, investable proposals. Concretely, the SCMM process helps identify which design features or interventions could catalyze shifts in resource flows, making a compelling case for investment in certain innovations. For example, designers might use SCMM to pinpoint a set of features (a new service model, a technology platform, a policy prototype) that demonstrate how capital could be reallocated to achieve systemic goals (such as carbon reduction or community wealth building). By doing so, design highlights leverage points where an infusion of funds would have outsized transformational impact. In fact, design can reveal opportunities to reconfigure capital flows in alignment with the coalition’s goals – for instance, showing how investing in a network of community microgrids advances a national energy agenda. In the capital and investment context, design also plays a crucial communicative role: visual narratives, models, and scenarios developed through SCMM help bridge the language of design and the language of finance, ensuring that all coalition members understand both the systemic vision and the business case behind it. This mediation increases the likelihood that bold ideas gain the financial backing they need. By engaging

with capital coalitions, SCMM effectively brings design into boardroom decisions, orchestrating not just form and function but funding. The result is an institutional environment where monetary resources are consciously steered by design insights, allowing large-scale transformations to be realized. Still, the designer's task in this context is to balance vision with viability – to push the envelope of innovation while meeting investors' imperatives – thereby negotiating the boundary between what is ideal and what is financially attainable in driving systemic change.

7.5 The Conclusion

Revisiting the dissertation's central thesis, this chapter has demonstrated that design's capacity to orchestrate large-scale transformation truly hinges on strategically shaping features that redirect resource flows across multiple forms of capital. By framing designers as systemic mediators, integrating macro-level considerations early in the innovation process, reforming design education for expanded competencies, and infusing institutional strategy with long-term vision, the SCMM provides a blueprint for purposeful innovation in complex environments. In highlighting these implications, we see that design no longer operates on the periphery of institutional strategy or societal change. Instead, guided by SCMM, design emerges as a pivotal mediator—bridging institutional purpose with external forces to catalyze sustainable growth, equity, and resilience at scale. Through this lens, what might have once been a simple product tweak or service improvement can evolve into a lever for reconfiguring entire ecosystems of value and meaning. Taken together, these implications reinforce the dissertation's overarching argument: design does not merely respond to existing infrastructures and

forces—it can reconfigure entire ecosystems. When empowered by a choice-making framework like SCMM, design aligns the intentions of institutions with the needs of society, yielding transformations capable of addressing 21st-century challenges. The final chapter will now synthesize the findings from across this research, summarize its contributions, and offer concluding reflections on how this work propels the evolution of design practice in infrastructure–platform mashups and beyond.

CHAPTER 8

STRATEGIC VALUE CREATION THROUGH FEATURES

This dissertation set out to address a core hypothesis: *features in complex spaces of innovation (infrastructure-platform mashups) can mediate between institutional purpose and macro forces to reorient resource flows and enable large-scale transformations*. In answering the research question—How can design offer strategic choice-making in these mashups to drive value creation for large-scale transformations?—the study advanced the Strategic Choice-Making Model (SCMM), proposing a design practice that transcends product-centric fixes and systematically reshapes multi-capital resource flows. Through iterative macro assessment cases, workshops, and controlled experiments (Chapters 3–5), it became evident that features, when conceived at the intersection of institutional intent and macro forces, unlock new models of alignment capable of bridging short-term objectives with long-term societal goals. It demonstrated that when design is treated as a choice-making practice, each feature can become a strategic asset that reorients resources.

The findings illustrate that design has a transformative potential when reframed to handle multi-resource flows and placed in conversation with institutional purpose (Chapters 1 & 2). By advancing an iterative funnel of Prompting, Prototyping, Archetyping, and Infrastructuring/Platforming (Chapters 4–6), the SCMM equips designers and decision-makers to navigate overlapping resource dependencies, negotiate ethical considerations, and ultimately foster holistic innovation. In doing so, design shifts

from artifact-centric improvements to strategic levers for orchestrating value in infrastructure-platform mashups.

For example, in Peru's Yape system, and throughout the project we conducted applying the SCMM, the integration of embedded finance has transformed everyday transactions by embedding financial services directly into digital platforms—reshaping economic participation by extending financial inclusion and expanding access for underserved populations. Similarly, in the in-flight meal project, features that optimize supply chains through real-time data have redefined operational efficiency and reduced waste, aligning sustainability goals with operational efficiency. These examples demonstrate how the SCMM provides a structured approach for designers and institutions to navigate systemic constraints while unlocking new value opportunities. The model not only informs design interventions at the feature level but also fosters large-scale transformations, as seen in the ongoing shifts in digital finance, mobility ecosystems, and sustainable aviation. As similar interventions are already emerging globally—in Japan, Saudi Arabia, UAE, and Peru—the SCMM serves as a guiding model and a framework for embedding institutional intent into infrastructure-platform mashups, ultimately shaping more adaptive and resilient systems.

Despite these contributions, the scope of empirical inquiry is inherently finite. Scaling SCMM across sectors—like global humanitarian efforts, policy-driven settings, or design-led analysis of the mashup artifacts—could drive future research agendas for further exploring the model. Moreover, while the research streams substantiate SCMM's viability, more quantitative or longitudinal studies measuring resource-flow impacts would reinforce its efficacy and further refine the model's applicability.

Looking ahead, deeper policy integration, cross-disciplinary educational reforms, and pilot programs that adopt SCMM principles may well expand its transformative reach. Future research can explore archetype configurations in climate-sensitive regions, public health systems, or inter-institutional collaborations, analyzing how these strategic features evolve over time. By embracing design as a knowledge broker and flows mediator, practitioners, educators, and policy-makers alike can co-create solutions that adapt to emerging macro forces—reinforcing design’s capacity to shape resilient infrastructures and inclusive economic models. Moreover, educational curricula could embed SCMM principles in cross-disciplinary coursework, cultivating design leaders adept at synthesizing data, stakeholder interests, and strategic foresight.

In closing, strategic value creation through features underscores the dissertation’s overarching argument: design, if structured by the SCMM, can integrate macro-level transformation forces with institutional purpose, reorienting resource flows for collective benefit. Such a perspective elevates design beyond product optimization, framing it as a catalyst for bridging diverse stakeholders, unlocking novel revenue streams and investments, and fostering resilient, equitable, and intelligent infrastructures. As the world’s challenges intensify, this design-led orientation emerges as a powerful blueprint for reimagining and harmonizing the interfaces between institutions, societies, and the planet itself.

APPENDIX A
MACRO ASSESSMENT CASE STUDIES

This appendix describes the five macro assessment case studies conducted as part of this research—Controlled Environment Agriculture (CEA) (Bataineh & Teixeira, 2024h), EV Charging Stations (Bataineh & Teixeira, 2024j), Embedded Finance (Bataineh & Teixeira, 2024i), 911 Emergency Response System (Bataineh & Teixeira, 2024g), and In-Flight Meal Services (Bataineh & Teixeira, 2024k)—to establish a systems-level perspective on “everyday infrastructures.” These cases provided essential context and foundational inputs to the Strategic Choice-Making Model by clarifying problem spaces, identifying leverage points, and highlighting opportunities for sustainability, equity, and value creation (See Fig A.1.).



Figure A.1. The Macro Assessment Structure.

A.1 Purpose and Structure

A.1.1 Purpose

- Offer *macro-level assessments* of real-world infrastructures that are undergoing rapid transformation. Through studying the offering at the intersection of systems, including their historical evolution, and the organizational coalitions they require.

- Examine *systemic constraints*, resource flows, and stakeholder relationships that influence large-scale transformation. Revealing potential leverage points and “hot spots” for design intervention that integrate sustainability, equity, and financial considerations.
- Surface *opportunity areas* where design interventions can shift resource flows toward more equitable and sustainable outcomes. Framing new agendas and research prompts that inform subsequent experiments and workshops in this dissertation.

A.1.2 Common Structure

Although each case study had domain-specific nuances, the analysis generally covered seven core elements:

1. *Introduction & Historical Context* – Tracing the evolution and current state of the offering.
2. *Offerings & Rituals* – Investigating everyday usage patterns, user expectations, and typical workflows.
3. *Offerings & Coalitions* – Mapping key interdependent stakeholders and alliances (public, private, nonprofit).
4. *Offerings & Organizational Boundaries* – Highlighting cross-institutional dependencies and the need for organizational cooperation (and sometimes blur their boundaries).
5. *Offerings & Bias* – Identifies potential inequities or systemic biases in current workflows (e.g., digital divides, discriminatory practices).

6. Offerings & Change – Suggesting the reorientation of resource flows at key leverage points, sometimes through adopting new technologies or policies to disrupt existing routines.
7. *Offerings & Network Activation* – Shows possibilities for broader ecosystem shifts when redesigning or repurposing existing offerings.

A.2 Overview of Each Case

1. *Controlled Environment Agriculture (CEA)*: The objective of this case was to understand how indoor farming solutions, including vertical farms, manage energy, water, and labor while striving for local, sustainable food production. In one example, a 60-acre Dutch greenhouse produced yields comparable to a 1,500–1,800 acre open-field farm while using about 90% less water, highlighting how *data-driven approaches* can optimize resource use. The implication of this research illustrated the interplay of multiple forms of capital—financial, natural, and digital—and showed how carefully designed feature-level interventions (such as sensor-based lighting or automated distribution apps) can reconfigure resource flows for more efficient energy utilization and resilient local supply chains (see Fig A.2.) (Bataineh & Teixeira, 2024h).

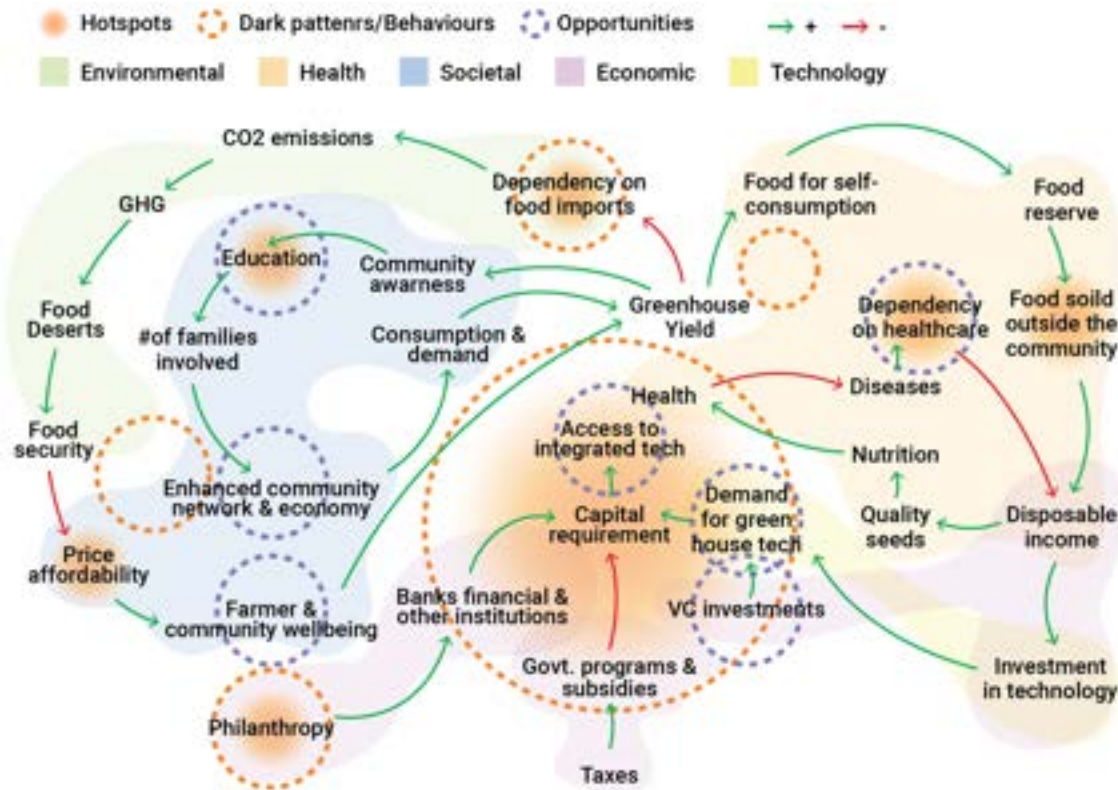


Figure A.2. An examination of the system dynamics of greenhouses highlighting problem areas and opportunity spaces.

2. *EV Charging Stations:* This case was set to examine how electric vehicle infrastructure expands beyond mere charging hardware to incorporate real-time data monitoring, digital payment systems, and strategic site planning. Across the United States, the annual installation rate of charging ports currently hovers at around 18,700, yet it must rise significantly to meet the 500,000-port national target by 2030. This points to an infrastructure in rapid transition. The implication underscores the necessity of bridging public and private coalitions—spanning utilities, local governments, and automotive manufacturers—to guide both investment and user adoption. By focusing on

feature-level interventions such as mobile apps for payment or analytics dashboards that monitor station utilization, design can help orchestrate how environmental and financial resources are deployed for maximum impact (see Fig A.3.) (Bataineh & Teixeira, 2024j).

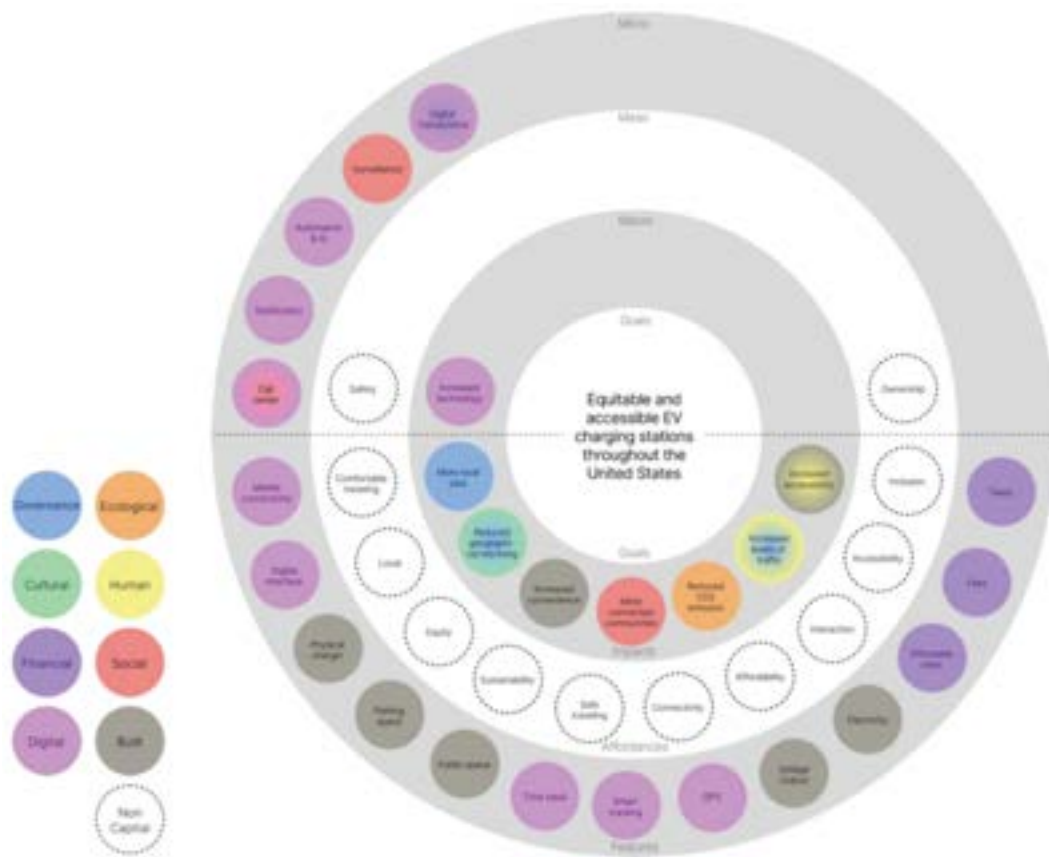


Figure A.3. Anatomy of infrastructure; highlighting the multiple agents and capitals across micro to macro levels.

3. *In-Flight Meal Services*: To investigate the entire service pipeline for on-board meals, from procurement and meal preparation to in-flight logistics and waste management, in order to identify design-driven improvements that

could lessen waste and enhance the passenger experience. Industry reports placed the global in-flight catering market at around \$15 billion in 2017, yet large amounts of this value end up as food waste due to inefficiencies and inaccurate demand forecasting. The key insight is that better integration of data—through real-time inventory tracking, user pre-order systems, and more robust forecasting tools—could significantly cut waste while optimizing costs and raising customer satisfaction. This underscores how strategic design features at the meal-service level can produce systemic benefits across supply chains (See Fig A.4.) (Bataineh & Teixeira, 2024k).

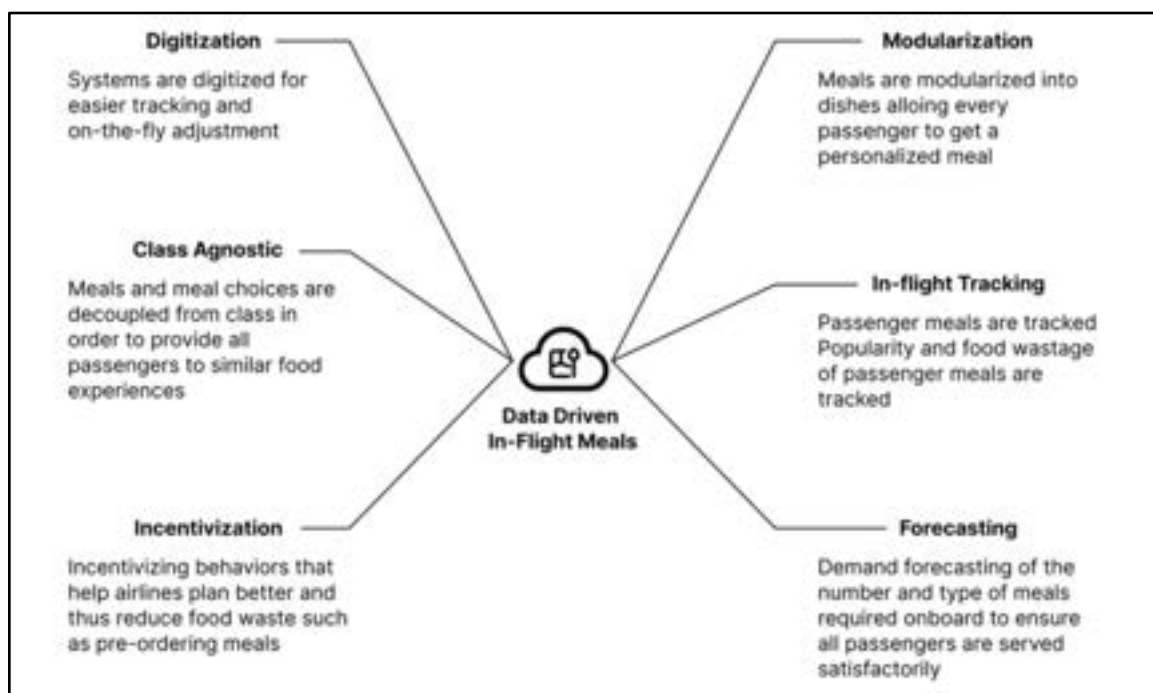


Figure A.4. A prescriptive archetype; highlighting data-driven interventions.

4. *Embedded Finance*: this case was to assess how financial services embedded in non-financial platforms can broaden access yet introduce new equity and

regulatory complexities. In Peru, for example, the mobile app Yape has over 11 million users and has enabled more than two million previously unbanked individuals to open their first bank accounts. This swift adoption shows how everyday transactions—such as peer-to-peer payments, gig-work payouts, or small loans—are being reshaped by app-based ecosystems that promise seamless convenience and lower barriers to credit. The implication is that embedded finance can boost economic inclusion but must address transparency gaps, data security risks, and biases in credit scoring. By designing interventions at the “feature” level—like robust user literacy prompts, simplified risk assessments, or clearer data-ownership controls—products can foster genuine financial empowerment. This ultimately calls for collective oversight by fintech innovators, regulators, and local communities to ensure that these rapidly evolving infrastructures uphold responsible finance and equitable resource flows (See Fig A.5.) (Bataineh & Teixeira, 2024i).



Figure A.5. A prescriptive System Dynamic Map; highlighting key opportunity areas envisioning the impact of equitable finance.

5. *911 Emergency Response System:* The objective was to explore how 911 coordinates multi-agency emergency responses while confronting legacy technology and issues of equity. In one illustration, the United States fields roughly 240 million 911 calls annually, yet many dispatch centers still rely on technology introduced in 1968, highlighting the need for modernized infrastructures and data-driven dispatching. This antiquated setup also raises questions of equitable service, as certain communities face over-policing or delayed responses, highlighting a system in urgent need of improvement. The implication is that bridging public and private technology tools can improve speed, accuracy, and transparency in emergency response. By focusing on

feature-level interventions—such as real-time location sharing, integrated apps for vulnerable populations, or wearable-alert systems—911 services can more effectively manage resources and address user trust and accountability concerns. This shift would expand the system’s capacity to deliver equitable support, reduce delays, and adapt more flexibly to local demands (See Fig A.6.) (Bataineh & Teixeira, 2024g).

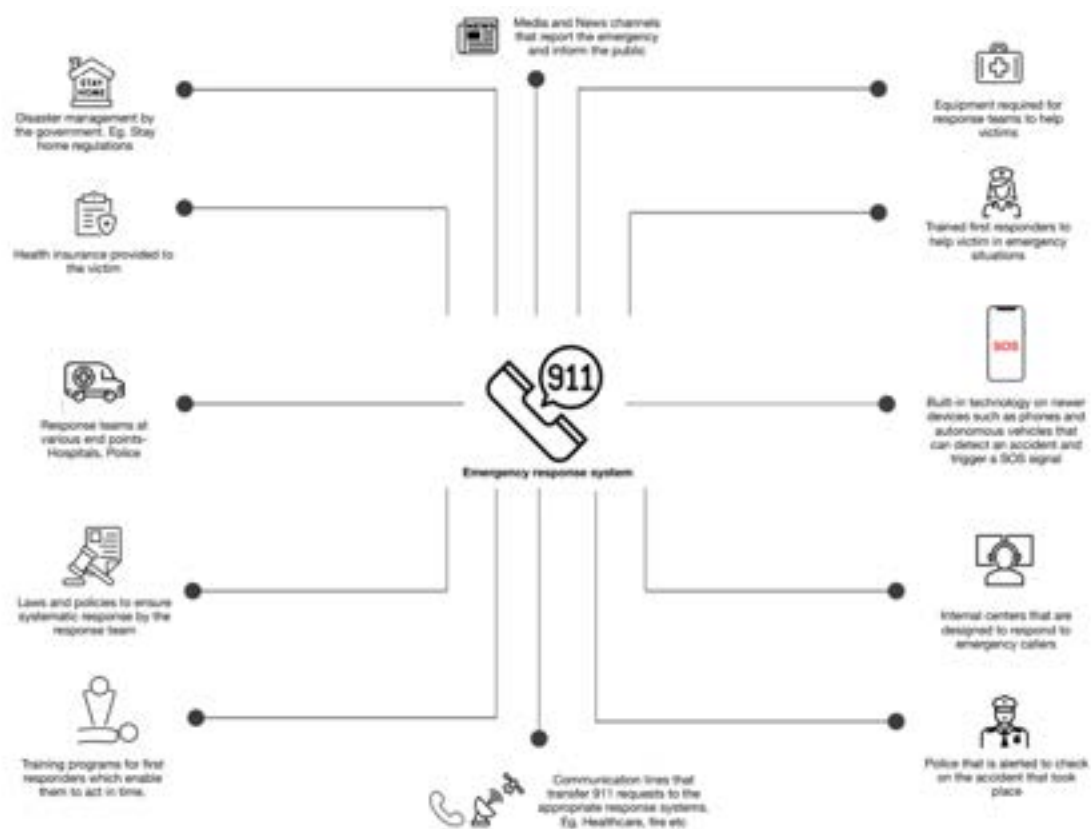


Figure A.6. Prescriptive Archetype of the 911 emergency response system.

A.3 Data Collection and Analysis Tools

- *Systems-Centric Design Frameworks:* A selection of design frameworks that investigate products and services at the intersection of systems—

Action Situations (Bataineh & Teixeira, 2024a), Causal Loops & System Dynamics (Bataineh & Teixeira, 2024d), and Anatomy of Infrastructure (Bataineh & Teixeira, 2024b), Archetype (Bataineh & Teixeira, 2024c), Intersectional Narratives (Bataineh & Teixeira, 2024f), and Hotspots & Dark Pattern (Bataineh & Teixeira, 2024e)—frameworks were used descriptively and prescriptively to visualize resource flows and agent interactions, identify power imbalances and potential biases, “dark patterns,” or structural inequities, and the identifying leverage points.

- *Data Collection:* Data were gathered mainly through Secondary Research—Industry reports, policy documents, and technical publications. Observation was also one of the methods in which reflections and further analysis were done through the synthesis using the design frameworks.

A.4 Implications for Strategic Choice-Making

1. *Reveling Resource Flow Constraints:* Focusing on how products/services operate at intersections of systems uncovered how financialization dominates decision-making, overshadowing sustainability and equity goals.
2. *Growth Opportunities:* Cases revealed specific leverage points and “hot spots” (e.g., data dashboards, user pre-order apps, AI-driven analytics) where targeted interventions can realign resource flows.
3. *Institutional and Agenda-driven interventions:* These macro assessments provided foundational agendas for sustainability, equity, and resilience, and how driving large-scale transformations depend on policy alignment, cross-

sector collaboration, and managing potential inequities or biases (particularly in 911 and embedded finance scenarios).

Collectively, these macro assessment case studies demonstrate how everyday infrastructures—ranging from food production to emergency services—are shaped by multi-level forces and stakeholder coalitions. By highlighting areas of inefficiency, inequity, and technological gaps, the cases guide the subsequent design phases and experiments of this dissertation. Their primary contribution is paving the way for strategic choice-making, wherein design interventions at the “feature” level can re-orient resources toward more sustainable, equitable, and value-generating outcomes.

APPENDIX B
SEMESTER-LONG WORKSHOP SERIES

This appendix summarizes the semester-long workshop series conducted in Spring 2023 and Spring 2024, designed to explore how design practitioners can embed strategic intent into “everyday infrastructures.” Each workshop ran for approximately 14 weeks and was anchored by two key projects per semester—examples include EV Charging Stations, Controlled Environment Agriculture, Embedded Finance, and In-Flight Meal Services. Graduate-level students with professional backgrounds collaborated in teams, supplemented by guidance from institutional partners specializing in business, engineering, or investment.

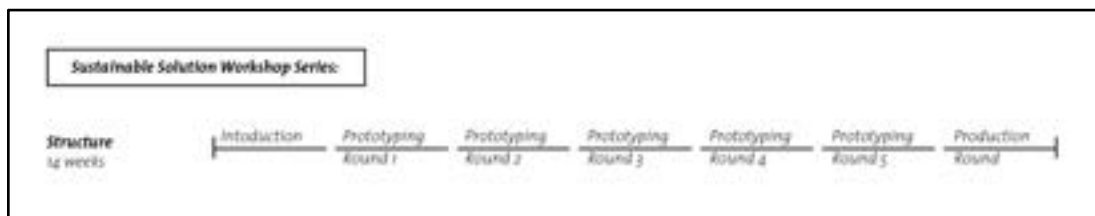


Figure B.1. Workshop Series Structure.

B.1 Structure and Objectives

As shown in (Fig B.1.), each workshop was divided into an introduction phase, five rounds of iterative prototyping, and a final “production” round. To onboard teams, a macro assessment case study for the project at hand and a design brief were distributed, and each prototyping round featured a structured prompt to direct design efforts, forcing participants to consider specific agendas such as equity, investability, or sustainability (See Fig 5.3. and Fig 5.4). Across multiple rounds, participants refined their ideas in response to critique sessions, weekly reading assignments, and discussions with industry collaborators. The overarching goal was to examine how strategic choice-making can manifest in new “archetypes” for reconfiguring resource flows. The central objectives

were (1) to embed strategic intent at the feature level and (2) to refine the Strategic Choice-Making Model (SCMM). Participants practiced systems thinking and design frameworks to envision new possibilities for reconfiguring resource flows in “everyday infrastructures.”

B.2 Data Collection

Participants delivered a final written report and prototyping cards capturing concepts sketches, findings, and emerging features and archetypes. The research team formed a “coach’s bench” to keep track of how the workshop evolved, developing prompts for each prototyping round and frameworks to be navigated and used by the participants, all to trace and evaluate how insights evolved from one iteration to the next. This mixed set of data—prototypes, reflection notes, prototyping cards, and reports—provided both qualitative depth and design artifacts, clarifying how participants negotiated constraints, tested assumptions, and arrived at actionable feature concepts.

B.3 Key Insights and Implications

Results from these workshops indicated that well-structured prompts help narrow participants’ attention to critical forces (e.g., “digitization,” “sustainability”) without constraining creativity, the prompts were essential in abstracting each infrastructure and platform, allowing participants to insert strategic intent into feature designs from the outset. The final outputs—future “archetypes”—illustrated how combining multiple features at different points in the ecosystem can realign resource flows and embed institutional purpose. Moreover, student teams continually revisited questions of

feasibility, stakeholder buy-in, and potential unintended consequences, showing that strategic choice-making requires balancing the big picture (macro agendas) with practical constraints (institutional context). This iterative, practice-based exploration validated the SCMM's capacity to incorporate macro-level agendas (digitization, equity, sustainability) at the micro-level of everyday service features.

Overall, the semester-long workshop series served as a dynamic test bed for refining the strategic choice-making model. The iterative blend of prompting, prototyping, and archotyping demonstrated how design can embed transformative agendas at the feature level. These findings informed subsequent refinements to both the methodology and the SCMM framework, as discussed in chapter five.

APPENDIX C

CONTROLLED EXPERIMENTS WITH SENIOR PRACTITIONERS

This appendix describes a set of three controlled experiments, each involving two experienced practitioners who explored how an institutional context shapes feature-level design choices. The experiments focused on 911 Emergency Response Systems (Bataineh & Teixeira, 2024g) and tested whether participants could use the Strategic Choice-Making Model (SCMM) to embed broader institutional purposes—such as equity, sustainability, or technological advancement—into the design of everyday features.

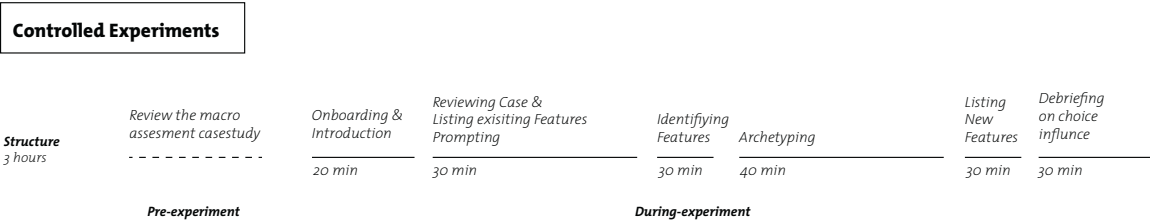


Figure C.1. Workshop Series Structure.

C.1 Experiment Structure and Participants

Each experiment lasted approximately three hours. Two practitioners with decision-making responsibilities from different sectors (e.g., public sector, tech-centric industry, private consulting) participated per session to represent distinct institutional vantage points. Before arriving, they reviewed a macro assessment of the 911 Emergency Response System, giving them baseline knowledge of systemic issues, stakeholder relationships, and resource-flow constraints. Figure C.1 illustrates the overall timeline, from reviewing the macro assessment to a structured debrief:

1. *Pre-Experiment (Self-Study)*: Participants read the 911 Emergency Response Systems Macro Assessment (Bataineh & Teixeira, 2024g), reviewed the “Project Brief” (see sample excerpt in Fig C.2), which summarized the 911

system’s legislative mandates (accountability, modernization, sustainability), and posed guiding questions (e.g., “How might we embed equity in emergency-response protocols?”).

The Brief Context: In recent years, a high-profile incident involving delayed emergency response in an underserved community led to widespread public outrage. Amplified by social media, this event triggered a national debate about the equity and effectiveness of the 911 Emergency Response System. In response, policymakers introduced legislation mandating the rethink of 911 emergency response systems

The legislative agenda aimed to achieve:

1. **Accountability and Transparency:** Addressing public demands for fairness and inclusivity by implementing systems to track and report response times and outcomes.
2. **Technological Modernization:** Leveraging advanced technologies to optimize response efficiency and minimize systemic biases.
3. **Sustainability:** Embedding environmentally conscious practices, such as electric responder fleets, into emergency operations to align with climate resilience goals.

The Brief Guiding Question:
How can the 911 Emergency Response System be effectively redesigned to enhance inclusivity, trust, and technological advancement, while addressing systemic biases, ensuring equitable access, and maintaining operational efficiency in the diverse communities it serves?

Figure C.2. Project Brief Sample Excerpt.

2. *Onboarding & Introduction (20 min):* The facilitator outlined the experiment’s objectives, presented the strategic choice-making model, and contextualized the research. They were introduced to the research context, the topic, and the structure of the experiment.
3. *Listing Existing Features (30 min):* Practitioners were then asked to list where 911 as a system fits on the infrastructure-platforms-mashups diagram (see Fig C.3).

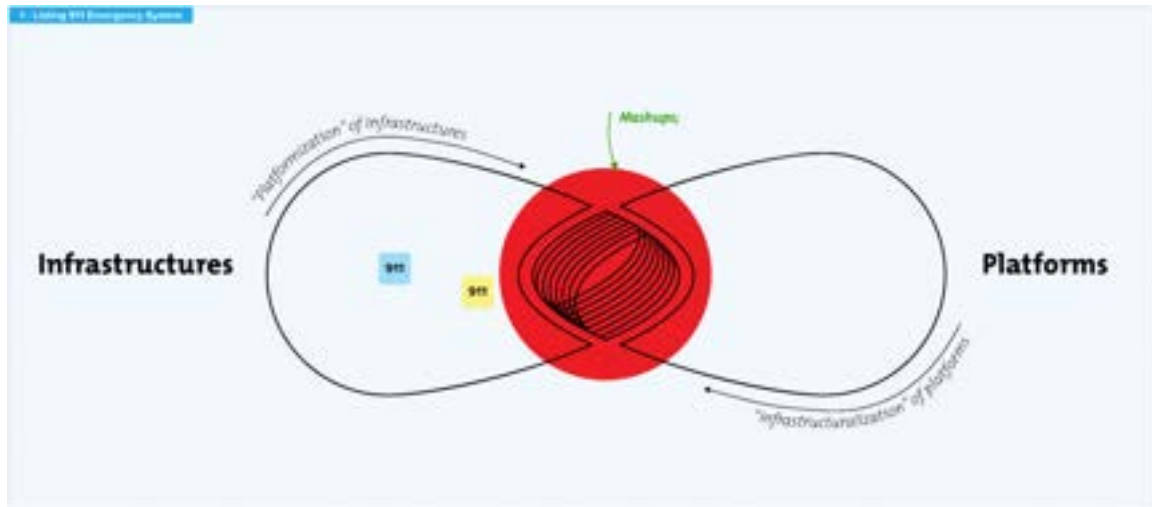


Figure C.3. 911 System Positioning to the Mashups.

4. *Identifying Archetypes and Features (30 min)*: Participants were then asked to select multiple archetypes (see Fig C.4.) and then take a deeper level to select the features (see Fig C.5.) from the selected archetypes that align with their institutional priorities.

	1	2	3	4	5	6	7	8	9	10
Archetypes	Emergency Response	Emergency Response	Emergency Response	Emergency Response	Emergency Response	Emergency Response	Emergency Response	Emergency Response	Emergency Response	Emergency Response
Description	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.
Impact	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.
Archetype Consideration	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.	Emergency response system that provides a central point of contact for emergency services.

Figure C.4. Pre-Generated Archetypes.

1

Archetypes	Infrastructure Platform Mashups	Features	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5	Feature 6	Feature 7	Feature 8	Feature 9	Feature 10	Feature 11	Feature 12
Description	Infrastructure Platform Mashups	Description	Feature 1 Description	Feature 2 Description	Feature 3 Description	Feature 4 Description	Feature 5 Description	Feature 6 Description	Feature 7 Description	Feature 8 Description	Feature 9 Description	Feature 10 Description	Feature 11 Description	Feature 12 Description
Impact	Infrastructure Platform Mashups	Impact	Feature 1 Impact	Feature 2 Impact	Feature 3 Impact	Feature 4 Impact	Feature 5 Impact	Feature 6 Impact	Feature 7 Impact	Feature 8 Impact	Feature 9 Impact	Feature 10 Impact	Feature 11 Impact	Feature 12 Impact
Archetype Consideration	Infrastructure Platform Mashups	Features Consideration	Feature 1 Consideration	Feature 2 Consideration	Feature 3 Consideration	Feature 4 Consideration	Feature 5 Consideration	Feature 6 Consideration	Feature 7 Consideration	Feature 8 Consideration	Feature 9 Consideration	Feature 10 Consideration	Feature 11 Consideration	Feature 12 Consideration

Figure C.5. Pre-Generated Features.

5. *List Identified Archetypes*: participants were then asked to position the pre-generated archetypes by visually mapping them across the boundary of infrastructure-platform mashups. Representing that through their institutional vantage point is how these archetypes are viewed (see Fig C.6.).

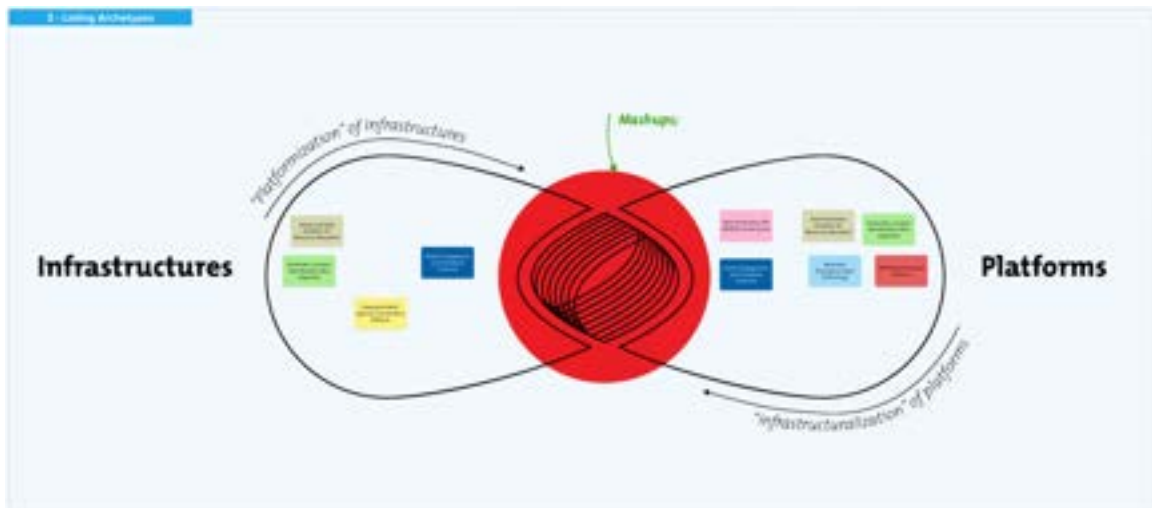


Figure C.6. Identified Archetypes Listed.

6. *Archotyping (40 min)*: Drawing on the multiple prompts shared with participants, they were again asked to select the prompts that align with their institutional priorities, then identified existing, new, or modified features that

could address macro-level concerns: for instance, a “Proactive Crisis Prediction with Big Data” approach. They assessed feasibility and alignment with the participant’s institutional context.

7. *Listing New Archetypes/Features (30 Min)*: Teams clustered their features into “archetypes,” visually mapping them across the boundary of infrastructures and platforms (see Fig C.7 for an example Archetype Map).

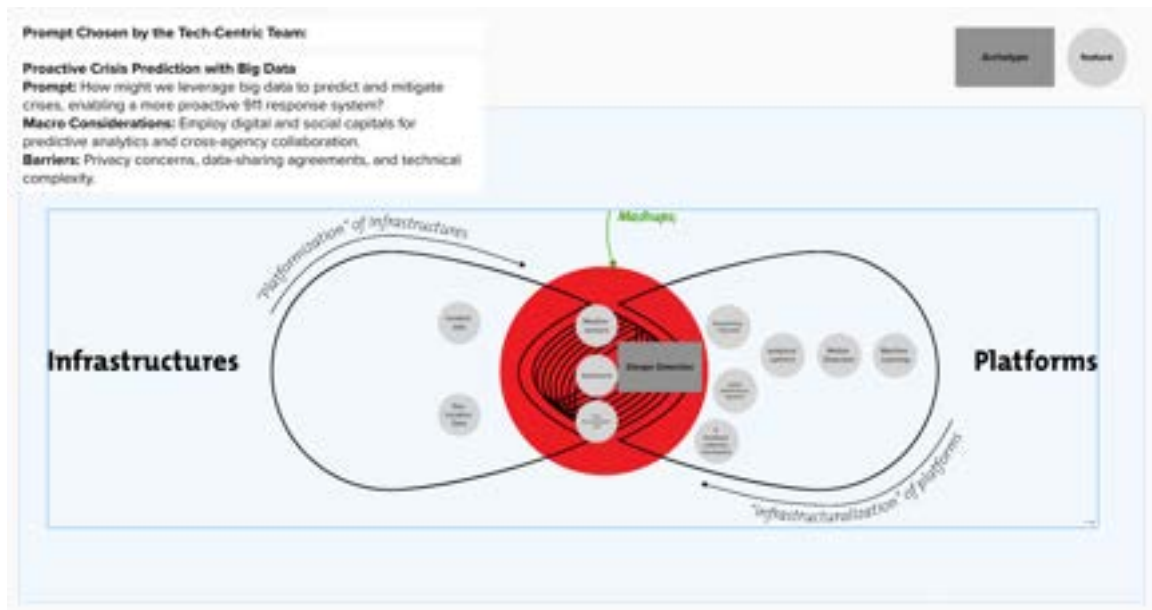


Figure C.7. Newly Formed Archetype.

8. *Listing New 911 System*: Having formed archetypes, participants were asked once again to position the 911 system after they had gone through the experiment.
9. *Debriefing on Choice Influence (30 min)*: Finally, participants reflected on how their institutional vantage point shaped the final solutions. One group from a civic context prioritized trust and public accountability, while a tech-centric consultancy prioritized data-driven solutions and algorithmic

scalability, and the traditional innovation-focused consultancy team emphasized usability and convenience

The overarching goal was to examine how the Strategic Choice-Making Model (SCMM) might help these practitioners align new “feature-level” interventions with broader institutional mandates, ranging from equity to modernization.

C.2 Flow of the Controlled Experiments

The Research through Design (RTD) (Gaver, 2012; P. J. Stappers et al., 2014; Pieter Jan Stappers & Giaccardi, 2017; Zimmerman et al., 2007) approach was applied to the controlled experiments. The discussions throughout the experiments were bi-directional, giving the opportunity to reflect on what is governing choice-making. While the experiment was structured and timed, it flows in a semi-structured way, allowing rapid “iterative prototyping” of specific elements of the research hypothesis. This approach highlighted how prompts and archetype formation served as core techniques for guiding participants’ decision-making. In practice, each experiment opened with a brief conversation that situationally reintroduced the 911 system context and outlined the SCMM. Practitioners then embarked on a series of design moves in which they proposed, debated, and iterated on new or modified 911 features. Underlying this structure was a common concern: how might an institution’s mission—public safety, technological leadership, ROI requirements—influence both the breadth of solutions considered and the final shape of those solutions?

Discussions were video and screen-recorded to capture not only the features participants selected but also the reasoning behind them. In particular, participants revealed motivations tied to funding streams, regulatory frameworks, community trust, or brand reputation. As the design exercise moved forward, these “real-world” constraints surfaced continually. This interplay between strategic vision and institutional practicality underscored how well the SCMM could (or could not) expand the range of feasible options. By the end of each session, practitioners had built a small set of archetypes, each representing a bundle of features layered onto existing emergency-response workflows—examples included big-data-enabled dispatching, wearable alert systems, or community-based feedback loops.

C.3 Emerging Themes

Across all three experiments, several themes recurred in how senior practitioners embraced or resisted strategic choice-making. One persistent tension appeared around equity vs. efficiency: participants from public-oriented institutions championed expansions of 911 services for vulnerable populations, while those from private or tech-focused contexts often gravitated toward sophisticated data tools promising efficiency gains. At the same time, each group recognized that major “infrastructural leaps” must tackle equity and data-security concerns simultaneously.

Another critical insight related to institutional readiness. Even when participants identified promising new features—like an AI-based call-routing system—some concluded their institutions lacked the legal clearance, data governance policies, or stakeholder alignment to adopt them swiftly. Nonetheless, the SCMM framework

appeared to encourage an expanded set of choices, prompting, for example, partnerships across institutions and stakeholders as means to navigate innovation challenges.

These controlled experiments demonstrated how senior practitioners could operationalize the Strategic Choice-Making Model to envision alternative futures for vital public services. By combining context-specific prompts, visual archotyping, and structured debriefs, the sessions revealed both opportunities (e.g., advanced data analytics for equity) and barriers (privacy, policy) in redesigning the 911 system. Future researchers may replicate or adapt this format—especially the purposeful use of macroscale briefs and archetype mapping—for other complex innovation scenarios.

APPENDIX D
STRATEGIC CHOICE-MAKING AS A FRAMEWORK

This appendix offers a more detailed, practical guide to enacting the Strategic Choice-Making Model (SCMM) introduced in Chapter 6. It elaborates on each of the four SCMM stages—Prompting, Prototyping, Archotyping, and Infrastructuring/Platforming—demonstrating how teams can iteratively move from broad agendas to implementable solutions in everyday infrastructures.

D.1 Overview of the SCMM

The SCMM is designed at the interplay of choice-making (expanding a range of possibilities) and decision-making (converging on feasible solutions) through an institutional context. Before beginning the first stage, teams typically perform a macro assessment, looking at both the problem space (e.g., existing constraints, stakeholder tensions) and the opportunity space (potential for equity or sustainability gains). This assessment ensures that any subsequent design actions consider multi-level forces—like digitization, financialization, equity, deglobalization, or sustainability—through a systems-thinking lens. Once the macro assessment is conceived as an input, the teams embarked on the four stages—Prompting, Prototyping, Archotyping, and Infrastructuring/Platforming—of the SCMM. These four stages encapsulate the design process of generating, evaluating, and implementing choices: features, products, and services (See Fig. D.1). The Stages transition through a cyclical, iterative layered approach.

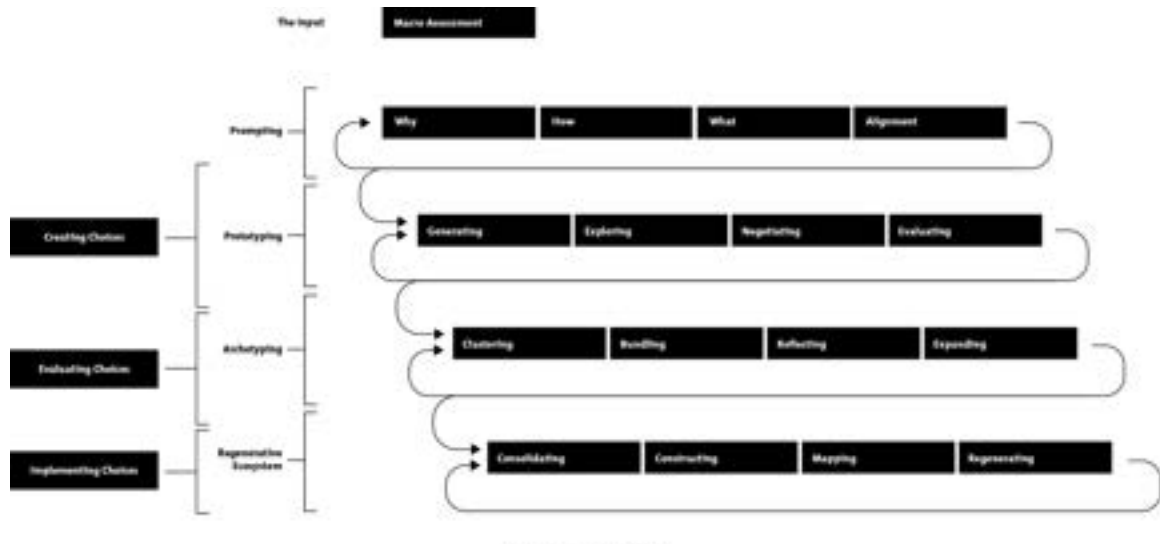


Figure D.1. Stages and Cycles of the SCMM.

D.2 SCMM Stages:

1. *The Prompting Stage:* This stage initiates the design process by defining the purpose, context, and constraints that shape strategic intent (see Fig D.2.).
 - a. *Formulate the Agenda (Why):* Teams integrate insights from the macro assessment (organizational goals, social or environmental mandates) to articulate the “why” behind their design work. These core agendas often respond to forces that shape decisions within the institution's context, such as sustainability, equity, financialization, digitization, and deglobalization. When transitioning toward these forces to affect system dynamics, considering the type of capital/s—natural, social, financial, cultural, manufactured, political, human, and digital—(Nogueira et

al., 2019) contributes to how and where change is taking place and determines the flow of resources. For example, addressing carbon reduction might become the guiding agenda in an EV-charging scenario.

- b. *How*: This stage relates to achieving a future state aligned with the strategic intent. Adapted from strategy choice cascades (Lafley & Martin, 2013), these questions pinpoint market or demographic segments, technologies, and competitive advantages. Further defining the unique value proposition of winning and leveraging the transformative force/s. In practice, a design team might decide to focus on a specific user base or regulatory environment to leverage transformative forces like digitization or deglobalization.
- c. *Context of Intervention (What)*: Teams clarify which aspect(s) of infrastructure or platform is relevant—such as “storing” or “monitoring” in the case of energy systems. They also note barriers (e.g., policy, funding, stakeholder conflict) that could limit solution adoption.
- d. *Alignment of Capabilities*: This stage is about realizing “what else must be considered.” The team pinpoints missing skills or partners necessary to enact the agenda. This might involve forming cross-industry coalitions, hiring data analytics experts, or building out internal knowledge about equitable design practices.



Figure D.2. The Prompting Stage.

2. The Prototyping Stage: Prototyping converts the strategic intent into tangible, testable concepts, facilitating iterative exploration and a dialogue between the generated prototypes (see Fig D.3.).
 - a. *Generating*: A wide range of early concepts are produced, each reflecting the strategic agenda defined in Prompting. Allowing the prototypes to function as smaller arguments within a broader debate, aiding in the critical process of design and development (Galey & Ruecker, 2010). Teams may brainstorm or sketch prototypes that address a specific predefined requirement in the prompt stage (e.g., a “smart charge management feature”) while embedding broader goals (like equity or sustainability).
 - b. *Exploring*: The team critically examines each prototype’s implications—technical feasibility, potential impacts on stakeholder relationships, and resource flow changes. Rapid feedback loops (workshop exercises, experts feedbacks) help refine or discard initial ideas. The role prototyping plays here is crucial in structuring explorations by Research through Design,

allowing for the understanding, exploring, and communicating of design concepts (P. J. Stappers et al., 2014). The iterative approach contributes to increasing the fidelity of prototypes, revealing tacit and explicit knowledge embedded within the prototype, and helps in addressing issues like the consequences and impacts of the choices being designed.

- c. *Negotiating*: Prototypes are iteratively aligned with real-world constraints, from financial viability to policy guidelines. Negotiating could involve balancing long-term gains (like ecological benefits) against immediate business metrics, ensuring final choices can survive organizational and market realities. Prototyping here plays a key role in embodying technical possibilities and actively convening new constituencies (Turner, 2014).
- d. *Evaluating*: Teams apply metrics related to their own institutional context against the strategic intent and the prompt—such as cost savings, user adoption potential, or resource-flow realignments—to assess each prototype’s performance. This helps confirm which ideas best serve the institution’s strategic intent, clearing the path for deeper development.



Figure D.3. The Prototyping Stage.

3. The Archetyping Stage: In Archetyping, teams cluster, refine, and combine prototype concepts into robust “archetypes” that reflect the new system’s strategic logic (see Fig D.4.).
 - a. *Clustering*: Design concepts that share similar goals or technical enablers are grouped together. For instance, multiple EV-charging features related to data analytics and user payment might form a single cluster addressing “intelligent station management.” This is where designers leverage the capability of probabilistic systems (Whitney, 2015) to see relationships among things that do not seem related yet supported by evidence. These clusters are mapped to realize the strategic intent and the infrastructure-platform functions.
 - b. *Bundling*: These clusters are assembled into coherent archetypes—new configurations of resources, offerings, and agent interactions, suggesting new value creation and growth (Romer, 2021). Each archetype aims to fulfill the strategic intent across an entire system, not just an isolated product.

- c. *Reflecting*: Teams examine each archetype against the broader flow of capital—natural, social, financial, cultural, manufactured, political, human, and digital—(Nogueira et al., 2019) to ensure it meets sustainability or equity mandates. This reflection step allows for iterative improvements as new data or user feedback emerges.
- d. *Expanding*: As new combinations disrupt the existing flow of resources and shape new ones, echoing Romer's belief in the limitless potential of combinatorial possibilities for allowing growth. Archetypes are pushed beyond existing institutional constraints to explore growth trajectories or alternative business models. The Archotyping stage is a dynamic process of creating and refining new combinations of strategic choices that can drive sustainable growth and innovation (Heskett, 2001; Owen, 2005). For example, a local energy-management archetype could be scaled to additional cities, bridging municipal collaboration for deeper carbon reductions.



Figure D.4. The Archotyping Stage.

4. *Infrastructuring/Platforming Stage*: Finally, teams consolidate and operationalize archetypes within an institutional fabric, building the ecosystem for sustained impact (see Fig D.5.).
 - a. *Consolidating*: Preliminary archetypes and features become well-defined. The design team solidifies which solutions will move forward, ensuring they can withstand shifts in policy or market conditions.
 - b. *Constructing*: Capabilities—technical, legal, human— needed to support the emerging innovation are built out to make each archetype functional. This can include forging partnerships, securing funding, or updating legacy systems.
 - c. *Mapping*: Stakeholders outline how these newly constructed solutions align with existing processes and structures. Where conflicts arise (e.g., data governance), teams define protocols or revise frameworks for improved synergy.
 - d. *Regenerating*: As the new system matures, the institution transitions from initial rollout to ongoing iteration. Continuous feedback loops measure performance against the original agenda (sustainability, equity, ROI), ensuring the ecosystem can evolve and adapt over time.



Figure D.5. Infrastructuring/Platforming Stage.

By following these four stages—Prompting, Prototyping, Archotyping, and Regenerative Ecosystem—teams can consistently align everyday design choices with broader strategic agendas. This cyclical, iterative approach helps maintain a focus on sustainability, equity, and user value while balancing institutional imperatives (like sustainability, budget constraints, or brand equity). Appendix D thus expands upon Chapter Six’s high-level explanation of the SCMM, detailing an operational blueprint for practitioners who aim to coordinate multi-stakeholder agendas in the complex domain of everyday infrastructures.

BIBLIOGRAPHY

Alexander, C. (1964). *Notes on the Synthesis of Form*. Harvard University Press.

AlphaBeta. (2017). *Valuing the SDG Prize: Unlocking Business Opportunities to Accelerate Sustainable and Inclusive Growth*.

Archer, B. (1995). The Nature of Research. *Co-Design*, 6–13.

Bataineh, D., & Teixeira, C. (2024a). *Action Situation Guide*.

<https://www.rethinkingsystems.design/>

Bataineh, D., & Teixeira, C. (2024b). *Anatomy of Infrastructures Guided*.

<https://www.rethinkingsystems.design/>

Bataineh, D., & Teixeira, C. (2024c). *Archetype Guide*.

<https://www.rethinkingsystems.design/>

Bataineh, D., & Teixeira, C. (2024d). *Causal Loops Guide*.

<https://www.rethinkingsystems.design/>

Bataineh, D., & Teixeira, C. (2024e). *Dark Patterns Hotspots & Dark Patterns Guide*.

<https://www.rethinkingsystems.design/>

Bataineh, D., & Teixeira, C. (2024f). *Intersectional Narratives Guide*.

<https://www.rethinkingsystems.design/>

Bataineh, D., & Teixeira, C. (2024g). *911 Emergency Response System; Infrastructuring Public Safety, Bridging Tech Gaps and Systemic Biases _ Case Study*.

<https://www.rethinkingsystems.design/>

- Bataineh, D., & Teixeira, C. (2024h). *Controlled Environment Agriculture: An Infrastructure for Sustainable Food Production _ Case Study*.
<https://www.rethinkingsystems.design/>
- Bataineh, D., & Teixeira, C. (2024i). *Embedded Finance: Infrastructuring Inclusive Finance _ Case Study*. <https://www.rethinkingsystems.design/>
- Bataineh, D., & Teixeira, C. (2024j). *EV Charging Stations: Infrastructuring Sustainable Energy Transitions _ Case Study*. <https://www.rethinkingsystems.design/>
- Bataineh, D., & Teixeira, C. (2024k). *Inflight Meals: Addressing Waste and Resource Inefficiencies _ Case Study*. <https://www.rethinkingsystems.design/>
- Bertola, P., & Teixeira, J. C. (2003). Design as a knowledge agent. *Design Studies*, 24(2), 181–194.
- Bonn, A., Akkermans, H., & Pini, M. (2019). *Culture, Citizenship, Participation - Comparative Perspectives from Latin America on Inclusive Education*. Pangea.
- Buchanan, R. (1992). Wicked Problems in Design Thinking. *Design Issues*, 8(2), 5–21.
- Buchanan, R. (1998). Branzi's Dilemma: Design in Contemporary Culture. *Design Issues*, 14(1), 3–20.
- Chachra, D. (2023). *How infrastructure works*. Riverhead Books.
- Chan, J. K. H. (2016). The ethics of working with wicked urban waste problems: The case of Singapore's Semakau Landfill. *Landscape and Urban Planning*, 154, 123–131.
- Chan, S., & Klareld, A.-S. (2022). Platform or infrastructure or both at once? Detangling the two concept's knotty cross-articulations. *SSRN Electronic Journal*.
<https://doi.org/10.2139/ssrn.4126285>

- Chapman, J. (2021). *Meaningful Stuff, Design That Lasts*. MIT Press.
- Churchman, C. W. (1967). Guest Editorial: Wicked Problems. *Management Science*, 14(4), B141–B142.
- Co, V. (2020). *Planet Centric Design*. Planet Centric Design.
<https://planetcentricdesign.com/>
- Collins, P. (1998). *Changing Ideals in Modern Architecture, 1750-1950*. McGill-Queen's Press - MQUP.
- Constantinides, P., Henfridsson, O., & Parker, G. G. (2018). Introduction—Platforms and Infrastructures in the Digital Age. *Information Systems Research*, 29(2), 381–400.
- Cooper, A., Reimann, R., Cronin, D., & Noessel, C. (2014). *About face: The essentials of interaction design* (4th ed.). Wiley.
- Cross, N. (1982). Designerly ways of knowing. *Design Studies*, 3(4), 221–227.
- Dalsgaard, P. (2014). Pragmatism and Design Thinking. *International Journal of Design*, 8. <https://www.researchgate.net/publication/288295454>
- Di Salvo, C. (2009). Design and the Construction of Publics. *Design Issue*.
- Dilnot, C. (Ed.). (2011). *A John Heskett Reader: Design, History, Economics*. Bloomsbury Academic.
- Dorst, K. (2011). The core of 'design thinking' and its application. *Design Studies*, 32(6), 521–532.
- Ehn, P. (October 1-4 2008). Participation in Design Things. *Proceedings of the Tenth Conference on Participatory Design (PDC 2008)*.
<https://doi.org/10.1145/1795234.1795248>

- Epstein, G. A. (Ed.). (2006). *Financialization and the World Economy*. Edward Elgar Publishing.
- Escobar, A. (2018). *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*. Duke University Press.
- Galey, A., & Ruecker, S. (2010). How a prototype argues. *Literary and Linguistic Computing*, 25(4), 405–424.
- Gaver, W. (2012, May 5). What should we expect from research through design? *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. CHI '12: CHI Conference on Human Factors in Computing Systems, Austin Texas USA. <https://doi.org/10.1145/2207676.2208538>
- Gerlach, R. (2020). *Tools for Sustainable Innovation*. Threebility.
<https://www.threebility.com/tools>
- Goldschmidt, G. (2014). *Linkography: Unfolding the design process*. Mit Press.
- Gottlieb, R. (2022). *Care-Centered Politics: from the home to the planet*. The MIT Press.
- Graham, S., & Marvin, S. (2002). *Splintering urbanism: networked infrastructures, technological mobilities and the urban condition*. Routledge.
- Haraway, D. (1991). *A Cyborg Manifesto: Science, Technology, and Socialist-Feminism in the Late 20th Century*.
- Hatcher, G., Ion, W., Maclachlan, R., Marlow, M., Simpson, B., Wilson, N., & Wodehouse, A. (2018). Using linkography to compare creative methods for group ideation. *Design Studies*, 58, 127–152.
- Heskett, J. (2001). Past, Present, and Future in Design for Industry. *Design Issues*, 17(1), 18–26.

- Hickman, L. (2001). *Philosophical tools for technological culture: Putting pragmatism to work*. Indiana University Press.
- Hickman, L. A., & Alexander, T. (Eds.). (1998). *The essential Dewey, volume 1*. Indiana University Press.
- Hjorth, P., & Bagheri, A. (2006). Navigating towards sustainable development: A system dynamics approach. *Futures*, 38(1), 74–92.
- Holm, K. L., & Larson, T. C. (1953). *The Diagram of The Inter-Related Factors Connected to Community Relationships, part of the “Development Index.”*
https://www.researchgate.net/publication/327337385_Story_systems_The_potential_of_transmedia_storytelling_as_material_embodiment_of_a_collective_enactment_of_place_and_identity
- IEA. (2021). *World Energy Outlook 2021* (No. 383). International Energy Agency.
www.iea.org
- Ioannou, S., & Wójcik, D. (2019). On financialization and its future. *Environment & Planning A*, 51(1), 263–271.
- IPCC. (2022). *Mitigation of Climate Change*. Intergovernmental Panel on Climate Change.
- Kenney, M., & Zysman, J. (2016, March 29). *The Rise of the Platform Economy*. Issues.
<https://issues.org/rise-platform-economy-big-data-work/>
- Koffka, K. (1935). *Principles Of Gestalt Psychology*. London.
- Kohler, W. (1938). *The Place of Value in a World of Facts*. Liveright.
- Lafley, A. G., & Martin, R. (2013). *Playing to Win: How Strategy Really Works*. Harvard Business Review Press.

- Latour, B. (1992). Where are the Missing Masses? A Sociology of Few Mundane Objects. In *Shaping Technology/Building Society. Studies in Sociotechnical Change* (pp. 151–180).
- Lemke, T. (2021). *The Government of Things*.
<https://doi.org/10.18574/nyu/9781479808816.001.0001>
- Lutz, D. (2022). *The Life-Centered Design Guide: Product Design Strategies and Tools Supporting a Thriving Planet for All*.
- Martin, R. L. (2020). *When More Is Not Better: Overcoming America's Obsession with Economic Efficiency*. Harvard Business Press.
- Mazzucato M. (2020). *The value of everything: Making and taking in the global economy*. Shi Bao Chu Ban.
- Meharry, J. (2022). *Sensemaking for Power Asymmetries in Anti-Oppressive Design Practice* (C. Teixeira, Ed.) [Doctor of Philosophy in Design]. Illinois Institute of Technology - Institute of Design.
- Morton, T. (2013). *Hyperobjects: Philosophy and Ecology after the End of the World*. Univ Of Minnesota Press.
- Neimand, A., Asorey, N., Christiano, A., & Wallace, Z. (September 29 2021). Why Intersectional Stories Are Key to Helping the Communities We Serve. *Stanford Social Innovation Review*.
https://ssir.org/articles/entry/why_intersectional_stories_are_key_to_helping_the_communities_we_serve

- Nogueira, A. (2019). *Sustainable Solutions in Complex Spaces of Innovation* (C. Teixeira, W. Ashton, & P. Whitney, Eds.) [Doctor of Philosophy in Design]. Illinois Institute of Technology.
- Nogueira, A., Ashton, W. S., & Teixeira, C. (2019). Expanding perceptions of the circular economy through design: Eight capitals as innovation lenses. *Resources, Conservation and Recycling*, 149, 566–576.
- OECD. (2019). *Under pressure: the squeezed middle class*. Organisation for Economic Co-operation and Development. <https://doi.org/10.1787/689afed1-en>
- Ostapchuk, J. (2024). *Capital Design: The Role of Design in Institutional Capital Allocation* (C. Teixeira, Ed.) [Doctor of Philosophy in Design]. Illinois Institute of Technology - Institute of Design.
- Ostrom, E. (2011). Background on the Institutional Analysis and Development Framework. *Policy Studies Journal: The Journal of the Policy Studies Organization*, 39, 7–27.
- Owen, C. L. (2005, May). *Societal Responsibilities. Growing the Role of Design*. The International Conference on Planning and Design Creativity, Interaction and Sustainable Development, National Cheng Kung University, Tainan, Taiwan.
- Ozenfant, A. (1952). *Foundations of Modern Art*. Dover Publications.
- Parker, G. G., Van Alstyne, M. W., & Choudary, S. P. (2016). *Platform Revolution: How Networked Markets Are Transforming the Economy and How to Make Them Work for You*. W. W. Norton & Company.

- Plantin, J.-C., Lagoze, C., Edwards, P. N., & Sandvig, C. (2018). Infrastructure studies meet platform studies in the age of Google and Facebook. *New Media & Society*, 20(1), 293–310.
- Poggenpohl, S. H. (2017). Blindspots in Economics and Design: A Review of John Heskett's Design and the Creation of Value. *She Ji: The Journal of Design, Economics, and Innovation*, 3(4), 251–261.
- Potter, M. (2017). *Story systems. The potential of transmedia storytelling as material embodiment of a collective enactment of place and identity.*
- Puig de la Bellacasa, M. (2017). *Matters of care : speculative ethics in more than human worlds.* University of Minnesota Press.
- Raworth, K. (2017). *Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist.* Chelsea Green Publishing.
- Rodrik, D. (2019). *Straight talk on trade.* Princeton University Press.
- Romer, P. [@ciebuniversitatbasel7103]. (2021, September 24). *The economics of innovation | Paul Romer | Innoscape Talk #2.* CIEB Universität Basel.
<https://www.youtube.com/watch?v=zyMa2y9dzBo>
- Sachs, J. D. (2015). *The Age of Sustainable Development.* Columbia University Press.
- Sevinç, D., Polat, A., Sevil, T., & Sevil, G. (2023). How does deglobalization affect economic growth? *Ekonomi Politika & Finans Araştırmaları Dergisi*, 8(1), 1–19.
- Srnicek, N. (2017). *Platform Capitalism.* Polity Press.
- Stappers, P. J., Sleeswijk Visser, F., & Keller, A. I. (2014). The role of prototypes and frameworks for structuring explorations by research through design. In *The Routledge Companion to Design Research* (pp. 163–174).

- Stappers, Pieter Jan. (2013). Prototypes as Central Vein for Knowledge Development. In L. Valentine (Ed.), *Prototyping: Design and Craft in the 21st Century* (pp. 85–98). Bloomsbury.
- Stappers, Pieter Jan, & Giaccardi, E. (2017). Research through Design. In *The Encyclopedia of Human-Computer Interaction*.
- Stern, A., & Siegelbaum, S. (2019). Special issue: Design and neoliberalism. *Design and Culture*, 11(3), 265–277.
- Stiglitz, J. E. (2019). *People, Power, and Profits: Progressive Capitalism for an Age of Discontent*. W. W. Norton & Company.
- Teixeira, C. (2002). *Design Knowledge and Business Innovation* [Doctoral]. Institute of Design at Illinois Tech .
- Teixeira, C. (2017). Transforming Design Matters. *She Ji: The Journal of Design, Economics, and Innovation*, 3(1), 1–2.
- Thackara, J. (2005). *In the Bubble: Designing in a Complex World*. The MIT Press.
- Tooze, A. (2018). *Crashed: How a Decade of Financial Crises Changed the World*. Penguin Books.
- Tricks, H. (2024, January 18). Why BlackRock is betting billions on infrastructure. *The Economist*. <https://www.economist.com/business/2024/01/18/why-blackrock-is-betting-billions-on-infrastructure?giftId=45df2db9-95f5-40a0-aedf-4c264e1cf6e5>
- Turner, F. (2014). Prototypes. In B. Peters (Ed.), *Digital Keywords A Vocabulary of Information Society and Culture* (pp. 256–268). Princeton University Press.

- UNDESA. (2021). *World social report 2021: Reconsidering rural development*. {United Nations Department of Economic and Social Affairs};
<https://doi.org/10.18356/9789216040628>
- UNDESA. (2023). *World population prospects 2022: summary of results*. United Nations: Department of Economic and Social Affairs: Population Division.
- Verganti, R., Vendraminelli, L., & Iansiti, M. (2020). Innovation and Design in the Age of Artificial Intelligence. *The Journal of Product Innovation Management*, 37(3), 212–227.
- Wakkary, R. (2021, December 1). *Things we could design*. MIT Press; The MIT Press, Massachusetts Institute of Technology.
<https://mitpress.mit.edu/9780262542999/things-we-could-design/>
- Wamba, S. F., Gunasekaran, A., Akter, S., Ren, S. J.-F., Dubey, R., & Childe, S. J. (2017). Big data analytics and firm performance: Effects of dynamic capabilities. *Journal of Business Research*, 70(C), 356–365.
- WEF. (2022). *The Global Risks Report 2022* (Edition, 17th;). World Economic Forum.
<https://www.weforum.org/publications/global-risks-report-2022/>
- Wendt, T. (2017, January 10). *Empathy as Faux Ethics - EPIC*.
<https://www.epicpeople.org/empathy-faux-ethics/>
- Whitney, P. (2015). Design and the Economy of Choice. *She Ji: The Journal of Design, Economics, and Innovation*, 1(1), 58–80.
- WHO. (2024). *World Health Statistics 2024: Monitoring health for the SDGs, Sustainable Development Goals*. World Health Organization.

Wizinsky, M. (2022). *Design After Capitalism; Transforming Design Today for an Equitable Tomorrow*. MIT Press.

Zimmerman, J., Forlizzi, J., & Evenson, S. (2007). Research Through Design as a Method for Interaction Design Research in HCI. *Human-Computer Interaction Institute*.

Zuboff, S. (2020, January 24). You Are Now Remotely Controlled: Surveillance capitalists control the science and the scientists, the secrets and the truth. *The New York Times*.